

Bioscara

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1 Bioscara C++ API Documentation	1
2 Todo List	3
3 Namespace Index	5
3.1 Namespace List	5
4 Hierarchical Index	7
4.1 Class Hierarchy	7
5 Class Index	9
5.1 Class List	9
6 File Index	11
6.1 File List	11
7 Namespace Documentation	13
7.1 bioscara hardware_drivers Namespace Reference	13
7.1.1 Enumeration Type Documentation	13
7.1.1.1 err_type_t	13
7.1.2 Function Documentation	14
7.1.2.1 error_to_string()	14
7.2 bioscara hardware_interfaces Namespace Reference	14
7.2.1 Variable Documentation	14
7.2.1.1 HW_IF_HOME	14
7.3 bioscara joint_firmware Namespace Reference	15
7.3.1 Detailed Description	16
7.3.2 Enumeration Type Documentation	16
7.3.2.1 stp_reg_t	16
7.3.3 Function Documentation	17
7.3.3.1 blocking_handler()	17
7.3.3.2 loop()	17
7.3.3.3 non_blocking_handler()	18
7.3.3.4 readValue()	18
7.3.3.5 receiveEvent()	18
7.3.3.6 requestEvent()	19
7.3.3.7 set_flags_for_blocking_handler()	19
7.3.3.8 setup()	19
7.3.3.9 stall_threshold()	19
7.3.3.10 writeValue()	20
7.3.4 Variable Documentation	20
7.3.4.1 blk_reg	20
7.3.4.2 reg	20
7.3.4.3 rx_buf	20

7.3.4.4 rx_data_ready	20
7.3.4.5 rx_length	20
7.3.4.6 stepper	21
7.3.4.7 tx_buf	21
7.3.4.8 tx_length	21
7.4 bioscara_rviz_plugin Namespace Reference	21
7.5 single_trigger_controller Namespace Reference	21
7.5.1 Typedef Documentation	21
7.5.1.1 CallbackReturn	21
7.5.1.2 CmdType	22
7.5.1.3 InterfacesNames	22
7.5.1.4 MapOfReferencesToCommandInterfaces	22
7.5.1.5 MapOfReferencesToStateInterfaces	22
7.5.1.6 StateInterfaces	22
7.5.1.7 StateType	22
8 Class Documentation	23
8.1 bioscara_hardware_drivers::BaseGripper Class Reference	23
8.1.1 Detailed Description	24
8.1.2 Constructor & Destructor Documentation	24
8.1.2.1 BaseGripper()	24
8.1.2.2 ~BaseGripper()	25
8.1.3 Member Function Documentation	25
8.1.3.1 deinit()	25
8.1.3.2 disable()	26
8.1.3.3 enable()	26
8.1.3.4 getPosition()	26
8.1.3.5 init()	27
8.1.3.6 retrieve_last_position()	27
8.1.3.7 save_last_position()	27
8.1.3.8 setOffset()	27
8.1.3.9 setPosition()	28
8.1.3.10 setReduction()	28
8.1.4 Member Data Documentation	28
8.1.4.1 _backup_init_pos	28
8.1.4.2 _max	28
8.1.4.3 _min	28
8.1.4.4 _new_cmd_time	29
8.1.4.5 _offset	29
8.1.4.6 _pos	29
8.1.4.7 _pos_get	29
8.1.4.8 _reduction	29

8.2 bioscara_hardware_drivers::BaseJoint Class Reference	29
8.2.1 Detailed Description	32
8.2.2 Member Enumeration Documentation	32
8.2.2.1 stp_reg_t	32
8.2.3 Constructor & Destructor Documentation	33
8.2.3.1 BaseJoint()	33
8.2.3.2 ~BaseJoint()	33
8.2.4 Member Function Documentation	33
8.2.4.1 _home()	33
8.2.4.2 checkOrientation()	34
8.2.4.3 deinit()	34
8.2.4.4 disable()	34
8.2.4.5 disableCL()	35
8.2.4.6 enable()	35
8.2.4.7 enableStallguard()	35
8.2.4.8 getCurrentBCmd()	36
8.2.4.9 getFlags() [1/2]	36
8.2.4.10 getFlags() [2/2]	36
8.2.4.11 getPosition()	37
8.2.4.12 getVelocity()	37
8.2.4.13 home()	37
8.2.4.14 init()	38
8.2.4.15 isBusy()	38
8.2.4.16 isEnabled()	38
8.2.4.17 isHomed()	39
8.2.4.18 isStalled()	39
8.2.4.19 moveSteps()	39
8.2.4.20 postHoming()	40
8.2.4.21 setBrakeMode()	40
8.2.4.22 setDriveCurrent()	40
8.2.4.23 setHoldCurrent()	41
8.2.4.24 setMaxAcceleration()	41
8.2.4.25 setMaxVelocity()	41
8.2.4.26 setPosition()	42
8.2.4.27 setVelocity()	42
8.2.4.28 startHoming()	42
8.2.4.29 stop()	43
8.2.4.30 wait_while_busy()	43
8.2.5 Member Data Documentation	43
8.2.5.1 current_b_cmd	43
8.2.5.2 flags	44
8.2.5.3 name	44

8.3 bioscara_hardware_interfaces::BioscaraArmHardwareInterface Class Reference	44
8.3.1 Detailed Description	46
8.3.2 Member Function Documentation	47
8.3.2.1 activate_joint()	47
8.3.2.2 deactivate_joint()	47
8.3.2.3 on_activate()	47
8.3.2.4 on_cleanup()	48
8.3.2.5 on_configure()	48
8.3.2.6 on_deactivate()	49
8.3.2.7 on_error()	49
8.3.2.8 on_init()	50
8.3.2.9 on_shutdown()	51
8.3.2.10 perform_command_mode_switch()	51
8.3.2.11 prepare_command_mode_switch()	52
8.3.2.12 read()	53
8.3.2.13 split_interface_string_to_joint_and_name()	53
8.3.2.14 start_homing()	54
8.3.2.15 stop_homing()	54
8.3.2.16 write()	54
8.3.3 Member Data Documentation	55
8.3.3.1 _joint_cfg	55
8.3.3.2 _joint_command_modes	55
8.3.3.3 _joints	56
8.3.3.4 _new_joint_command_modes	56
8.3.3.5 _ordered_joint_state_interfaces_ptr	56
8.3.3.6 mtx	57
8.4 bioscara_hardware_interfaces::BioscaraGripperHardwareInterface Class Reference	57
8.4.1 Detailed Description	58
8.4.2 Member Function Documentation	58
8.4.2.1 on_activate()	58
8.4.2.2 on_cleanup()	59
8.4.2.3 on_configure()	59
8.4.2.4 on_deactivate()	59
8.4.2.5 on_error()	60
8.4.2.6 on_init()	60
8.4.2.7 on_shutdown()	61
8.4.2.8 read()	61
8.4.2.9 write()	61
8.4.3 Member Data Documentation	62
8.4.3.1 _gripper	62
8.4.3.2 _gripper_cfg	62
8.4.3.3 _last_pos	62

8.4.3.4 _vel	62
8.5 bioscara_rviz_plugin::BioscaraPanel Class Reference	63
8.5.1 Detailed Description	65
8.5.2 Constructor & Destructor Documentation	65
8.5.2.1 BioscaraPanel()	65
8.5.2.2 ~BioscaraPanel()	65
8.5.3 Member Function Documentation	66
8.5.3.1 arm_en_btn_cb	66
8.5.3.2 check_activation_conditions()	66
8.5.3.3 cm_state_callback()	66
8.5.3.4 configure_controller()	66
8.5.3.5 ctrl_en_btn_cb()	67
8.5.3.6 dynamic_joint_state_msg_to_map()	67
8.5.3.7 ensure_jsb_is_active()	67
8.5.3.8 gripper_en_btn_cb	67
8.5.3.9 homing_cmd()	68
8.5.3.10 joint_state_callback()	68
8.5.3.11 named_lcs_msg_to_map()	68
8.5.3.12 onInitialize()	68
8.5.3.13 prepopulate_joint_state_map()	69
8.5.3.14 prepopulate_state_map()	69
8.5.3.15 print_cm_map()	70
8.5.3.16 set_controller_state()	70
8.5.3.17 set_en_btn()	70
8.5.3.18 set_hardware_component_state()	70
8.5.3.19 set_homing_state_label()	71
8.5.3.20 set_state_label()	71
8.5.3.21 switch_controllers()	71
8.5.3.22 target_state_from_current()	72
8.5.3.23 update_homing_grp_state()	72
8.5.3.24 update_homing_state_labels()	72
8.5.3.25 update_state_label_and_btn()	73
8.5.3.26 update_state_labels_and_btns()	73
8.5.4 Member Data Documentation	73
8.5.4.1 cm_state_subscription_	73
8.5.4.2 configure_controller_client_	73
8.5.4.3 controller_states_	74
8.5.4.4 hardware_state_client_	74
8.5.4.5 hardware_states_	74
8.5.4.6 homing_publisher_	74
8.5.4.7 joint_state_subscription_	74
8.5.4.8 joint_states_	74

8.5.4.9 node_	75
8.5.4.10 prune_timer_	75
8.5.4.11 switch_controller_client_	75
8.5.4.12 ui_	75
8.6 bioscara_hardware_drivers::Gripper Class Reference	75
8.6.1 Detailed Description	77
8.6.2 Constructor & Destructor Documentation	77
8.6.2.1 Gripper()	77
8.6.3 Member Function Documentation	77
8.6.3.1 disable()	77
8.6.3.2 enable()	78
8.6.3.3 setPosition()	78
8.6.3.4 setServoPosition()	78
8.6.4 Member Data Documentation	79
8.6.4.1 _freq	79
8.6.4.2 _pwm	79
8.7 bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t Struct Reference	79
8.7.1 Detailed Description	79
8.7.2 Member Data Documentation	80
8.7.2.1 init_pos	80
8.7.2.2 max	80
8.7.2.3 min	80
8.7.2.4 offset	80
8.7.2.5 reduction	80
8.8 bioscara_hardware_drivers::Joint Class Reference	80
8.8.1 Detailed Description	83
8.8.2 Constructor & Destructor Documentation	84
8.8.2.1 Joint()	84
8.8.2.2 ~Joint()	84
8.8.3 Member Function Documentation	85
8.8.3.1 _home()	85
8.8.3.2 checkCom()	85
8.8.3.3 checkOrientation()	85
8.8.3.4 deinit()	86
8.8.3.5 disableCL()	86
8.8.3.6 enable()	86
8.8.3.7 enableStallguard()	87
8.8.3.8 getFlags()	87
8.8.3.9 getHomingOffset()	87
8.8.3.10 getPosition()	88
8.8.3.11 getVelocity()	88
8.8.3.12 init()	88

8.8.3.13 moveSteps()	89
8.8.3.14 postHoming()	89
8.8.3.15 read()	89
8.8.3.16 setBrakeMode()	90
8.8.3.17 setDriveCurrent()	90
8.8.3.18 setHoldCurrent()	90
8.8.3.19 setHomingOffset()	92
8.8.3.20 setMaxAcceleration()	92
8.8.3.21 setMaxVelocity()	92
8.8.3.22 setPosition()	93
8.8.3.23 setVelocity()	93
8.8.3.24 stop()	93
8.8.3.25 write()	94
8.8.4 Member Data Documentation	94
8.8.4.1 address	94
8.8.4.2 handle	94
8.8.4.3 max	95
8.8.4.4 min	95
8.8.4.5 offset	95
8.8.4.6 reduction	95
8.9 bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t Struct Reference	95
8.9.1 Detailed Description	96
8.9.2 Member Data Documentation	96
8.9.2.1 drive_current	96
8.9.2.2 hold_current	96
8.9.2.3 homing	97
8.9.2.4 i2c_address	97
8.9.2.5 max	97
8.9.2.6 max_acceleration	97
8.9.2.7 max_velocity	97
8.9.2.8 min	98
8.9.2.9 reduction	98
8.9.2.10 stall_threshold	98
8.10 bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t Struct Reference	98
8.10.1 Detailed Description	99
8.10.2 Member Data Documentation	99
8.10.2.1 acceleration	99
8.10.2.2 current	99
8.10.2.3 speed	99
8.10.2.4 threshold	100
8.11 bioscara_joint_firmware::Lowpass Class Reference	100

8.11.1 Detailed Description	100
8.11.2 Constructor & Destructor Documentation	100
8.11.2.1 Lowpass()	100
8.11.3 Member Function Documentation	101
8.11.3.1 resetState()	101
8.11.3.2 updateState()	101
8.11.4 Member Data Documentation	101
8.11.4.1 K	101
8.11.4.2 tau	101
8.11.4.3 Ts	101
8.11.4.4 x	102
8.12 bioscara_hardware_drivers::MockGripper Class Reference	102
8.12.1 Detailed Description	103
8.12.2 Constructor & Destructor Documentation	103
8.12.2.1 MockGripper()	103
8.13 bioscara_hardware_drivers::MockJoint Class Reference	104
8.13.1 Detailed Description	106
8.13.2 Constructor & Destructor Documentation	107
8.13.2.1 MockJoint()	107
8.13.3 Member Function Documentation	107
8.13.3.1 _home()	107
8.13.3.2 checkOrientation()	107
8.13.3.3 disable()	108
8.13.3.4 enable()	108
8.13.3.5 getDeltaT()	108
8.13.3.6 getFlags()	110
8.13.3.7 getPosition()	110
8.13.3.8 getVelocity()	110
8.13.3.9 isHomed()	111
8.13.3.10 setPosition()	111
8.13.3.11 setVelocity()	111
8.13.3.12 stop()	112
8.13.4 Member Data Documentation	112
8.13.4.1 async_start_time	112
8.13.4.2 last_set_position	112
8.13.4.3 last_set_velocity	112
8.13.4.4 op_mode	112
8.13.4.5 q	113
8.13.4.6 qd	113
8.14 bioscara_joint_firmware::MovMax Class Reference	113
8.14.1 Detailed Description	113
8.14.2 Constructor & Destructor Documentation	113

8.14.2.1 MovMax()	113
8.14.3 Member Function Documentation	114
8.14.3.1 updateState()	114
8.14.4 Member Data Documentation	114
8.14.4.1 cb_data	114
8.14.4.2 cb_index	114
8.14.4.3 M	114
8.15 RPI_PWM Class Reference	114
8.15.1 Detailed Description	115
8.15.2 Constructor & Destructor Documentation	115
8.15.2.1 ~RPI_PWM()	115
8.15.3 Member Function Documentation	115
8.15.3.1 disable()	115
8.15.3.2 enable()	116
8.15.3.3 setDutyCycle()	116
8.15.3.4 setDutyCycleNS()	116
8.15.3.5 setPeriod()	116
8.15.3.6 start()	116
8.15.3.7 stop()	117
8.15.3.8 writeSYS()	117
8.15.4 Member Data Documentation	117
8.15.4.1 chippath	117
8.15.4.2 per	117
8.15.4.3 pwmpath	117
8.16 single_trigger_controller::SingleTriggerController Class Reference	117
8.16.1 Detailed Description	119
8.16.2 Constructor & Destructor Documentation	119
8.16.2.1 SingleTriggerController()	119
8.16.3 Member Function Documentation	119
8.16.3.1 apply_command()	119
8.16.3.2 apply_state_value()	120
8.16.3.3 check_if_configured_interfaces_matches_received()	120
8.16.3.4 command_interface_configuration()	121
8.16.3.5 create_map_of_references_to_interfaces()	121
8.16.3.6 get_state_interfaces_names()	121
8.16.3.7 initialize_state_msg()	122
8.16.3.8 on_activate()	122
8.16.3.9 on_configure()	122
8.16.3.10 on_deactivate()	123
8.16.3.11 on_init()	123
8.16.3.12 state_interface_configuration()	123
8.16.3.13 store_command_interface_types()	124

8.16.3.14 store_state_interface_types()	124
8.16.3.15 update()	124
8.16.3.16 update_commands()	124
8.16.3.17 update_dynamic_map_parameters()	125
8.16.3.18 update_states()	125
8.16.4 Member Data Documentation	125
8.16.4.1 command_interface_types_	125
8.16.4.2 command_interfaces_map_	125
8.16.4.3 command_subscriber_	126
8.16.4.4 param_listener_	126
8.16.4.5 params_	126
8.16.4.6 realtime_state_publisher_	126
8.16.4.7 state_interface_types_	126
8.16.4.8 state_interfaces_map_	126
8.16.4.9 state_msg_	127
8.16.4.10 state_publisher_	127
9 File Documentation	129
9.1 docs/doxygen/index.md File Reference	129
9.2 lib/joint_firmware/joint/configuration.h File Reference	129
9.2.1 Detailed Description	129
9.2.2 Macro Definition Documentation	130
9.2.2.1 ADR	130
9.2.2.2 MAXACCEL	130
9.2.2.3 MAXVEL	130
9.3 configuration.h	130
9.4 lib/joint_firmware/joint/filters.h File Reference	131
9.4.1 Detailed Description	131
9.5 filters.h	132
9.6 lib/joint_firmware/joint/joint.h File Reference	132
9.6.1 Detailed Description	134
9.6.2 Macro Definition Documentation	134
9.6.2.1 ACK	134
9.6.2.2 DUMP_BUFFER	134
9.6.2.3 MAX_BUFFER	135
9.6.2.4 NACK	135
9.6.2.5 RFLAGS_SIZE	135
9.7 joint.h	135
9.8 lib/joint_firmware/joint/joint.ino File Reference	136
9.8.1 Detailed Description	137
9.8.2 Macro Definition Documentation	138
9.8.2.1 J1	138

9.8.3 Function Documentation	138
9.8.3.1 loop()	138
9.8.3.2 setup()	138
9.9 lib/joint_firmware/joint/stall.h File Reference	138
9.9.1 Detailed Description	139
9.10 stall.h	139
9.11 lib/ros2_ws/src/bioscara_rviz_plugin/include/bioscara_rviz_plugin/bioscara_panel.hpp File Reference	139
9.12 bioscara_panel.hpp	140
9.13 lib/ros2_ws/src/bioscara_rviz_plugin/src/bioscara_panel.cpp File Reference	141
9.14 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/common.h File Reference	142
9.14.1 Detailed Description	142
9.14.2 Macro Definition Documentation	142
9.14.2.1 DUMP_BUFFER	142
9.15 common.h	143
9.16 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/mBaseJoint.h File Reference	143
9.16.1 Detailed Description	144
9.17 mBaseJoint.h	144
9.18 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/mJoint.h File Reference	146
9.18.1 Detailed Description	146
9.19 mJoint.h	147
9.20 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/mJoint.hpp File Reference	148
9.20.1 Detailed Description	148
9.21 mJoint.hpp	149
9.22 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/mMockJoint.h File Reference	149
9.22.1 Detailed Description	150
9.23 mMockJoint.h	150
9.24 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/uErr.h File Reference	151
9.24.1 Detailed Description	151
9.24.2 Macro Definition Documentation	152
9.24.2.1 RETURN_ON_ERROR	152
9.24.2.2 RETURN_ON_FALSE	152
9.24.2.3 RETURN_ON_NEGATIVE	152
9.25 uErr.h	153
9.26 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/ul2C.h File Reference	154
9.26.1 Detailed Description	154
9.26.2 Macro Definition Documentation	155
9.26.2.1 ACK	155

9.26.2.2 MAX_BUFFER	155
9.26.2.3 NACK	155
9.26.2.4 RFLAGS_SIZE	155
9.26.3 Function Documentation	155
9.26.3.1 closeI2CDevHandle()	155
9.26.3.2 openI2CDevHandle()	156
9.26.3.3 readFromI2CDev()	156
9.26.3.4 writeToI2CDev()	156
9.27 ul2C.h	157
9.28 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_↔ hardware_driver/uTransmission.h File Reference	157
9.28.1 Macro Definition Documentation	158
9.28.1.1 ACTUATOR2JOINT	158
9.28.1.2 DEG2RAD	158
9.28.1.3 JOINT2ACTUATOR	158
9.28.1.4 M_PI	159
9.28.1.5 RAD2DEG	159
9.29 uTransmission.h	159
9.30 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/joint_comm_node.cpp File Reference	159
9.30.1 Function Documentation	160
9.30.1.1 INT_handler()	160
9.30.1.2 main()	160
9.30.2 Variable Documentation	160
9.30.2.1 J1	160
9.30.2.2 J2	160
9.30.2.3 J3	160
9.30.2.4 J4	160
9.31 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mBaseJoint.cpp File Refer- ence	161
9.32 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mJoint.cpp File Reference	161
9.33 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mMockJoint.cpp File Refer- ence	161
9.34 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/uErr.cpp File Reference	161
9.35 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/ul2C.cpp File Reference	162
9.35.1 Function Documentation	162
9.35.1.1 closeI2CDevHandle()	162
9.35.1.2 openI2CDevHandle()	162
9.35.1.3 readFromI2CDev()	163
9.35.1.4 writeToI2CDev()	163
9.36 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_↔ hardware_interface/arm_hardware.hpp File Reference	164
9.37 arm_hardware.hpp	165

9.38	lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/src/arm_hardware.cpp File Reference	166
9.39	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mBaseGripper.h File Reference	167
9.39.1	Detailed Description	167
9.40	mBaseGripper.h	168
9.41	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mGripper.h File Reference	168
9.41.1	Detailed Description	169
9.42	mGripper.h	169
9.43	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mMockGripper.h File Reference	170
9.43.1	Detailed Description	170
9.44	mMockGripper.h	171
9.45	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uPWM.h File Reference	171
9.45.1	Detailed Description	171
9.46	uPWM.h	172
9.47	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mBaseGripper.cpp File Reference	173
9.48	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mGripper.cpp File Reference	173
9.49	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mMockGripper.cpp File Reference	173
9.50	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/include/bioscara_gripper_hardware_interface/gripper_hardware.hpp File Reference	174
9.51	gripper_hardware.hpp	174
9.52	lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/src/gripper_hardware.cpp File Reference	175
9.53	lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/include/single_trigger_controller/single_trigger_controller.hpp File Reference	176
9.54	single_trigger_controller.hpp	177
9.55	lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/src/single_trigger_controller.cpp File Reference	178
	Index	179

Chapter 1

Bioscara C++ API Documentation

This is the API documentation for the Bioscara robotic control middleware based on the ROS2 framework.

Please find the key components below:

- [bioscara_joint_firmware](#): contains the joint firmware and its associated helper classes.
- [bioscara_hardware_drivers](#): contains the BaseJoint, Joint and MockJoint drivers as well as the BaseGripper, Gripper and MockGripper drivers.
- [bioscara_hardware_interfaces](#): contains the ros2_control hardware interfaces that implement the Bioscara arm and gripper hardware components in the BioscaraArmHardwareInterface and BioscaraGripper↔HardwareInterface respectively.
- [bioscara_rviz_plugin](#): contains the BioscaraPanel Rviz GUI panel.
- [single_trigger_controller::SingleTriggerController](#): contains the documentation for the SingleTriggerController.

Chapter 2

Todo List

Member [bioscara_hardware_interfaces::BioscaraArmHardwareInterface::on_init](#) (const hardware_↵
interface::HardwareComponentInterfaceParams ¶ms) override
threshold and current are uint8_t, if a number larger outside $0 < n < 255$ is passed as a parameters it will
overflow.

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all namespaces with brief descriptions:

bioscara_hardware_drivers	13
bioscara_hardware_interfaces	14
bioscara_joint_firmware	
Joint firmware	15
bioscara_rviz_plugin	21
single_trigger_controller	21

Chapter 4

Hierarchical Index

4.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

bioscara_hardware_drivers::BaseGripper	23
bioscara_hardware_drivers::Gripper	75
bioscara_hardware_drivers::MockGripper	102
bioscara_hardware_drivers::BaseJoint	29
bioscara_hardware_drivers::Joint	80
bioscara_hardware_drivers::MockJoint	104
controller_interface::ControllerInterface	
single_trigger_controller::SingleTriggerController	117
bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t	79
bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t	95
bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t	98
bioscara_joint_firmware::Lowpass	100
bioscara_joint_firmware::MovMax	113
rviz_common::Panel	
bioscara_rviz_plugin::BioscaraPanel	63
RPI_PWM	114
hardware_interface::SystemInterface	
bioscara_hardware_interfaces::BioscaraArmHardwareInterface	44
bioscara_hardware_interfaces::BioscaraGripperHardwareInterface	57

Chapter 5

Class Index

5.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

bioscara_hardware_drivers::BaseGripper	
Generic base class for gripper control implementations	23
bioscara_hardware_drivers::BaseJoint	
Generic base class to control a single joint	29
bioscara_hardware_interfaces::BioscaraArmHardwareInterface	
The bioscara arm hardware interface class	44
bioscara_hardware_interfaces::BioscaraGripperHardwareInterface	
The bioscara gripper hardware interface class	57
bioscara_rviz_plugin::BioscaraPanel	
RViz Panel to control the hardware specific functions of the Bioscara robot	63
bioscara_hardware_drivers::Gripper	
Child class implementing control of the hardware gripper	75
bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t	
Configuration structure holding the passed paramters from the ros2_control urdf	79
bioscara_hardware_drivers::Joint	
Representing a single hardware joint connected via I2C	80
bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t	
Configuration structure holding the passed paramters from the ros2_control urdf	95
bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t	
Configuration structure holding the passed homing paramters from the ros2_control urdf	98
bioscara_joint_firmware::Lowpass	
Simple discrete IIR lowpass filter	100
bioscara_hardware_drivers::MockGripper	
Child class for to mock the gripper hardware	102
bioscara_hardware_drivers::MockJoint	
Representing a single joint mocking the hardware joint	104
bioscara_joint_firmware::MovMax	
Simple FIR moving maximum filter	113
RPI_PWM	
Class to create a Pulse Width Modulated (PWM) signal on the Raspberry PI 4 and 5	114
single_trigger_controller::SingleTriggerController	
A general purpose controller which can be used to trigger command interfaces	117

Chapter 6

File Index

6.1 File List

Here is a list of all files with brief descriptions:

lib/joint_firmware/joint/ configuration.h	
Configuration definitions for Joint 1 to Joint 4	129
lib/joint_firmware/joint/ filters.h	
Helper classes for FIR and IIR filters	131
lib/joint_firmware/joint/ joint.h	
Joint firmware header	132
lib/joint_firmware/joint/ joint.ino	
Joint firmware	136
lib/joint_firmware/joint/ stall.h	
Helper functions for improved stall detection	138
lib/ros2_ws/src/bioscara_rviz_plugin/include/bioscara_rviz_plugin/ bioscara_panel.hpp	139
lib/ros2_ws/src/bioscara_rviz_plugin/src/ bioscara_panel.cpp	141
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ common.h	
A file containing utility macros and functions	142
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ mBaseJoint.h	
File including the BaseJoint class	143
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ mJoint.h	
File including the Joint class	146
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ mJoint.hpp	
Templated functions for the Joint class	148
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ mMockJoint.h	
File including the MockJoint class	149
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ uErr.h	
Defining common return types	151
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ uI2C.h	
Low level utility for I2C communication on Raspberry Pi using I2C library	154
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔ driver/ uTransmission.h	
	157

lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/joint_comm_node.cpp	159
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mBaseJoint.cpp	161
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mJoint.cpp	161
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mMockJoint.cpp	161
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/uErr.cpp	161
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/ul2C.cpp	162
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_hardware_↵ _interface/arm_hardware.hpp	164
lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/src/arm_hardware.cpp	166
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↵ hardware_driver/mBaseGripper.h	167
File containing the BaseGripper class	
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↵ hardware_driver/mGripper.h	168
File containing the Gripper class	
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↵ hardware_driver/mMockGripper.h	170
File containing the MockGripper class	
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↵ hardware_driver/uPWM.h	171
Includes source code for Hardware PWM generation on Raspberry Pi 4	
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mBaseGripper.cpp	173
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mGripper.cpp	173
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mMockGripper.cpp	173
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/include/bioscara_↵ gripper_hardware_interface/gripper_hardware.hpp	174
lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/src/gripper_hardware.cpp	175
lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/include/single_trigger_controller/single_trigger_controller.hpp	176
lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/src/single_trigger_controller.cpp	178

Chapter 7

Namespace Documentation

7.1 bioscara_hardware_drivers Namespace Reference

Classes

- class [BaseGripper](#)
Generic base class for gripper control implementations.
- class [BaseJoint](#)
Generic base class to control a single joint.
- class [Gripper](#)
Child class implementing control of the hardware gripper.
- class [Joint](#)
Representing a single hardware joint connected via I2C.
- class [MockGripper](#)
Child class for to mock the gripper hardware.
- class [MockJoint](#)
Representing a single joint mocking the hardware joint.

Enumerations

- enum class [err_type_t](#) {
 [OK](#) = 0 , [ERROR](#) = -1 , [NOT_HOMED](#) = -2 , [NOT_ENABLED](#) = -3 ,
 [STALLED](#) = -4 , [NOT_INIT](#) = -5 , [COMM_ERROR](#) = -6 , [INVALID_ARGUMENT](#) = -101 ,
 [INCORRECT_STATE](#) = -109 }
Enum defining common error types.

Functions

- `std::string error\_to\_string (err\_type\_t err)`
Converts an error code to a string and returns it.

7.1.1 Enumeration Type Documentation

7.1.1.1 [err_type_t](#)

```
enum class bioscara\_hardware\_drivers::err\_type\_t [strong]
```

Enum defining common error types.

Enumerator

OK	Success.
ERROR	Generic Error.
NOT_HOMED	Joint not homed or failed to home.
NOT_ENABLED	Joint not enabled or failed to enable.
STALLED	Joint stalled.
NOT_INIT	Joint not initialized.
COMM_ERROR	Communication error.
INVALID_ARGUMENT	Argument violates conditions.
INCORRECT_STATE	Joint is busy executing another blocking command.

7.1.2 Function Documentation

7.1.2.1 error_to_string()

```
std::string bioscara_hardware_drivers::error_to_string (
    err_type_t err )
```

Converts an error code to a string and returns it.

Parameters

<i>err</i>	
------------	--

Returns

std::string

7.2 bioscara_hardware_interfaces Namespace Reference

Classes

- class [BioscaraArmHardwareInterface](#)
The bioscara arm hardware interface class.
- class [BioscaraGripperHardwareInterface](#)
The bioscara gripper hardware interface class.

Variables

- constexpr char [HW_IF_HOME](#) [] = "home"

7.2.1 Variable Documentation

7.2.1.1 HW_IF_HOME

```
constexpr char bioscara_hardware_interfaces::HW_IF_HOME[] = "home" [constexpr]
```

7.3 bioscara_joint_firmware Namespace Reference

Joint firmware.

Classes

- class [Lowpass](#)
Simple discrete IIR lowpass filter.
- class [MovMax](#)
Simple FIR moving maximum filter.

Enumerations

- enum [stp_reg_t](#) {
[PING](#) = 0x0f , [SETUP](#) = 0x10 , [SETRPM](#) = 0x11 , [GETDRIVERRPM](#) = 0x12 ,
[MOVESTEPS](#) = 0x13 , [MOVEANGLE](#) = 0x14 , [MOVETOANGLE](#) = 0x15 , [GETMOTORSTATE](#) = 0x16 ,
[RUNCOTINOUS](#) = 0x17 , [ANGLEMOVED](#) = 0x18 , [SETCURRENT](#) = 0x19 , [SETHOLDCURRENT](#) = 0x1A ,
[SETMAXACCELERATION](#) = 0x1B , [SETMAXDECELERATION](#) = 0x1C , [SETMAXVELOCITY](#) = 0x1D ,
[ENABLESTALLGUARD](#) = 0x1E ,
[DISABLESTALLGUARD](#) = 0x1F , [CLEARSTALL](#) = 0x20 , [SETBRAKEMODE](#) = 0x22 , [ENABLEPID](#) = 0x23 ,
[DISABLEPID](#) = 0x24 , [ENABLECLOSEDLOOP](#) = 0x25 , [DISABLECLOSEDLOOP](#) = 0x26 , [SETCONTROLTHRESHOLD](#)
= 0x27 ,
[MOVETOEND](#) = 0x28 , [STOP](#) = 0x29 , [GETPIDERROR](#) = 0x2A , [CHECKORIENTATION](#) = 0x2B ,
[GETENCODERRPM](#) = 0x2C , [HOME](#) = 0x2D , [HOMEOFFSET](#) = 0x2E }
register and command definitions

Functions

- template<typename T >
void [readValue](#) (T &val, uint8_t *rxBuf, size_t rx_length)
Reads a value from a buffer to a value of the specified type.
- template<typename T >
int [writeValue](#) (const T val, uint8_t *txBuf, size_t &tx_length)
Writes a value of the specified type to a buffer.
- void [blocking_handler](#) (uint8_t reg)
Handles commands received via I2C.
- void [non_blocking_handler](#) (uint8_t reg)
Handles read request received via I2C.
- void [set_flags_for_blocking_handler](#) (uint8_t reg)
prepare flags to initiate the blocking handling of the received comman.
- void [receiveEvent](#) (int n)
I2C receive event Handler.
- void [requestEvent](#) ()
I2C request event Handler.
- void [setup](#) (void)
Setup Peripherals.
- void [loop](#) (void)
Main loop.
- float [stall_threshold](#) (float qd_rad, float offset)
computes the speed adaptive threshold.

Variables

- UstepperS32 [stepper](#)
The core UstepperS32 stepper object to control the motor.
- uint8_t [reg](#) = 0
- uint8_t [blk_reg](#) = 0
- uint8_t [rx_buf](#) [MAX_BUFFER] = {0}
- uint8_t [tx_buf](#) [MAX_BUFFER+RFLAGS_SIZE] = {0}
- bool [rx_data_ready](#) = 0
- size_t [tx_length](#) = 0
- size_t [rx_length](#) = 0

7.3.1 Detailed Description

Joint firmware.

The joint firmware is implemented in the Arduino framework. Initially the [setup\(\)](#) function is called and subsequently the [loop\(\)](#) function in an infinite loop. The [receiveEvent\(\)](#) is executed when a I2C message is received, and the [requestEvent\(\)](#) when data is requested. This always happens sequentially because at least always the #state flags are returned to the robot controller

7.3.2 Enumeration Type Documentation

7.3.2.1 stp_reg_t

```
enum bioscara_joint_firmware::stp_reg_t
```

register and command definitions

a register can be read (R) or written (W), each register has a size in bytes. The payload can be split into multiple values or just be a single value. Note that not all functions are implemented.

Enumerator

PING	R; Size: 1; [(char) ACK].
SETUP	W; Size: 2; [(uint8) holdCurrent, (uint8) driveCurrent].
SETRPM	W; Size: 4; [(float) RPM].
GETDRIVERRPM	
MOVESTEPS	W; Size: 4; [(int32) steps].
MOVEANGLE	
MOVETOANGLE	W; Size: 4; [(float) degrees].
GETMOTORSTATE	
RUNCOTINOUS	
ANGLEMOVED	R; Size: 4; [(float) degrees].
SETCURRENT	W; Size: 1; [(uint8) driveCurrent].
SETHOLDCURRENT	W; Size: 1; [(uint8) holdCurrent].
SETMAXACCELERATION	W; Size: 4; [(float) deg/s ²].
SETMAXDECELERATION	
SETMAXVELOCITY	W; Size: 4; [(float) deg/s].
ENABLESTALLGUARD	W; Size: 1; [(uint8) threshold].

Enumerator

DISABLESTALLGUARD	
CLEARSTALL	
SETBRAKEMODE	W; Size: 1; [(uint8) mode].
ENABLEPID	
DISABLEPID	
ENABLECLOSEDLOOP	
DISABLECLOSEDLOOP	W; Size: 1; [(uint8) 0].
SETCONTROLTHRESHOLD	
MOVETOEND	
STOP	W; Size: 1; [(uint8) mode].
GETPIDERROR	
CHECKORIENTATION	W; Size: 4; [(float) degrees].
GETENCODERRPM	R; Size: 4; [(float) RPM].
HOME	W; Size: 4; [(uint8) current, (uint8) sensitivity, (uint8) speed, (uint8) direction].
HOMEOFFSET	R/W; Size: 4; [(float) -].

7.3.3 Function Documentation

7.3.3.1 blocking_handler()

```
void bioscara_joint_firmware::blocking_handler (
    uint8_t reg )
```

Handles commands received via I2C.

Warning

This is a blocking function which may take some time to execute. This function must not be called from an ISR or callback! Call from main loop instead.

The registers handled in this handler are those whose implementation can take time and can thereby not be called directly from the request handler.

Parameters

<i>reg</i>	command that should be executed.
------------	----------------------------------

7.3.3.2 loop()

```
void bioscara_joint_firmware::loop (
    void )
```

Main loop.

Executes the following:

1. if `isStallguardEnabled`: compares `stepper.getPidError()` with `stallguardThreshold` and sets `isStalled` flag.
2. if `rx_data_ready`: set `isBusy` flag to indicate device is busy. Invoke `blocking_handler`. Clear `isBusy` flag to indicate device is no longer busy

7.3.3.3 `non_blocking_handler()`

```
void bioscara_joint_firmware::non_blocking_handler (
    uint8_t reg )
```

Handles read request received via I2C.

Can be invoked from the I2C ISR since reads from the stepper are non-blocking. Also Handling reads and the subsequent `wire.write()`, did not work from the main loop.

Parameters

<i>reg</i>	command to execute/register to read.
------------	--------------------------------------

7.3.3.4 `readValue()`

```
template<typename T >
void bioscara_joint_firmware::readValue (
    T & val,
    uint8_t * rxBuf,
    size_t rx_length )
```

Reads a value from a buffer to a value of the specified type.

Parameters

<i>val</i>	Reference to output variable
<i>rxBuf</i>	Buffer to read value from
<i>rx_length</i>	Length of the buffer

7.3.3.5 `receiveEvent()`

```
void bioscara_joint_firmware::receiveEvent (
    int n )
```

I2C receive event Handler.

Reads the content of the received message. Saves the register so it can be used in the main loop. If the master invokes the `read()` function the message contains only the register byte and no payload. If the master invokes the `write()` the message has a payload of appropriate size for the command. Every I2C transaction starts with a receive event when the command is sent and is immediately followed by a request since at minimum the flags need to be transmitted back. This means that the receive handler and request handler are always executed sequentially. The

main loop is not executed since both handlers are ISRs. For a read request the message looks like this:

```
< [REG]
> [TXBUFn]...[TXBUF2][TXBUF1][TXBUF0][FLAGS]
```

For a command the message looks like this:

```
< [REG][RXBUFn]...[RXBUF2][RXBUF1][RXBUF0]
> [FLAGS]
```

The payload is read into the rx_buf, rx_length is set to the payload length.

Parameters

<i>n</i>	the number of bytes read from the controller device: MAX_BUFFER
----------	---

7.3.3.6 requestEvent()

```
void bioscara_joint_firmware::requestEvent ( )
```

I2C request event Handler.

Sends the response data to the master. Every transaction begins with a receive event. The request event is always triggered since at a minimum the status flags are returned to the master. Hence this function is only invoked after the [receiveEvent\(\)](#) handler has been called. The function calls the [non_blocking_handler\(\)](#) which is non-blocking. Since most Ustepper functions are non-blocking as they just read/write registers to the stepper driver/encoder they can be handled directly in the ISR. The [non_blocking_handler\(\)](#) populates the tx_buf with relevant data, the current state flags are appended to the tx_buf and then it is send to the master.

7.3.3.7 set_flags_for_blocking_handler()

```
void bioscara_joint_firmware::set_flags_for_blocking_handler (
    uint8_t reg )
```

prepare flags to initiate the blocking handling of the received comman.

Sets the blk_reg to reg, this makes sure that even if a new command is received the blocking hanlder access the latest blocking command. Sets the isBusy flag, to indicate immediatly that the blocking has not been finished. This is if a status request follows immediatly the command, before the context back to the main loop has happened. Sets the rx_data_ready flag to indicate to the main loop that there is data to read in the blocking handler. Sets the tx_length to 0 because blocking commands can not return a payload.

7.3.3.8 setup()

```
void bioscara_joint_firmware::setup (
    void )
```

Setup Peripherals.

Setup I2C with the address ADR, and begin Serial for debugging with baudrate 9600. Setup the stepper, perform orientation check to check wiring and disable the stepper again.

7.3.3.9 stall_threshold()

```
float bioscara_joint_firmware::stall_threshold (
    float qd_rad,
    float offset )
```

computes the speed adaptive threshold.

Parameters

<i>qd_rad</i>	speed in rad/s. Should be measured speed because set speed is only available in velocity mode.
---------------	--

7.3.3.10 writeValue()

```
template<typename T >
int bioscara_joint_firmware::writeValue (
    const T val,
    uint8_t * txBuf,
    size_t & tx_length )
```

Writes a value of the specified type to a buffer.

Parameters

<i>val</i>	Reference to input variable
<i>txBuf</i>	pointer to tx buffer
<i>tx_length</i>	Length of the buffer returned

Returns

0 On success

7.3.4 Variable Documentation**7.3.4.1 blk_reg**

```
uint8_t bioscara_joint_firmware::blk_reg = 0
```

7.3.4.2 reg

```
uint8_t bioscara_joint_firmware::reg = 0
```

7.3.4.3 rx_buf

```
uint8_t bioscara_joint_firmware::rx_buf[MAX_BUFFER] = {0}
```

7.3.4.4 rx_data_ready

```
bool bioscara_joint_firmware::rx_data_ready = 0
```

7.3.4.5 rx_length

```
size_t bioscara_joint_firmware::rx_length = 0
```

7.3.4.6 stepper

```
UstepperS32 bioscara_joint_firmware::stepper
```

The core UstepperS32 stepper object to control the motor.

7.3.4.7 tx_buf

```
uint8_t bioscara_joint_firmware::tx_buf[MAX_BUFFER+RFLAGS_SIZE] = {0}
```

7.3.4.8 tx_length

```
size_t bioscara_joint_firmware::tx_length = 0
```

7.4 bioscara_rviz_plugin Namespace Reference

Classes

- class [BioscaraPanel](#)
RViz Panel to control the hardware specific functions of the Bioscara robot.

7.5 single_trigger_controller Namespace Reference

Classes

- class [SingleTriggerController](#)
A general purpose controller which can be used to trigger command interfaces.

Typedefs

- using [CmdType](#) = control_msgs::msg::DynamicInterfaceGroupValues
- using [StateType](#) = control_msgs::msg::DynamicInterfaceGroupValues
- using [CallbackReturn](#) = controller_interface::CallbackReturn
- using [InterfacesNames](#) = std::vector< std::string >
- using [MapOfReferencesToCommandInterfaces](#) = std::unordered_map< std::string, std::reference_wrapper< hardware_interface::LoanedCommandInterface > >
- using [MapOfReferencesToStateInterfaces](#) = std::unordered_map< std::string, std::reference_wrapper< hardware_interface::LoanedStateInterface > >
- using [StateInterfaces](#) = std::vector< std::reference_wrapper< hardware_interface::LoanedStateInterface > >

7.5.1 Typedef Documentation

7.5.1.1 CallbackReturn

```
using single_trigger_controller::CallbackReturn = typedef controller_interface::CallbackReturn
```

7.5.1.2 CmdType

```
using single_trigger_controller::CmdType = typedef control_msgs::msg::DynamicInterfaceGroup↵  
Values
```

7.5.1.3 InterfacesNames

```
using single_trigger_controller::InterfacesNames = typedef std::vector<std::string>
```

7.5.1.4 MapOfReferencesToCommandInterfaces

```
using single_trigger_controller::MapOfReferencesToCommandInterfaces = typedef std::unordered↵  
_map< std::string, std::reference_wrapper<hardware_interface::LoanedCommandInterface> >
```

7.5.1.5 MapOfReferencesToStateInterfaces

```
using single_trigger_controller::MapOfReferencesToStateInterfaces = typedef std::unordered↵  
map<std::string, std::reference_wrapper<hardware_interface::LoanedStateInterface> >
```

7.5.1.6 StateInterfaces

```
using single_trigger_controller::StateInterfaces = typedef std::vector<std::reference_wrapper↵  
_interface::LoanedStateInterface> >
```

7.5.1.7 StateType

```
using single_trigger_controller::StateType = typedef control_msgs::msg::DynamicInterface↵  
GroupValues
```

Chapter 8

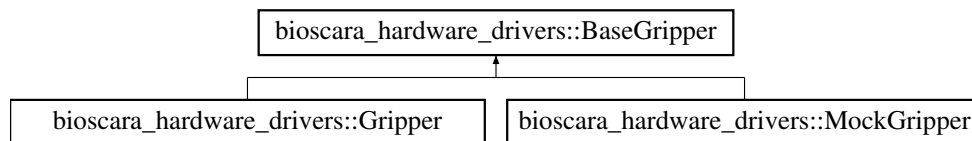
Class Documentation

8.1 bioscara_hardware_drivers::BaseGripper Class Reference

Generic base class for gripper control implementations.

```
#include <mBaseGripper.h>
```

Inheritance diagram for bioscara_hardware_drivers::BaseGripper:



Public Member Functions

- [BaseGripper](#) (float reduction, float offset, float min, float max, float backup_init_pos)
Construct a new [BaseGripper](#) object.
- [~BaseGripper](#) (void)
Destroy the [BaseGripper](#) object.
- virtual [err_type_t init](#) (void)
Initializes the gripper.
- virtual [err_type_t deinit](#) (void)
Deinitializes the gripper.
- virtual [err_type_t enable](#) (void)
Enables the gripper.
- virtual [err_type_t disable](#) (void)
Disables the gripper.
- virtual [err_type_t setPosition](#) (float width)
Sets the gripper width in m.
- virtual [err_type_t getPosition](#) (float &width)
Gets the gripper width.
- virtual void [setReduction](#) (float reduction)
Manually set reduction.
- virtual void [setOffset](#) (float offset)
Manually set offset.

Protected Member Functions

- `err_type_t save_last_position` (float pos)
Stores the latest position to the buffer file.
- `err_type_t retrieve_last_position` (float &pos)
Retrieves the stored position from the buffer file.

Protected Attributes

- float `_reduction` = 1
gripper width to actuator reduction ratio
- float `_offset` = 0
gripper width position offset
- float `_min` = 0
gripper width lower limit
- float `_max` = 0
gripper width upper limit
- float `_backup_init_pos` = 0.0
Initial position assumed if none can be retrieved from the buffer file.
- float `_pos` = `_backup_init_pos`
stored position
- float `_pos_get` = `_pos`
reported position

Private Attributes

- `std::chrono::_V2::system_clock::time_point _new_cmd_time` = `std::chrono::_V2::system_clock::time_point()`

8.1.1 Detailed Description

Generic base class for gripper control implementations.

This class is a wrapper function to interact with the robot gripper either through a [MockGripper](#) or hardware [Gripper](#) object.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 BaseGripper()

```
bioscara_hardware_drivers::BaseGripper::BaseGripper (
    float reduction,
    float offset,
    float min,
    float max,
    float backup_init_pos )
```

Construct a new [BaseGripper](#) object.

The gripper has the reduction r and offset o parameters which are used to translate from a desired gripper width to the servo angle. The relationship between gripper width w and actuator angle α is as follows:

$$\alpha = r(w - o)$$

To determine these parameters execute the following steps:

1. Manually set the gripper to an open position by setting a actuator angle. Be carefull to not exceed the physical limits of the gripper since the actuator is strong enough to break PLA before stalling.
2. Measure the gripper width w_1 and note the set actuator angle α_1 .
3. Move the gripper to a more closed position that still allows you to accurately measure the width
4. Measure the second width w_2 and note the corresponding angle α_2
5. Calculate the offset o :

$$o = \frac{\alpha_1 w_2 - \alpha_2 w_1}{\alpha_1 - \alpha_2}$$

6. Calculate the reduction r :

$$r = \frac{\alpha_1}{w_1 - o}$$

Parameters

<i>reduction</i>	the gripper width to actuator reduction ratio
<i>offset</i>	gripper width to actuator zero offset
<i>min</i>	lower limit in m
<i>max</i>	upper limit in m
<i>backup_init_pos</i>	initial position assumed if none can be retrieved from the buffer file

8.1.2.2 ~BaseGripper()

```
bioscara_hardware_drivers::BaseGripper::~~BaseGripper (
    void )
```

Destroy the [BaseGripper](#) object.

invokes the [disable\(\)](#) and [deinit\(\)](#) method to clean up.

8.1.3 Member Function Documentation

8.1.3.1 deinit()

```
err_type_t bioscara_hardware_drivers::BaseGripper::deinit (
    void ) [virtual]
```

Deinitializes the gripper.

Returns

[err_type_t::OK](#)

8.1.3.2 disable()

```
err_type_t bioscara_hardware_drivers::BaseGripper::disable (
    void ) [virtual]
```

Disables the gripper.

As described in [enable\(\)](#)

Returns

[err_type_t::OK](#)

Reimplemented in [bioscara_hardware_drivers::Gripper](#).

8.1.3.3 enable()

```
err_type_t bioscara_hardware_drivers::BaseGripper::enable (
    void ) [virtual]
```

Enables the gripper.

Since the gripper has no position feedback its position is initially unknown. For this reason we try to retrieve the latest commanded position from a buffer file using [retrieve_last_position\(\)](#). If this fails the [_backup_init_pos](#) is used.

Returns

[err_type_t::OK](#)

Reimplemented in [bioscara_hardware_drivers::Gripper](#).

8.1.3.4 getPosition()

```
err_type_t bioscara_hardware_drivers::BaseGripper::getPosition (
    float & width ) [virtual]
```

Gets the gripper width.

Note

Since the PWM servo has no position feedback, the latest position command is returned.

Parameters

<i>width</i>	width in m.
--------------	-------------

Returns

[err_type_t::OK](#)

8.1.3.5 init()

```
err_type_t bioscara_hardware_drivers::BaseGripper::init (
    void ) [virtual]
```

Initializes the gripper.

Returns

[err_type_t::OK](#)

8.1.3.6 retrieve_last_position()

```
err_type_t bioscara_hardware_drivers::BaseGripper::retrieve_last_position (
    float & pos ) [protected]
```

Retrieves the stored position from the buffer file.

Parameters

<i>pos</i>	position value that is retrieved
------------	----------------------------------

Returns

[err_type_t::OK](#) on success, [err_type_t::ERROR](#) if parsing buffer file fails.

8.1.3.7 save_last_position()

```
err_type_t bioscara_hardware_drivers::BaseGripper::save_last_position (
    float pos ) [protected]
```

Stores the latest position to the buffer file.

Parameters

<i>pos</i>	position value to store
------------	-------------------------

Returns

[err_type_t::OK](#) on success, [err_type_t::ERROR](#) if writing buffer file fails.

8.1.3.8 setOffset()

```
void bioscara_hardware_drivers::BaseGripper::setOffset (
    float offset ) [virtual]
```

Manually set offset.

8.1.3.9 setPosition()

```
err_type_t bioscara_hardware_drivers::BaseGripper::setPosition (
    float width ) [virtual]
```

Sets the gripper width in m.

Arguments outside the allowed range are bounded to `_min` and `_max`.

Parameters

<i>width</i>	width in m.
--------------	-------------

Returns

`err_type_t::OK`

Reimplemented in `bioscara_hardware_drivers::Gripper`.

8.1.3.10 setReduction()

```
void bioscara_hardware_drivers::BaseGripper::setReduction (
    float reduction ) [virtual]
```

Manually set reduction.

Parameters

<i>reduction</i>	
------------------	--

8.1.4 Member Data Documentation

8.1.4.1 _backup_init_pos

```
float bioscara_hardware_drivers::BaseGripper::_backup_init_pos = 0.0 [protected]
```

Initial position assumed if none can be retrieved from the buffer file.

8.1.4.2 _max

```
float bioscara_hardware_drivers::BaseGripper::_max = 0 [protected]
```

gripper width upper limit

8.1.4.3 _min

```
float bioscara_hardware_drivers::BaseGripper::_min = 0 [protected]
```

gripper width lower limit

8.1.4.4 _new_cmd_time

```
std::chrono::_V2::system_clock::time_point bioscara_hardware_drivers::BaseGripper::_new_cmd_time ←
time = std::chrono::_V2::system_clock::time_point() [private]
```

8.1.4.5 _offset

```
float bioscara_hardware_drivers::BaseGripper::_offset = 0 [protected]
```

gripper width position offset

8.1.4.6 _pos

```
float bioscara_hardware_drivers::BaseGripper::_pos = _backup_init_pos [protected]
```

stored position

8.1.4.7 _pos_get

```
float bioscara_hardware_drivers::BaseGripper::_pos_get = _pos [protected]
```

reported position

8.1.4.8 _reduction

```
float bioscara_hardware_drivers::BaseGripper::_reduction = 1 [protected]
```

gripper width to actuator reduction ratio

The documentation for this class was generated from the following files:

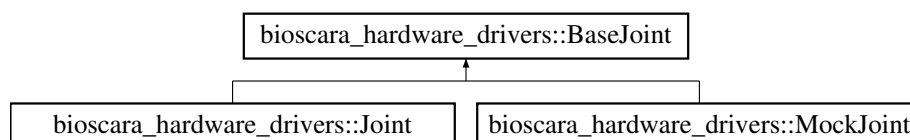
- [lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mBaseGripper.h](#)
- [lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mBaseGripper.cpp](#)

8.2 bioscara_hardware_drivers::BaseJoint Class Reference

Generic base class to control a single joint.

```
#include <mBaseJoint.h>
```

Inheritance diagram for bioscara_hardware_drivers::BaseJoint:



Public Types

- enum `stp_reg_t` {
`NONE` = 0x00 , `PING` = 0x0f , `SETUP` = 0x10 , `SETRPM` = 0x11 ,
`GETDRIVERRPM` = 0x12 , `MOVESTEPS` = 0x13 , `MOVEANGLE` = 0x14 , `MOVETOANGLE` = 0x15 ,
`GETMOTORSTATE` = 0x16 , `RUNCOTINOUS` = 0x17 , `ANGLEMOVED` = 0x18 , `SETCURRENT` = 0x19 ,
`SETHOLDCURRENT` = 0x1A , `SETMAXACCELERATION` = 0x1B , `SETMAXDECELERATION` = 0x1C ,
`SETMAXVELOCITY` = 0x1D ,
`ENABLESTALLGUARD` = 0x1E , `DISABLESTALLGUARD` = 0x1F , `CLEARSTALL` = 0x20 , `SETBRAKEMODE` = 0x22 ,
`ENABLEPID` = 0x23 , `DISABLEPID` = 0x24 , `ENABLECLOSEDLOOP` = 0x25 , `DISABLECLOSEDLOOP` = 0x26 ,
`SETCONTROLTHRESHOLD` = 0x27 , `MOVETOEND` = 0x28 , `STOP` = 0x29 , `GETPIDERROR` = 0x2A ,
`CHECKORIENTATION` = 0x2B , `GETENCODERRPM` = 0x2C , `HOME` = 0x2D , `HOMEOFFSET` = 0x2E }
register and command definitions

Public Member Functions

- `BaseJoint` (const std::string name)
Create a `Joint` object.
- `~BaseJoint` (void)
Destroy the `BaseJoint` object.
- virtual `err_type_t` `init` (void)
Initialize the joint communication.
- virtual `err_type_t` `deinit` (void)
Denitalize the joint communication.
- virtual `err_type_t` `enable` (u_int8_t driveCurrent, u_int8_t holdCurrent)
Engages the joint.
- virtual `err_type_t` `disable` (void)
disengages the joint
- virtual `err_type_t` `home` (float velocity, u_int8_t sensitivity, u_int8_t current)
Blocking implementation to home the joint.
- virtual `err_type_t` `startHoming` (float velocity, u_int8_t sensitivity, u_int8_t current)
non-blocking implementation to home the joint
- virtual `err_type_t` `postHoming` (void)
perform tasks after a non-blocking homing
- virtual `err_type_t` `getPosition` (float &pos)=0
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- virtual `err_type_t` `setPosition` (float pos)
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- virtual `err_type_t` `moveSteps` (int32_t steps)
Move full steps.
- virtual `err_type_t` `getVelocity` (float &vel)=0
get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- virtual `err_type_t` `setVelocity` (float vel)
Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- virtual `err_type_t` `checkOrientation` (float angle=2.0)
Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.
- virtual `err_type_t` `stop` (void)
Stops the motor.
- virtual `err_type_t` `disableCL` (void)
- virtual `err_type_t` `setDriveCurrent` (u_int8_t current)

- Set the Drive Current.*

 - virtual [err_type_t setHoldCurrent](#) (u_int8_t current)
- Set the Hold Current.*

 - virtual [err_type_t setBrakeMode](#) (u_int8_t mode)
- Set Brake Mode.*

 - virtual [err_type_t setMaxAcceleration](#) (float maxAccel)

Set the maximum permitted joint acceleration (and deceleration) in rad/s² or m/s² for cylindrical and prismatic joints respectively.
- virtual [err_type_t setMaxVelocity](#) (float maxVel)

Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.
- virtual [err_type_t enableStallguard](#) (u_int8_t sensitivity)

Enable encoder stall detection of the joint.
- virtual bool [isHomed](#) (void)

Checks the state if the motor is homed.
- virtual bool [isEnabled](#) (void)

Checks the state if the motor is enabled.
- virtual bool [isStalled](#) (void)

Checks if the motor is stalled.
- virtual bool [isBusy](#) (void)

Checks if the joint controller is busy processing a blocking command.
- virtual [err_type_t getFlags](#) (u_int8_t &flags)

get the latest driver state [flags](#) from the joint
- virtual [err_type_t getFlags](#) (void)

Overload of [getFlags\(u_int8_t &flags\)](#)
- virtual [stp_reg_t getCurrentBCmd](#) (void)

get the currently active blocking command

Public Attributes

- std::string [name](#)
- [Joint](#) name for logging.*

Protected Member Functions

- virtual void [wait_while_busy](#) (const float period_ms)
- Blocking loop waiting for BUSY flag to reset.*
- virtual [err_type_t _home](#) (float velocity, u_int8_t sensitivity, u_int8_t current)=0
- Call to start the homing sequence of a joint.*

Protected Attributes

- u_int8_t [flags](#) = 0b00001100
- State [flags](#) transmitted with every I2C transaction.*
- [stp_reg_t current_b_cmd](#) = NONE
- Keeps track if a blocking command is being executed.*

8.2.1 Detailed Description

Generic base class to control a single joint.

This class is a wrapper function to interact with a robot joint either through a [MockJoint](#) or hardware [Joint](#) object.

8.2.2 Member Enumeration Documentation

8.2.2.1 stp_reg_t

```
enum bioscara_hardware_drivers::BaseJoint::stp_reg_t
```

register and command definitions

a register can be read (R) or written (W), each register has a size in bytes. The payload can be split into multiple values or just be a single value. Note that not all functions are implemented.

Enumerator

NONE	Used for signalling purposes.
PING	R; Size: 1; [(char) ACK].
SETUP	W; Size: 2; [(uint8) holdCurrent, (uint8) driveCurrent].
SETRPM	W; Size: 4; [(float) RPM].
GETDRIVERRPM	
MOVESTEPS	W; Size: 4; [(int32) steps].
MOVEANGLE	
MOVETOANGLE	W; Size: 4; [(float) degrees].
GETMOTORSTATE	
RUNCOTINOUS	
ANGLEMOVED	R; Size: 4; [(float) degrees].
SETCURRENT	W; Size: 1; [(uint8) driveCurrent].
SETHOLDCURRENT	W; Size: 1; [(uint8) holdCurrent].
SETMAXACCELERATION	
SETMAXDECELERATION	
SETMAXVELOCITY	
ENABLESTALLGUARD	W; Size: 1; [(uint8) threshold].
DISABLESTALLGUARD	
CLEARSTALL	
SETBRAKEMODE	W; Size: 1; [(uint8) mode].
ENABLEPID	
DISABLEPID	
ENABLECLOSEDLOOP	
DISABLECLOSEDLOOP	W; Size: 1; [(uint8) 0].
SETCONTROLTHRESHOLD	
MOVETOEND	
STOP	W; Size: 1; [(uint8) mode].
GETPIDERROR	
CHECKORIENTATION	W; Size: 4; [(float) degrees].
GETENCODERRPM	R; Size: 4; [(float) RPM].
HOME	W; Size: 4; [(uint8) current, (int8) sensitivity, (uint8) speed, (uint8) direction].
HOMEOFFSET	R/W; Size: 4; [(float) -].

8.2.3 Constructor & Destructor Documentation

8.2.3.1 BaseJoint()

```
bioscara_hardware_drivers::BaseJoint::BaseJoint (
    const std::string name )
```

Create a [Joint](#) object.

The [Joint](#) object represents a single joint.

Parameters

<i>name</i>	string device name for logging.
-------------	---------------------------------

8.2.3.2 ~BaseJoint()

```
bioscara_hardware_drivers::BaseJoint::~~BaseJoint (
    void )
```

Destroy the [BaseJoint](#) object.

Invokes the [disable\(\)](#) and [deinit\(\)](#) method to clean up.

8.2.4 Member Function Documentation

8.2.4.1 _home()

```
virtual err\_type\_t bioscara_hardware_drivers::BaseJoint::_home (
    float velocity,
    u_int8_t sensitivity,
    u_int8_t current ) [protected], [pure virtual]
```

Call to start the homing sequence of a joint.

The joint will start rotating with the specified speed until a resistance which drives the PID error above the specified threshold is encountered. At this point the stepper stops and zeros the encoder.

Parameters

<i>velocity</i>	signed velocity in rad/s or m/s. Must be between $1.0 < \text{RAD2DEG}(\text{JOINT2ACTUATOR}(\text{velocity}, \text{reduction}, 0)) / 6 < 250.0$
<i>sensitivity</i>	Encoder pid error threshold 0 to 255.
<i>current</i>	homing current, determines how easy it is to stop the motor and thereby provoke a stall

Returns

[err_type_t](#)

Implemented in [bioscara_hardware_drivers::MockJoint](#), and [bioscara_hardware_drivers::Joint](#).

8.2.4.2 checkOrientation()

```
err_type_t bioscara_hardware_drivers::BaseJoint::checkOrientation (
    float angle = 2.0 ) [virtual]
```

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

<i>angle</i>	degrees how much the motor should turn. A few degrees is sufficient.
--------------	--

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.3 deinit()

```
err_type_t bioscara_hardware_drivers::BaseJoint::deinit (
    void ) [virtual]
```

Denitalize the joint communication.

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.4 disable()

```
err_type_t bioscara_hardware_drivers::BaseJoint::disable (
    void ) [virtual]
```

disenganges the joint

invokes [stop\(\)](#), sets hold and drive current to 0 and sets the the joint brake mode to freewheeling

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::MockJoint](#).

8.2.4.5 disableCL()

```
err_type_t bioscara_hardware_drivers::BaseJoint::disableCL (
    void ) [virtual]
```

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.6 enable()

```
err_type_t bioscara_hardware_drivers::BaseJoint::enable (
    u_int8_t driveCurrent,
    u_int8_t holdCurrent ) [virtual]
```

Engages the joint.

This function prepares the motor for movement. After successful execution the joint is ready to accept [setPosition\(\)](#) and [setVelocity\(\)](#) commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

<i>driveCurrent</i>	drive current in 0-100 % of 2.5A output (check uStepper doc.)
<i>holdCurrent</i>	hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.7 enableStallguard()

```
err_type_t bioscara_hardware_drivers::BaseJoint::enableStallguard (
    u_int8_t sensitivity ) [virtual]
```

Enable encoder stall detection of the joint.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. A detected stall can be reset by homing or reenabling the joint using [enable\(\)](#).

Note

If stall detection shall be enabled, invoke this method AFTER enabling the joint with [enable\(\)](#).

Parameters

<i>sensitivity</i>	value of threshold. 0 - 255 where lower is more sensitive.
--------------------	--

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.8 getCurrentBCmd()

```
BaseJoint::stp_reg_t bioscara_hardware_drivers::BaseJoint::getCurrentBCmd (
    void ) [virtual]
```

get the currently active blocking command

Returns

The the command of type `stp_reg_t`

8.2.4.9 getFlags() [1/2]

```
err_type_t bioscara_hardware_drivers::BaseJoint::getFlags (
    u_int8_t & flags ) [virtual]
```

get the latest driver state [flags](#) from the joint

Parameters

<i>flags</i>	if succesfull, populated with the latest flags
--------------	--

Returns

`err_type_t`

8.2.4.10 getFlags() [2/2]

```
err_type_t bioscara_hardware_drivers::BaseJoint::getFlags (
    void ) [virtual]
```

Overload of [getFlags\(u_int8_t &flags\)](#)

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.11 getPosition()

```
virtual err\_type\_t bioscara_hardware_drivers::BaseJoint::getPosition (
    float & pos ) [pure virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Warning

If the joint is not homed this method does not return an error. Instead `pos` will be 0.0.

Parameters

<code>pos</code>	the current joint position in rad or m.
------------------	---

Returns

`err_type_t`

Implemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.12 getVelocity()

```
virtual err\_type\_t bioscara_hardware_drivers::BaseJoint::getVelocity (
    float & vel ) [pure virtual]
```

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<code>vel</code>	the current joint velocity in rad/s or m/s.
------------------	---

Returns

`err_type_t`

Implemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.13 home()

```
err\_type\_t bioscara_hardware_drivers::BaseJoint::home (
    float velocity,
    u_int8_t sensitivity,
    u_int8_t current ) [virtual]
```

Blocking implementation to home the joint.

Homing the joint is necessary after its controller has been powered off. The joint can be moved while the encoders are inactive and hence the position at startup is unknown. See [_home\(\)](#) for information on implementation and description of the parameters.

This is a blocking implementation which only returns after the joint is no longer busy. First [startHoming\(\)](#) is called, and subsequently waits with [wait_while_busy\(\)](#) to finish homing. Lastly [postHoming\(\)](#) is called.

Returns

err_type_t

8.2.4.14 init()

```
err_type_t bioscara_hardware_drivers::BaseJoint::init (
    void ) [virtual]
```

Initialize the joint communication.

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.15 isBusy()

```
bool bioscara_hardware_drivers::BaseJoint::isBusy (
    void ) [virtual]
```

Checks if the joint controller is busy processing a blocking command.

Reads the internal state [flags](#) from the last transmission. If an update is necessary call [getFlags\(\)](#) before invoking this function.

Returns

true if a blocking command is currently executing, false if not.

8.2.4.16 isEnabled()

```
bool bioscara_hardware_drivers::BaseJoint::isEnabled (
    void ) [virtual]
```

Checks the state if the motor is enabled.

Reads the internal state [flags](#) from the last transmission. If an update is necessary call [getFlags\(\)](#) before invoking this function. If the motor actually can move depends on the state of the STALLED flag which can be checked using [Joint::isStalled\(\)](#).

Returns

true if the motor is enabled, false if not.

8.2.4.17 isHomed()

```
bool bioscara_hardware_drivers::BaseJoint::isHomed (
    void ) [virtual]
```

Checks the state if the motor is homed.

Reads the internal state [flags](#) from the last transmission. If an update is necessary call [getFlags\(\)](#) before invoking this function.

Returns

true if the motor is homed, false if not.

Reimplemented in [bioscara_hardware_drivers::MockJoint](#).

8.2.4.18 isStalled()

```
bool bioscara_hardware_drivers::BaseJoint::isStalled (
    void ) [virtual]
```

Checks if the motor is stalled.

Reads the internal state [flags](#) from the last transmission. If an update is necessary call [getFlags\(\)](#) before invoking this function.

Returns

true if the motor is stalled, false if not.

8.2.4.19 moveSteps()

```
err_type_t bioscara_hardware_drivers::BaseJoint::moveSteps (
    int32_t steps ) [virtual]
```

Move full steps.

This function can be called even when not homed.

Parameters

<i>steps</i>	number of full steps
--------------	----------------------

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.20 postHoming()

```
err_type_t bioscara_hardware_drivers::BaseJoint::postHoming (
    void ) [virtual]
```

perform tasks after a non-blocking homing

This method resets the `current_b_cmd` to `stp_reg_t::NONE`, checks if the joint is homed, and saves the homing offset to the joint.

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.21 setBrakeMode()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setBrakeMode (
    u_int8_t mode ) [virtual]
```

Set Brake Mode.

Parameters

<i>mode</i>	Freewheel: 0, Coolbrake: 1, Hardbrake: 2
-------------	--

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.22 setDriveCurrent()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setDriveCurrent (
    u_int8_t current ) [virtual]
```

Set the Drive Current.

Parameters

<i>current</i>	0% - 100% of driver current
----------------	-----------------------------

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.23 setHoldCurrent()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setHoldCurrent (
    u_int8_t current ) [virtual]
```

Set the Hold Current.

Parameters

<i>current</i>	0% - 100% of driver current
----------------	-----------------------------

Returns

err_type_t -1 on communication error,

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.24 setMaxAcceleration()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setMaxAcceleration (
    float maxAccel ) [virtual]
```

Set the maximum permitted joint acceleration (and deceleration) in rad/s² or m/s² for cylindrical and prismatic joints respectively.

Parameters

<i>maxAccel</i>	maximum joint acceleration.
-----------------	-----------------------------

Returns

err_type_t

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.25 setMaxVelocity()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setMaxVelocity (
    float maxVel ) [virtual]
```

Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<i>maxVel</i>	maximum joint velocity.
---------------	-------------------------

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#).

8.2.4.26 setPosition()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setPosition (
    float pos ) [virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters

<i>pos</i>	the commanded joint position in rad or m.
------------	---

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.27 setVelocity()

```
err_type_t bioscara_hardware_drivers::BaseJoint::setVelocity (
    float vel ) [virtual]
```

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<i>vel</i>	the commanded joint velocity in rad/s or m/s.
------------	---

Returns

`err_type_t`

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.28 startHoming()

```
err_type_t bioscara_hardware_drivers::BaseJoint::startHoming (
    float velocity,
    u_int8_t sensitivity,
    u_int8_t current ) [virtual]
```

non-blocking implementation to home the joint

Homing the joint is necessary after its controller has been powered off. The joint can be moved while the encoders are inactive and hence the position at startup is unknown. See [_home\(\)](#) for information on implementation and description of the parameters.

This method returns immediately after starting the homing sequence and should be used when the blocking implementation is not acceptable, for example in a realtime loop.

This method sets the [current_b_cmd](#) flag to [stp_reg_t::HOME](#)

Returns

[err_type_t](#)

8.2.4.29 stop()

```
err\_type\_t bioscara_hardware_drivers::BaseJoint::stop (
    void ) [virtual]
```

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

[err_type_t](#)

Reimplemented in [bioscara_hardware_drivers::Joint](#), and [bioscara_hardware_drivers::MockJoint](#).

8.2.4.30 wait_while_busy()

```
void bioscara_hardware_drivers::BaseJoint::wait_while_busy (
    const float period_ms ) [protected], [virtual]
```

Blocking loop waiting for BUSY flag to reset.

Parameters

period_ms	time in ms between polls.
---------------------------	---------------------------

8.2.5 Member Data Documentation

8.2.5.1 current_b_cmd

```
stp\_reg\_t bioscara_hardware_drivers::BaseJoint::current_b_cmd = NONE [protected]
```

Keeps track if a blocking command is being executed.

8.2.5.2 flags

```
u_int8_t bioscara_hardware_drivers::BaseJoint::flags = 0b00001100 [protected]
```

State [flags](#) transmitted with every I2C transaction.

The transmission flags purpose are to transmit the joints current state. Note: They can not be used as error indication of the execution of a transmitted write command, since commands are executed after the I2C transaction is completed. The status flags are one byte with following structure:

BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
reserved	reserved	reserved	reserved	NOTENABLED	NOTHOMED	BUSY	STALL

STALL is set if a stall from the stall detection is sensed and the joint is stopped. The flag is cleared when the joint is homed or the Stallguard enabled.

BUSY is set if the slave is busy processing a previous command.

NOTHOMED is cleared if the joint is homed. Movement is only allowed if this flag is clear

NOTENABLED is cleared if the joint is enabled after calling [enable\(\)](#)

8.2.5.3 name

```
std::string bioscara_hardware_drivers::BaseJoint::name
```

[Joint](#) name for logging.

The documentation for this class was generated from the following files:

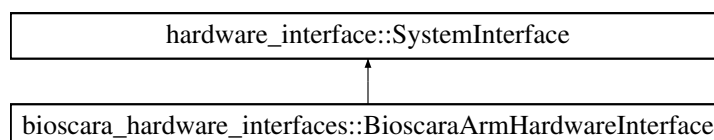
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↵
driver/[mBaseJoint.h](#)
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/[mBaseJoint.cpp](#)

8.3 bioscara_hardware_interfaces::BioscaraArmHardwareInterface Class Reference

The bioscara arm hardware interface class.

```
#include <arm_hardware.hpp>
```

Inheritance diagram for bioscara_hardware_interfaces::BioscaraArmHardwareInterface:



Classes

- struct [joint_config_t](#)
configuration structure holding the passed paramters from the ros2_control urdf
- struct [joint_homing_config_t](#)
configuration structure holding the passed homing paramters from the ros2_control urdf

Public Member Functions

- hardware_interface::CallbackReturn [on_init](#) (const hardware_interface::HardwareComponentInterface↔ Params ¶ms) override
*Called on initialization to the *unconfigured* state.*
- hardware_interface::CallbackReturn [on_shutdown](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *inactive*, *unconfigured* and *active* to the *finalized* state.*
- hardware_interface::CallbackReturn [on_configure](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *unconfigured* to the *inactive* state.*
- hardware_interface::CallbackReturn [on_cleanup](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *inactive* to the *unconfigured* state.*
- hardware_interface::CallbackReturn [on_activate](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *inactive* to the *active* state.*
- hardware_interface::CallbackReturn [on_deactivate](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *active* to the *inactive* state.*
- hardware_interface::return_type [read](#) (const rclcpp::Time &time, const rclcpp::Duration &period) override
Reads from the hardware and populates the state interfaces.
- hardware_interface::return_type [write](#) (const rclcpp::Time &time, const rclcpp::Duration &period) override
Writes commands to the hardware from the command interfaces.
- hardware_interface::return_type [prepare_command_mode_switch](#) (const std::vector< std::string > &start↔ _interfaces, const std::vector< std::string > &stop_interfaces) override
Performs checks and book keeping of the active control mode when changing controllers in non-realtime context.
- hardware_interface::return_type [perform_command_mode_switch](#) (const std::vector< std::string > &start↔ _interfaces, const std::vector< std::string > &stop_interfaces) override
Perform the mode-switching for the new command interface combination in realtime context.
- hardware_interface::CallbackReturn [on_error](#) (const rclcpp_lifecycle::State &previous_state) override
Called when an error in any state or state transition is thrown.

Private Member Functions

- [bioscara_hardware_drivers::err_type_t start_homing](#) (const std::string name, float velocity)
wrapper method to start homing.
- [bioscara_hardware_drivers::err_type_t stop_homing](#) (const std::string name)
wrapper method to stop homing.
- void [split_interface_string_to_joint_and_name](#) (std::string interface, std::string &joint_name, std::string &interface_name)
Split a interface string like "<joint_name>/<interface_name>" to "<joint_name>" and "<interface_name>".
- [bioscara_hardware_drivers::err_type_t activate_joint](#) (const std::string name)
Enables each joint, enables the stall detection and sets the maximum acceleration.
- [bioscara_hardware_drivers::err_type_t deactivate_joint](#) (const std::string name)
Disables each joint.

Private Attributes

- `std::unordered_map< std::string, std::unique_ptr< bioscara_hardware_drivers::BaseJoint > > _joints`
unordered map storing the pointers to BaseJoint objects. This will either be a MockJoint or Joint.
- `std::unordered_map< std::string, joint_config_t > _joint_cfg`
unordered map storing the configuration struct of the joints.
- `std::unordered_map< std::string, std::set< std::string > > _joint_command_modes`
unordered map of sets storing the active command interfaces for each joint.
- `std::unordered_map< std::string, std::set< std::string > > _new_joint_command_modes`
Temporary cache of new joint_command_modes when switching controllers.
- `std::vector< std::pair< std::string, hardware_interface::InterfaceDescription * > > _ordered_joint_state_interfaces_ptr`
A vector of a pair of interface name and pointer of the hardware's state interface which is ordered by joint and state interface type.
- `std::mutex mtx`
A mutex that is used to prevent concurrent access to hardware and `_joint_command_modes`.

8.3.1 Detailed Description

The bioscara arm hardware interface class.

The hardware interface serves to wrap custom hardware interaction with the arm joints in the standardized ros2_↔ control architecture.

Hardware Lifecycle

The hardware follows the ros2_control hardware interface lifecycle which intern is following the **ROS2 managed node lifecycle**.

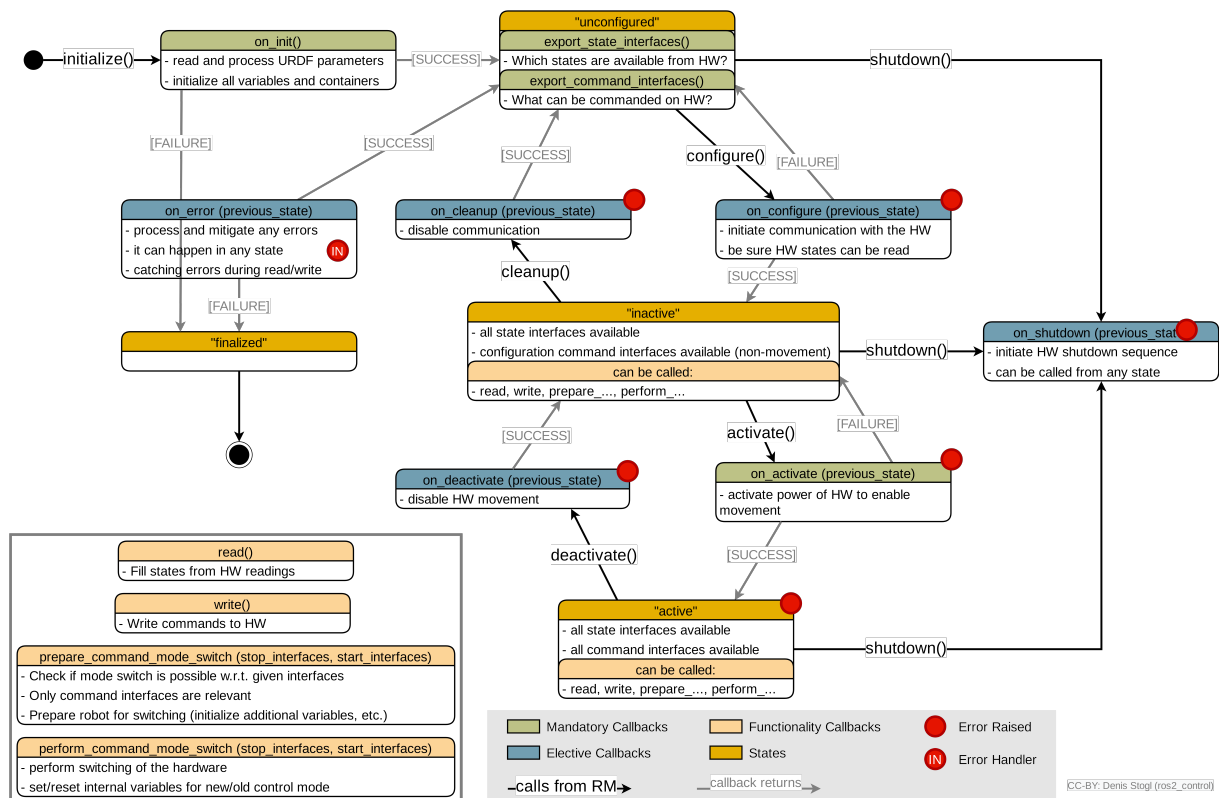


Figure 8.1 Hardware interface lifecycle

8.3.2 Member Function Documentation

8.3.2.1 activate_joint()

```

bioscara_hardware_drivers::err_type_t bioscara_hardware_interfaces::BioscaraArmHardware↵
Interface::activate_joint (
    const std::string name ) [private]

```

Enables each joint, enables the stall detection and sets the maximum acceleration.

Parameters

<i>name</i>	joint name to enable
-------------	----------------------

Returns

[bioscara_hardware_drivers::err_type_t](#)

8.3.2.2 deactivate_joint()

```

bioscara_hardware_drivers::err_type_t bioscara_hardware_interfaces::BioscaraArmHardware↵
Interface::deactivate_joint (
    const std::string name ) [private]

```

Disables each joint.

Parameters

<i>name</i>	joint name to disable
-------------	-----------------------

Returns

[bioscara_hardware_drivers::err_type_t](#)

8.3.2.3 on_activate()

```

hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵
::on_activate (
    const rclcpp_lifecycle::State & previous_state ) [override]

```

Called on the transition from the `inactive` to the `active` state.

Calls [activate_joint\(\)](#) to enable the joints.

It is allowed to activate the hardware even if it is not homed. To home the joint the homing_controller must be activated, but generally a hardware component must be active in order for controllers to become active.

To prohibit movement on activation the set point for each position command interface is set equal to the current measured position, and the velocity command is set to 0.0 for each command interface. The current values are obtained by calling the [read\(\)](#) method once which populates the state interfaces with values.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

Below a workaround to force a read cycle of all joints to get initial values for the state interfaces. These will be copied to the command interface to prevent movement at startup.

8.3.2.4 on_cleanup()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface←
::on_cleanup (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the *inactive* to the *unconfigured* state.

Disconnect from the joints.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

Disconnect from the joints and throw error if it fails

8.3.2.5 on_configure()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface←
::on_configure (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the *unconfigured* to the *inactive* state.

Establish and test connection to each joint.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.3.2.6 on_deactivate()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface↔
::on_deactivate (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transision from the `active` to the `inactive` state.

Disables all joints and thereby allows backdriving. State interfaces continue to be updated.

Parameters

<code>previous_state</code>	
-----------------------------	--

Returns

`hardware_interface::CallbackReturn`

disable the joints and throw error if it fails

8.3.2.7 on_error()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface↔
::on_error (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called when an error in any state or state transition is thrown.

According to the [ros2_control documentation](#):

Error handling follows the node lifecycle. If successful `CallbackReturn::SUCCESS` is returned and hardware is again in `UNCONFIGURED` state, if any `ERROR` or `FAILURE` happens the hardware ends in `FINALIZED` state and can not be recovered. The only option is to reload the complete plugin, but there is currently no service for this in the Controller Manager.

Since the hardware will immediatly return to the `unconfigured` state (`source`) if the error could be handled we manually call the transition functions which would normally be called to this state. Those are:

- **Previous state:** `active`
 - Deactivate hardware ([on_deactivate\(\)](#)) -> `inactive`
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> `unconfigured`
- **Previous state:** `inactive`
 - Deactivate hardware ([on_deactivate\(\)](#)) -> `inactive`
 - * call the deactivate function anyway regardless if state was active or inactive. For example if the [on_activate\(\)](#) function fails on [bioscara_hardware_drivers::Joint::enableStallguard\(\)](#) the joint will have been enabled, to disable it invoke [on_deactivate\(\)](#).
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> `unconfigured`

In particular the deactivation is important. For example if a joint stalls the [read\(\)](#) or [write\(\)](#) methods throw an error, which will be handled here and allow the hardware to be deactivated, disabling the joints to allow backdriving.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.3.2.8 on_init()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface↔
::on_init (
    const hardware_interface::HardwareComponentInterfaceParams & params ) [override]
```

Called on initialization to the `unconfigured` state.

Performs the following checks on the configures joints parsed form the URDF description:

- Each joint must have the 3 command interfaces (in this order): 'position', 'velocity', 'home'
- Each joint must have the 3 state interfaces (in this order): 'position', 'velocity', 'home'

Stores the configuration parameters for each joint in the [_joint_cfg](#) map. Each joint must have these parameters:

- `i2c_address` (int, HEX)
- `reduction` (float)
- `min` (float)
- `max` (float)
- `stall_threshold` (int, DEC)
- `hold_current` (int, DEC)
- `drive_current` (int, DEC)
- `max_acceleration` (float)
- `max_velocity` (float)
- `homing`
 - `speed` (float)
 - `threshold` (int, DEC)
 - `current` (int, DEC)
 - `acceleration` (float)

Adds each joint to the internal `_joints` map. Creates a `MockJoint` object if the `use_mock_hardware` parameter is 'True' or 'true', or else a hardware Joint.

Parameters

<i>params</i>	
---------------	--

Returns

hardware_interface::CallbackReturn

Loop over all joints described in the hardware description file, check if they have the position and velocity command and state interface defined and finally add them to the internal `_joints` list

Todo threshold and current are `uint8_t`, if a number larger outside $0 < n < 255$ is passed as a parameters it will overflow.

8.3.2.9 on_shutdown()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵
::on_shutdown (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the `inactive`, `unconfigured` and `active` to the `finalized` state.

When transitioning directly from `active` to `finalized` `on_deactivate()` is automatically called before [Source Code](#). If the previous state is either `inactive` or `active` the `on_cleanup()` method is called first. Then regardless of the previous state, the `_joints` map is cleared.

Parameters

<code>previous_state</code>	
-----------------------------	--

Returns

hardware_interface::CallbackReturn

8.3.2.10 perform_command_mode_switch()

```
hardware_interface::return_type bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵
::perform_command_mode_switch (
    const std::vector< std::string > & start_interfaces,
    const std::vector< std::string > & stop_interfaces ) [override]
```

Perform the mode-switching for the new command interface combination in realtime context.

Performs the following actions:

- acquires the [mtx](#) lock to protect the critical members.
- Copies the cached `_new_joint_command_modes` to the `_joint_command_modes`
- **On activation:**
 - **home** interface:
 - * Reset command to 0.0. This clears any remaining commands that have been written to the command interface while the hardware was unable to act on it. For example if it was inactive or the homing command was not the active command mode.

Parameters

<i>start_interfaces</i>	vector of string identifiers for the command interfaces starting.
<i>stop_interfaces</i>	vector of string identifiers for the command interfaces stopping.

Returns

return_type::OK if the new command interface combination can be switched to (or) if the interface key is not relevant to this system. Returns return_type::ERROR otherwise.

8.3.2.11 prepare_command_mode_switch()

```
hardware_interface::return_type bioscara_hardware_interfaces::BioscaraArmHardwareInterface↔
::prepare_command_mode_switch (
    const std::vector< std::string > & start_interfaces,
    const std::vector< std::string > & stop_interfaces ) [override]
```

Performs checks and book keeping of the active control mode when changing controllers in non-realtime context.

For safe operation only one controller may interact with the hardware at the time. For example if the velocity JTC is active and has claimed the velocity command interfaces it is technically possible to activate the position JTC (or a homing controller, or others) that claim a different command interface (position in this case). However if both controllers are active they start writing to the hardware simultaneously which is to be avoided. For this reason a book keeping mechanism has been implemented which stores the currently active command interfaces for each joint in the [_joint_command_modes](#) member. Each joint has a set of active command interfaces. When a controller switch is performed the interfaces that should be stopped are removed from each joint set, then the one that should be started are added, if they are already present an error is thrown. Lastly a validation is performed. Currently the validation is simple since each joint may only have one command interface. The validation can be expanded for future use cases that require a combination of active command interfaces per joint for example.

The following basic checks are implemented:

- **On deactivation:**

- [ERROR] Homing command interfaces may only be deactivated if no current homing process is ongoing ([bioscara_hardware_drivers::Joint::getCurrentBCmd\(\)](#) != [bioscara_hardware_drivers::Joint::HOME](#))
- [WARN] Deactivating a velocity command interface if the velocity set point is 0.0.
- [WARN] Deactivating a command interface that has not been started. This should not happen.

- **On activation:**

- [ERROR] Activating a command interface that is already started. This should not happen.
- [ERROR] Activating a second command interface for a joint.
- [ERROR] Activating 'position' or 'velocity' command interface if the joint is not homed ([bioscara_hardware_drivers::Joint::isHomed\(\)](#) == false).

Since this method operates in non-realtime context it must not access critical members ([_joint_command_modes](#) and [_joints](#)) to avoid priority inversion. Therefore the new command modes are first saved to a cache [_new_joint_command_modes](#). This will then be applied to [_joint_command_modes](#) in the [perform_command_mode_switch\(\)](#) which is executed in RT context.

Parameters

<i>start_interfaces</i>	command interfaces that should be started in the form "joint/interface"
<i>stop_interfaces</i>	command interfaces that should be stopped in the form "joint/interface"

Returns

hardware_interface::return_type

8.3.2.12 read()

```
hardware_interface::return_type bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵
::read (
    const rclcpp::Time & time,
    const rclcpp::Duration & period ) [override]
```

Reads from the hardware and populates the state interfaces.

Iterates over all state interfaces and calls the corresponding Joint method.

- State interface "position" -> [bioscara_hardware_drivers::Joint::getPosition\(\)](#)
- State interface "velocity" -> [bioscara_hardware_drivers::Joint::getVelocity\(\)](#)
- State interface "home" -> [bioscara_hardware_drivers::Joint::isHomed\(\)](#)
 - This does not actually trigger a communication, instead it relies on the return flags of the previous transmissions. Since position and velocity have been called immediatly before the return flags are assumed to be valid.
 - If the the homing of a joint has been activated through the command interface ([bioscara_hardware_drivers::Joint::getCurrentPosition\(\)](#) == [bioscara_hardware_drivers::Joint::HOME](#)) the device signals BUSY ([bioscara_hardware_drivers::Joint::isBusy\(\)](#)) as long as it is still homing. If the BUSY flag is reset while the current command is still [bioscara_hardware_drivers::Joint::HOME](#) we can assume the homing has finished. Then the "home" command interface of the joint is reset to 0.0, which will stop the homing (perform cleanup tasks) at the next write cycle.

Parameters

<i>time</i>	
<i>period</i>	

Returns

hardware_interface::return_type

8.3.2.13 split_interface_string_to_joint_and_name()

```
void bioscara_hardware_interfaces::BioscaraArmHardwareInterface::split_interface_string_to↵
joint_and_name (
    std::string interface,
    std::string & joint_name,
    std::string & interface_name ) [private]
```

Split a interface string like "<joint_name>/<interface_name>" to "<joint_name>" and "<interface_name>".

Parameters

<i>interface</i>	
<i>joint_name</i>	
<i>interface_name</i>	

8.3.2.14 start_homing()

```

bioscara_hardware_drivers::err_type_t bioscara_hardware_interfaces::BioscaraArmHardware↵
Interface::start_homing (
    const std::string name,
    float velocity ) [private]

```

wrapper method to start homing.

Activate the joint, set homing acceleration and start homing.

Parameters

<i>name</i>	
<i>velocity</i>	

Returns

[bioscara_hardware_drivers::err_type_t](#)

8.3.2.15 stop_homing()

```

bioscara_hardware_drivers::err_type_t bioscara_hardware_interfaces::BioscaraArmHardware↵
Interface::stop_homing (
    const std::string name ) [private]

```

wrapper method to stop homing.

Stop the homing. Reset acceleration and velocity and perform the postHoming cleanup, then deactivate the joint.

Parameters

<i>name</i>	
-------------	--

Returns

[bioscara_hardware_drivers::err_type_t](#)

8.3.2.16 write()

```

hardware_interface::return_type bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵
::write (

```

```
const rclcpp::Time & time,
const rclcpp::Duration & period ) [override]
```

Writes commands to the hardware from the command interfaces.

In contrast to the [read\(\)](#) method the [write\(\)](#) method only loops over the command interfaces that are currently active defined by the [_joint_command_modes](#) map. See [prepare_command_mode_switch\(\)](#) for a detailed reasoning why this approach has been chosen.

- Command interface "position" -> [bioscara_hardware_drivers::Joint::setPosition\(\)](#)
- Command interface "velocity" -> [bioscara_hardware_drivers::Joint::setVelocity\(\)](#)
- Command interface "home" -> [bioscara_hardware_drivers::Joint::startHoming\(\)](#)
 - If the commanded value in "home" is != 0.0 the and the joint is currently executing a blocking function, for example homing ([bioscara_hardware_drivers::Joint::getCurrentBCmd\(\)](#) == [bioscara_hardware_drivers::Joint::NONE](#)), the homing sequence is started with the speed, sensitivity, current and acceleration defined in the [_joint_cfg](#) which is polulated from the hardware description urdf. The direction of the homing is determined by the sign of the command interface value.
 - If the commanded value in "home" is = 0.0 and the joint is currently executing homing, the homing is stopped. This can either happen prematurely through user input or when the homing is completed which is registered in [read\(\)](#).

Parameters

<i>time</i>	
<i>period</i>	

Returns

hardware_interface::return_type

8.3.3 Member Data Documentation

8.3.3.1 _joint_cfg

```
std::unordered_map<std::string, joint_config_t> bioscara_hardware_interfaces::BioscaraArm↔
HardwareInterface::_joint_cfg [private]
```

unordered map storing the configuration struct of the joints.

An unordered map is chosen to simplify acces via the joint name, as this conforms well with the ROS2_control hardware interface The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

8.3.3.2 _joint_command_modes

```
std::unordered_map<std::string, std::set<std::string> > bioscara_hardware_interfaces::↔
BioscaraArmHardwareInterface::_joint_command_modes [private]
```

unordered map of sets storing the active command interfaces for each joint.

Each joint can have a set of active command interfaces. This type of structure is chosen to group interfaces by joint. In the [write\(\)](#) function the interface name can simply be constructed by concatenating joint name with interface name. Although currently only one active command interface is allowed at the time, a set can be used to store multiple command interfaces that are acceptable to be combined, for example it would be acceptable to set velocity and driver current and hence that would be an allowable combination.

An unordered map is chosen to simplify access via the joint name, as this conforms well with the ROS2_control hardware interface. The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

8.3.3.3 `_joints`

```
std::unordered_map<std::string, std::unique_ptr<bioscara_hardware_drivers::BaseJoint> > bioscara_↵
hardware_interfaces::BioscaraArmHardwareInterface::_joints [private]
```

unordered map storing the pointers to BaseJoint objects. This will either be a MockJoint or Joint.

An unordered map is chosen to simplify access via the joint name, as this conforms well with the ROS2_control hardware interface. The map does not need to be ordered. Search, insertion, and removal of elements have average constant-time complexity.

Since the BaseJoint methods are implemented as virtual, dynamic method dispatch can be utilized to call the correct implementation of a method. So either BaseJoint::foo() or Joint::foo()/MockJoint::foo() if foo() is overwritten in Joint or MockJoint. A smart pointer is used to guarantee destruction when the pointer is destructed. A unique pointer is used to prevent copying of the object.

8.3.3.4 `_new_joint_command_modes`

```
std::unordered_map<std::string, std::set<std::string> > bioscara_hardware_interfaces::↵
BioscaraArmHardwareInterface::_new_joint_command_modes [private]
```

Temporary cache of new joint_command_modes when switching controllers.

Since the [prepare_command_mode_switch\(\)](#) is executed in a non-RT context we save the new joint command modes to this cache first to avoid needing to lock the [_joint_command_modes](#).

8.3.3.5 `_ordered_joint_state_interfaces_ptr`

```
std::vector<std::pair<std::string, hardware_interface::InterfaceDescription *> > bioscara_↵
hardware_interfaces::BioscaraArmHardwareInterface::_ordered_joint_state_interfaces_ptr [private]
```

A vector of a pair of interface name and pointer of the hardware's state interface which is ordered by joint and state interface type.

A vector is chosen since it guarantees the correct order by insertion. The vector holds pointers to the actual interface structs stored in the parents HardwareComponentInterface::joint_state_interfaces_ but in the desired order. The order is:

1. position
2. velocity
3. home

This order guarantees that when reading the state interfaces in [read\(\)](#) that the 'home' interface is read last and has the latest state flags from the joint.

8.3.3.6 mtx

```
std::mutex bioscara_hardware_interfaces::BioscaraArmHardwareInterface::mtx [private]
```

A mutex that is used to prevent concurrent access to hardware and [_joint_command_modes](#).

The mutex prevents two things:

- Modifying the [_joint_command_modes](#) concurrently.
- Concurrent access to the hardware via the [_joints](#) map. The Joint hardware is not thread safe. In particular the [read\(\)](#) and [write\(\)](#) methods are executed in one RT thread while the [perform_command_mode_switch\(\)](#) is called from another RT thread. The latter also tries to modify the Joint object via [activate_joint\(\)](#) and [deactivate_joint\(\)](#) which must not happen concurrently with a [read\(\)](#) or [write\(\)](#) call.

All methods that need to acquire this lock need to be performed in RT context to avoid being preempted by other lower priority threads potentially blocking the other RT threads from continuing.

The documentation for this class was generated from the following files:

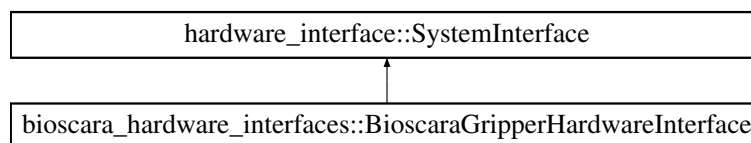
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_hardware_↔
interface/arm_hardware.hpp
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/src/arm_hardware.cpp

8.4 bioscara_hardware_interfaces::BioscaraGripperHardwareInterface Class Reference

The bioscara gripper hardware interface class.

```
#include <gripper_hardware.hpp>
```

Inheritance diagram for bioscara_hardware_interfaces::BioscaraGripperHardwareInterface:



Classes

- struct [gripper_config_t](#)
configuration structure holding the passed paramters from the ros2_control urdf

Public Member Functions

- `hardware_interface::CallbackReturn` [on_init](#) (const `hardware_interface::HardwareComponentInterface`↔
`Params ¶ms`) override
- `hardware_interface::CallbackReturn` [on_shutdown](#) (const `rcpp_lifecycle::State &previous_state`) override
Called on the transition from the `inactive`, `unconfigured` and `active` to the `finalized` state.
- `hardware_interface::CallbackReturn` [on_configure](#) (const `rcpp_lifecycle::State &previous_state`) override
Called on the transition from the `unconfigured` to the `inactive` state.
- `hardware_interface::CallbackReturn` [on_cleanup](#) (const `rcpp_lifecycle::State &previous_state`) override
Called on the transition from the `inactive` to the `unconfigured` state.
- `hardware_interface::CallbackReturn` [on_activate](#) (const `rcpp_lifecycle::State &previous_state`) override
Called on the transition from the `inactive` to the `active` state.
- `hardware_interface::CallbackReturn` [on_deactivate](#) (const `rcpp_lifecycle::State &previous_state`) override
Called on the transition from the `active` to the `inactive` state.
- `hardware_interface::return_type` [read](#) (const `rcpp::Time &time`, const `rcpp::Duration &period`) override
Reads from the hardware and populates the state interfaces.
- `hardware_interface::return_type` [write](#) (const `rcpp::Time &time`, const `rcpp::Duration &period`) override
Writes commands to the hardware from the command interfaces.
- `hardware_interface::CallbackReturn` [on_error](#) (const `rcpp_lifecycle::State &previous_state`) override
Called when an error in any state or state transition is thrown.

Private Attributes

- `std::unique_ptr< bioscara_hardware_drivers::BaseGripper > _gripper`
Smart pointer to the local `BaseGripper`.
- `gripper_config_t _gripper_cfg`
configuration struct of the gripper.
- `float _last_pos` = `std::numeric_limits<double>::quiet_NaN()`
- `float _vel` = `std::numeric_limits<double>::quiet_NaN()`

8.4.1 Detailed Description

The bioscara gripper hardware interface class.

System interface has been chosen to allow future modifications for grippers that provide feedback. refer to the [BioscaraArmHardwareInterface](#) class for a more detailed description of the hardware life-cycle.

8.4.2 Member Function Documentation

8.4.2.1 on_activate()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↔
Interface::on_activate (
    const rcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the `inactive` to the `active` state.

Enables PWM generation.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.2 on_cleanup()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↔
Interface::on_cleanup (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transistion from the `inactive` to the `unconfigured` state.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.3 on_configure()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↔
Interface::on_configure (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transistion from the `unconfigured` to the `inactive` state.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.4 on_deactivate()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↔
Interface::on_deactivate (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transistion from the `active` to the `inactive` state.

Disables PWM generation.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.5 on_error()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↵
Interface::on_error (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called when an error in any state or state transition is thrown.

According to the [ros2_control documentation](#):

Error handling follows the node lifecycle. If successful CallbackReturn::SUCCESS is returned and hardware is again in UNCONFIGURED state, if any ERROR or FAILURE happens the hardware ends in FINALIZED state and can not be recovered. The only option is to reload the complete plugin, but there is currently no service for this in the Controller Manager.

Since the hardware will immediatly return to the unconfigured state ([source](#)) if the error could be handled we manually call the transition functions which would normally be called to this state. Those are:

- **Previous state:** active
 - Deactivate hardware ([on_deactivate\(\)](#)) -> inactive
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> unconfigured
- **Previous state:** inactive
 - Deactivate hardware ([on_deactivate\(\)](#)) -> inactive
 - * call the deactivate function anyway regardless if state was active or inactive.
 - Clean-Up hardware ([on_cleanup\(\)](#)) -> unconfigured

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.6 on_init()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardware↵
Interface::on_init (
    const hardware_interface::HardwareComponentInterfaceParams & params ) [override]
```

check that only one joint is defined, more mimic joints may be defined.

8.4.2.7 on_shutdown()

```
hardware_interface::CallbackReturn bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::on_shutdown (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the inactive, unconfigured and active to the finalized state.

When transitioning directly from active to finalized [on_deactivate\(\)](#) is automatically called before [Source Code](#) If the previous state is either inactive or active the [on_cleanup\(\)](#) method is called first.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

hardware_interface::CallbackReturn

8.4.2.8 read()

```
hardware_interface::return_type bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::read (
    const rclcpp::Time & time,
    const rclcpp::Duration & period ) [override]
```

Reads from the hardware and populates the state interfaces.

Invokes the [bioscara_hardware_drivers::Gripper::getPosition\(\)](#) method to read the gripper position.

Parameters

<i>time</i>	
<i>period</i>	

Returns

hardware_interface::return_type

8.4.2.9 write()

```
hardware_interface::return_type bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::write (
    const rclcpp::Time & time,
    const rclcpp::Duration & period ) [override]
```

Writes commands to the hardware from the command interfaces.

Invokes the [bioscara_hardware_drivers::Gripper::setPosition\(\)](#) method to write the command to the gripper.

Parameters

<i>time</i>	
<i>period</i>	

Returns

hardware_interface::return_type

8.4.3 Member Data Documentation

8.4.3.1 _gripper

```
std::unique_ptr<bioscara_hardware_drivers::BaseGripper> bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::_gripper [private]
```

Smart pointer to the local BaseGripper.

This will be used to either interact with the hardware or mock hardware. A smart pointer is used to guarantee destruction when the pointer is destructed. A unique pointer is used to prevent copying of the object.

8.4.3.2 _gripper_cfg

```
gripper_config_t bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::_gripper_cfg [private]
```

configuration struct of the gripper.

8.4.3.3 _last_pos

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::_last_pos = std::numeric_limits<double>::quiet_NaN() [private]
```

8.4.3.4 _vel

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::_vel = std::numeric_limits<double>::quiet_NaN() [private]
```

The documentation for this class was generated from the following files:

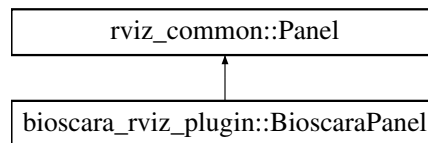
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/include/bioscara_gripper_hardware_interface/gripper_hardware.hpp
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/src/gripper_hardware.cpp

8.5 bioscara_rviz_plugin::BioscaraPanel Class Reference

RViz Panel to control the hardware specific functions of the Bioscara robot.

```
#include <bioscara_panel.hpp>
```

Inheritance diagram for bioscara_rviz_plugin::BioscaraPanel:



Public Member Functions

- [BioscaraPanel](#) (QWidget *parent=0)
Construct a new Bioscara Panel object.
- [~BioscaraPanel](#) () override
Destroy the Bioscara Panel object.
- void [onInitialize](#) () override
Automatically invoked by the RViz parent class. Initializes the panel.

Protected Member Functions

- void [cm_state_callback](#) (const ControllerManagerActivity &msg)
callback on reported controller or hardware state changes.
- void [joint_state_callback](#) (const DynamicJointState &msg)
callback on joint state update.
- void [ensure_jsb_is_active](#) (void)
Ensure the joint state broadcaster is active.
- void [set_hardware_component_state](#) (const std::string component, const lifecycle_msgs::msg::State target_state)
Makes a service request to change a hardware component state.
- void [configure_controller](#) (const std::string controller)
Makes a service request to configure a controller state.
- void [switch_controllers](#) (const std::vector< std::string > &activate_controllers, const std::vector< std::string > &deactivate_controllers, const int32_t=SwitchController::Request::BEST_EFFORT, const bool activate_asap=false, const builtin_interfaces::msg::Duration timeout=builtin_interfaces::msg::Duration())
Makes a service request to change controller states.
- void [set_controller_state](#) (const std::string controller, const lifecycle_msgs::msg::State target_state)
calls the correct functions to bring a controller to the correct state.
- void [dynamic_joint_state_msg_to_map](#) (const DynamicJointState &dynamic_joint_state_in, std::unordered_map< std::string, InterfaceValue > &map_out)
simple conversion message type conversion function
- void [named_lcs_msg_to_map](#) (const std::vector< NamedLifecycleState > &named_lcs_in, std::unordered_map< std::string, NamedLifecycleState > &map_out)
simple conversion message type conversion function
- void [print_cm_map](#) (const std::string prefix, const std::unordered_map< std::string, NamedLifecycleState > &map_in)
Print life-cycle state message to console.

- void [prepopulate_joint_state_map](#) (std::unordered_map< std::string, InterfaceValue > &state_map, std::vector< std::string > augment_vec)
Prepopulates a state map with a default NamedLifecycleState for every specified key.
- void [prepopulate_state_map](#) (std::unordered_map< std::string, NamedLifecycleState > &state_map, std::vector< std::string > augment_vec)
Prepopulates a state map with a default NamedLifecycleState for every specified key.
- void [update_homing_grp_state](#) (void)
Enabl/Disables the homing group depending on the homing controller state.
- void [update_homing_state_labels](#) (void)
updates all homing state labels.
- void [set_homing_state_label](#) (QLabel *label, const InterfaceValue &state)
Set state label text and color of the homing state label.
- void [update_state_labels_and_btns](#) (void)
Update all hardware and controller state lables.
- void [update_state_label_and_btn](#) (const std::unordered_map< std::string, NamedLifecycleState > &state_map, QLabel *state_label, QPushButton *en_button, const std::string &state_key)
updates component state label and control button.
- void [set_state_label](#) (QLabel *label, const NamedLifecycleState &state)
Set the state label.
- void [set_en_btn](#) (QPushButton *button, const NamedLifecycleState &state)
Set the control button state.
- bool [check_activation_conditions](#) (const std::string component_key)
Check under which conditions a component is allowed to be activated.
- lifecycle_msgs::msg::State [target_state_from_current](#) (lifecycle_msgs::msg::State current_state)
returns the next logical state from the current one.
- void [homing_cmd](#) (const std::string joint, const int cmd)
callback function called when a homing button is pressed
- void [ctrl_en_btn_cb](#) (const std::string controller)
callback on controller enable button

Protected Attributes

- rclcpp::Node::SharedPtr [node_](#)
Pointer to the parent node. Used for access to the executor.
- rclcpp::Subscription< ControllerManagerActivity >::SharedPtr [cm_state_subscription_](#)
Subscription to the the controller_manager state publisher.
- rclcpp::Subscription< DynamicJointState >::SharedPtr [joint_state_subscription_](#)
Subscription to the the joint state publisher state publisher.
- rclcpp::Publisher< DynamicInterfaceGroupValues >::SharedPtr [homing_publisher_](#)
Publisher to send commands to the homing_controller.
- rclcpp::Client< SwitchController >::SharedPtr [switch_controller_client_](#)
Service client to the "/controller_manager/switch_controller" service.
- rclcpp::Client< ConfigureController >::SharedPtr [configure_controller_client_](#)
Service client to the "/controller_manager/configure_controller" service.
- rclcpp::Client< SetHardwareComponentState >::SharedPtr [hardware_state_client_](#)
Service client to the "/controller_manager/set_hardware_component_state" service.
- rclcpp::TimerBase::SharedPtr [prune_timer_](#)
Timer triggered every 5 s to prune unanswered service requests.
- std::unordered_map< std::string, InterfaceValue > [joint_states_](#)
Map storing all joint state interface values by each joint.

- `std::unordered_map< std::string, NamedLifecycleState >` [hardware_states_](#)
Map correlating hardware component name to its lifecycle_msgs/State.
- `std::unordered_map< std::string, NamedLifecycleState >` [controller_states_](#)
Map correlating controller name to its lifecycle_msgs/State.
- `Ui::BioscaraUI *` [ui_](#)
The QT object containing all UI elements.

Private Slots

- `void` [arm_en_btn_cb](#) (`void`)
Tries to set the bioscara_arm hardware component to the active state and tries to enable the velocity_joint_↔ trajectory_controller.
- `void` [gripper_en_btn_cb](#) (`void`)
Tries to set the bioscara_gripper_128 hardware component to the active state and tries to enable the gripper_↔ controller.

8.5.1 Detailed Description

RViz Panel to control the hardware specific functions of the Bioscara robot.

Through the panel it is possible to modify the robot's hardware's and controller states and to execute the homing procedures.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 BioscaraPanel()

```
bioscara_rviz_plugin::BioscaraPanel::BioscaraPanel (
    QWidget * parent = 0 ) [explicit]
```

Construct a new Bioscara Panel object.

Creates a new [ui_](#) object from the ui object autogenerated from the "bioscara_rviz_plugin_frame.ui". This object contains all UI elements. Each button gets assigned a callback function.

Parameters

<i>parent</i>	
---------------	--

8.5.2.2 ~BioscaraPanel()

```
bioscara_rviz_plugin::BioscaraPanel::~~BioscaraPanel ( ) [override]
```

Destroy the Bioscara Panel object.

8.5.3 Member Function Documentation

8.5.3.1 arm_en_btn_cb

```
void bioscara_rviz_plugin::BioscaraPanel::arm_en_btn_cb (
    void ) [private], [slot]
```

Tries to set the bioscara_arm hardware component to the `active` state and tries to enable the velocity_joint_↵ trajectory_controller.

8.5.3.2 check_activation_conditions()

```
bool bioscara_rviz_plugin::BioscaraPanel::check_activation_conditions (
    const std::string component_key ) [protected]
```

Check under which conditions a component is allowed to be activated.

Some controllers will only activate under certain conditions. This functions disables the activation button if the conditions are met.

Parameters

<i>component_key</i>	
----------------------	--

Returns

true
false

8.5.3.3 cm_state_callback()

```
void bioscara_rviz_plugin::BioscaraPanel::cm_state_callback (
    const ControllerManagerActivity & msg ) [protected]
```

callback on reported controller or hardware state changes.

Updates the [controller_states_](#) and [hardware_states_](#) state maps. Reactivates the Joint State Broadcaster if it has become inactive. Updates controller and hardware state labels and buttons by invoking [update_state_labels_and_btns\(\)](#) Updates homing group state by invoking [update_homing_grp_state\(\)](#).

Parameters

<i>msg</i>	received controller states
------------	----------------------------

8.5.3.4 configure_controller()

```
void bioscara_rviz_plugin::BioscaraPanel::configure_controller (
    const std::string controller ) [protected]
```

Makes a service request to configure a controller state.

Request issued through the [configure_controller_client_](#). Asynchronous call, method exits after having issued the request.

Parameters

<i>controller</i>	name of controller to activate
-------------------	--------------------------------

8.5.3.5 ctrl_en_btn_cb()

```
void bioscara_rviz_plugin::BioscaraPanel::ctrl_en_btn_cb (
    const std::string controller ) [protected]
```

callback on controller enable button

Parameters

<i>controller</i>	string name of controller to change state
-------------------	---

8.5.3.6 dynamic_joint_state_msg_to_map()

```
void bioscara_rviz_plugin::BioscaraPanel::dynamic_joint_state_msg_to_map (
    const DynamicJointState & dynamic_joint_state_in,
    std::unordered_map< std::string, InterfaceValue > & map_out ) [protected]
```

simple conversion message type conversion function

Parameters

<i>dynamic_joint_state_in</i>	message in
<i>map_out</i>	joint states map out

8.5.3.7 ensure_jsb_is_active()

```
void bioscara_rviz_plugin::BioscaraPanel::ensure_jsb_is_active (
    void ) [protected]
```

Ensure the joint state broadcaster is active.

The JSB can become inactive, ensure that it comes active again.

Calls the [set_controller_state\(\)](#) method

8.5.3.8 gripper_en_btn_cb

```
void bioscara_rviz_plugin::BioscaraPanel::gripper_en_btn_cb (
    void ) [private], [slot]
```

Tries to set the bioscara_gripper_128 hardware component to the active state and tries to enable the gripper_↔ controller.

8.5.3.9 homing_cmd()

```
void bioscara_rviz_plugin::BioscaraPanel::homing_cmd (
    const std::string joint,
    const int cmd ) [protected]
```

callback function called when a homing button is pressed

publishes on [homing_publisher_](#) to initiate/stop homing

Parameters

<i>joint</i>	string name of joint
<i>cmd</i>	-1, 0, +1 to start homing to negative direction, stop homing, start homing to positive direction.

8.5.3.10 joint_state_callback()

```
void bioscara_rviz_plugin::BioscaraPanel::joint_state_callback (
    const DynamicJointState & msg ) [protected]
```

callback on joint state update.

Updates [joint_states_](#) with the latest values using [dynamic_joint_state_msg_to_map\(\)](#). Updates homing group state by invoking [update_homing_grp_state\(\)](#).

Parameters

<i>msg</i>	received joint states
------------	-----------------------

8.5.3.11 named_lcs_msg_to_map()

```
void bioscara_rviz_plugin::BioscaraPanel::named_lcs_msg_to_map (
    const std::vector< NamedLifecycleState > & named_lcs_in,
    std::unordered_map< std::string, NamedLifecycleState > & map_out ) [protected]
```

simple conversion message type conversion function

Parameters

<i>named_lcs_in</i>	NamedLifecycleState message in
<i>map_out</i>	state map out

8.5.3.12 onInitialize()

```
void bioscara_rviz_plugin::BioscaraPanel::onInitialize ( ) [override]
```

Automatically invoked by the RViz parent class. Initializes the panel.

Retrieves the pointer to the parent node and saves it in [node_](#). This node and its executor is used for all clients and publishers.

Additionally a the [prune_timer_](#) is created which periodically cleans all dangling service requests. All subscribers, publishers and clients are created and saved in their corresponding handler. Also the [#controller_states_state](#) and [hardware_states_](#) state maps are prepopulated by invoking [prepopulate_state_map\(\)](#)

8.5.3.13 prepopulate_joint_state_map()

```
void bioscara_rviz_plugin::BioscaraPanel::prepopulate_joint_state_map (
    std::unordered_map< std::string, InterfaceValue > & state_map,
    std::vector< std::string > augment_vec ) [protected]
```

Prepopulates a state map with a default NamedLifecycleState for every specified key.

This is done to avoid having to check for the existence of key when accessing the map. Keys could be missing if hardware interfaces are not specified or named differently.

For every joint the state interfaces are initialized as follows:

- *position*: 0.0
- *velocity*: 0.0
- *home*: 0.0

Some joints (the gripper) dont have all state interfaces, but they are initialized regardless.

Parameters

<i>state_map</i>	The state map to prepopulate
<i>augment_vec</i>	keys that are added/overwritten

8.5.3.14 prepopulate_state_map()

```
void bioscara_rviz_plugin::BioscaraPanel::prepopulate_state_map (
    std::unordered_map< std::string, NamedLifecycleState > & state_map,
    std::vector< std::string > augment_vec ) [protected]
```

Prepopulates a state map with a default NamedLifecycleState for every specified key.

This is done to avoid having to check for the existence of key when accessing the map. Keys could be missing if an expected controller or hardware component has not been loaded or has a different name.

Parameters

<i>state_map</i>	The state map to prepopulate
<i>augment_vec</i>	keys that are added/overwritten

8.5.3.15 print_cm_map()

```
void bioscara_rviz_plugin::BioscaraPanel::print_cm_map (
    const std::string prefix,
    const std::unordered_map< std::string, NamedLifecycleState > & map_in ) [protected]
```

Print life-cycle state message to console.

Parameters

<i>prefix</i>	string prefix
<i>map_in</i>	map to be printed

8.5.3.16 set_controller_state()

```
void bioscara_rviz_plugin::BioscaraPanel::set_controller_state (
    const std::string controller,
    const lifecycle_msgs::msg::State target_state ) [protected]
```

calls the correct functions to bring a controller to the correct state.

Checks if [configure_controller\(\)](#) needs to be called before [switch_controllers\(\)](#) can be called to bring the controller to the desired state.

Parameters

<i>controller</i>	controller name
<i>target_state</i>	target state

8.5.3.17 set_en_btn()

```
void bioscara_rviz_plugin::BioscaraPanel::set_en_btn (
    QPushButton * button,
    const NamedLifecycleState & state ) [protected]
```

Set the control button state.

Sets control buttons color, text and active status based on the component state.

Parameters

<i>button</i>	pointer to button to modify
<i>state</i>	state of the component

8.5.3.18 set_hardware_component_state()

```
void bioscara_rviz_plugin::BioscaraPanel::set_hardware_component_state (
```

```
const std::string component,
const lifecycle_msgs::msg::State target_state ) [protected]
```

Makes a service request to change a hardware component state.

Request issued through the [hardware_state_client_](#). Asynchronous call, method exits after having issued the request.

Parameters

<i>component</i>	name of hardware component to modify
<i>target_state</i>	target state

8.5.3.19 set_homing_state_label()

```
void bioscara_rviz_plugin::BioscaraPanel::set_homing_state_label (
    QLabel * label,
    const InterfaceValue & state ) [protected]
```

Set state label text and color of the homing state label.

Sets color and text of label.

Parameters

<i>label</i>	pointer to label
<i>state</i>	joint state

8.5.3.20 set_state_label()

```
void bioscara_rviz_plugin::BioscaraPanel::set_state_label (
    QLabel * label,
    const NamedLifecycleState & state ) [protected]
```

Set the state label.

sets color and text based on the component state.

Parameters

<i>label</i>	pointer to label to modify
<i>state</i>	state of the component

8.5.3.21 switch_controllers()

```
void bioscara_rviz_plugin::BioscaraPanel::switch_controllers (
    const std::vector< std::string > & activate_controllers,
    const std::vector< std::string > & deactivate_controllers,
```

```

        const int32_t strictness = SwitchController::Request::BEST_EFFORT,
        const bool activate_asap = false,
        const builtin_interfaces::msg::Duration timeout = builtin_interfaces::msg::↵
:Duration() ) [protected]

```

Makes a service request to change controller states.

Activates and deactivates a combination of controllers.

Parameters

<i>activate_controllers</i>	list of controllers to activate
<i>deactivate_controllers</i>	list of controllers to deactivate
<i>activate_asap</i>	if true controller switch happens in the realtime loop
<i>timeout</i>	switch timeout

8.5.3.22 target_state_from_current()

```

lifecycle_msgs::msg::State bioscara_rviz_plugin::BioscaraPanel::target_state_from_current (
    lifecycle_msgs::msg::State current_state ) [protected]

```

returns the next logical state from the current one.

Parameters

<i>current_state</i>	current state
----------------------	---------------

Returns

`lifecycle_msgs::msg::State`

8.5.3.23 update_homing_grp_state()

```

void bioscara_rviz_plugin::BioscaraPanel::update_homing_grp_state (
    void ) [protected]

```

Enabl/Disables the homing group depending on the homing controller state.

8.5.3.24 update_homing_state_labels()

```

void bioscara_rviz_plugin::BioscaraPanel::update_homing_state_labels (
    void ) [protected]

```

updates all homing state labels.

Calls [set_homing_state_label\(\)](#) for each joint state label.

8.5.3.25 update_state_label_and_btn()

```
void bioscara_rviz_plugin::BioscaraPanel::update_state_label_and_btn (
    const std::unordered_map< std::string, NamedLifecycleState > & state_map,
    QLabel * state_label,
    QPushButton * en_button,
    const std::string & state_key ) [protected]
```

updates component state label and control button.

Calls [set_state_label\(\)](#) for state_label and [set_en_btn\(\)](#) for en_button.

Parameters

<i>state_map</i>	component state map
<i>state_label</i>	pointer to state label
<i>en_button</i>	pointer to enable button
<i>state_key</i>	string key to access in state_map

8.5.3.26 update_state_labels_and_btns()

```
void bioscara_rviz_plugin::BioscaraPanel::update_state_labels_and_btns (
    void ) [protected]
```

Update all hardware and controller state lables.

calls [update_state_label_and_btn\(\)](#) for each controller and hardware component.

8.5.4 Member Data Documentation

8.5.4.1 cm_state_subscription_

```
rclcpp::Subscription<ControllerManagerActivity>::SharedPtr bioscara_rviz_plugin::Bioscara↵
Panel::cm_state_subscription_ [protected]
```

Subscription to the the controller_manager state publisher.

Used to retrieve information about hardware and controller state changes.

8.5.4.2 configure_controller_client_

```
rclcpp::Client<ConfigureController>::SharedPtr bioscara_rviz_plugin::BioscaraPanel::configure↵
_controller_client_ [protected]
```

Service client to the "/controller_manager/configure_controller" service.

Used to request configuration and cleanup of controllers.

8.5.4.3 controller_states_

```
std::unordered_map<std::string, NamedLifecycleState> bioscara_rviz_plugin::BioscaraPanel↔
::controller_states_ [protected]
```

Map correlating controller name to its lifecycle_msgs/State.

Stored as NamedLifecycleState to also simply access controller name from map value.

8.5.4.4 hardware_state_client_

```
rclcpp::Client<SetHardwareComponentState>::SharedPtr bioscara_rviz_plugin::BioscaraPanel↔
::hardware_state_client_ [protected]
```

Service client to the "/controller_manager/set_hardware_component_state" service.

Used to request activation and deactivation of hardware components.

8.5.4.5 hardware_states_

```
std::unordered_map<std::string, NamedLifecycleState> bioscara_rviz_plugin::BioscaraPanel↔
::hardware_states_ [protected]
```

Map correlating hardware component name to its lifecycle_msgs/State.

Stored as NamedLifecycleState to also simply access component name from map value.

8.5.4.6 homing_publisher_

```
rclcpp::Publisher<DynamicInterfaceGroupValues>::SharedPtr bioscara_rviz_plugin::Bioscara↔
Panel::homing_publisher_ [protected]
```

Publisher to send commands to the homing_controller.

Used to start and stop homing of joints.

8.5.4.7 joint_state_subscription_

```
rclcpp::Subscription<DynamicJointState>::SharedPtr bioscara_rviz_plugin::BioscaraPanel::joint↔
_state_subscription_ [protected]
```

Subscription to the joint state publisher state publisher.

Uses the "/dynamic_joint_states" to receive updates of all joints states.

8.5.4.8 joint_states_

```
std::unordered_map<std::string, InterfaceValue> bioscara_rviz_plugin::BioscaraPanel::joint↔
states_ [protected]
```

Map storing all joint state interface values by each joint.

8.5.4.9 node_

```
rcldcpp::Node::SharedPtr bioscara_rviz_plugin::BioscaraPanel::node_ [protected]
```

Pointer to the parent node. Used for access to the executor.

8.5.4.10 prune_timer_

```
rcldcpp::TimerBase::SharedPtr bioscara_rviz_plugin::BioscaraPanel::prune_timer_ [protected]
```

Timer triggered every 5 s to prune unanswered service requests.

8.5.4.11 switch_controller_client_

```
rcldcpp::Client<SwitchController>::SharedPtr bioscara_rviz_plugin::BioscaraPanel::switch_↵  
controller_client_ [protected]
```

Service client to the "/controller_manager/switch_controller" service.

Used to request activation and deactivation of controllers.

8.5.4.12 ui_

```
Ui::BioscaraUI* bioscara_rviz_plugin::BioscaraPanel::ui_ [protected]
```

The QT object containing all UI elements.

Autogenerated from the "bioscara_rviz_plugin_frame.ui" file.

The documentation for this class was generated from the following files:

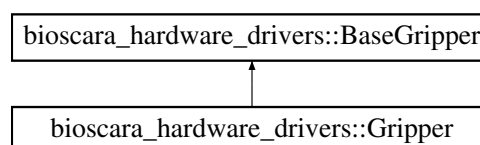
- lib/ros2_ws/src/bioscara_rviz_plugin/include/bioscara_rviz_plugin/bioscara_panel.hpp
- lib/ros2_ws/src/bioscara_rviz_plugin/src/bioscara_panel.cpp

8.6 bioscara_hardware_drivers::Gripper Class Reference

Child class implementing control of the hardware gripper.

```
#include <mGripper.h>
```

Inheritance diagram for bioscara_hardware_drivers::Gripper:



Public Member Functions

- [Gripper](#) (float reduction, float offset, float min, float max, float backup_init_pos)
Construct a new [Gripper](#) object.
- [err_type_t enable](#) (void) override
Activates the the hardware.
- [err_type_t disable](#) (void) override
Deactivates the hardware.
- [err_type_t setPosition](#) (float width) override
Sets the gripper width in m.
- [err_type_t setServoPosition](#) (float angle)
Sets the servo position of the gripper actuator in degrees.

Public Member Functions inherited from [bioscara_hardware_drivers::BaseGripper](#)

- [BaseGripper](#) (float reduction, float offset, float min, float max, float backup_init_pos)
Construct a new [BaseGripper](#) object.
- [~BaseGripper](#) (void)
Destroy the [BaseGripper](#) object.
- virtual [err_type_t init](#) (void)
Initializes the gripper.
- virtual [err_type_t deinit](#) (void)
Deinitializes the gripper.
- virtual [err_type_t getPosition](#) (float &width)
Gets the gripper width.
- virtual void [setReduction](#) (float reduction)
Manually set reduction.
- virtual void [setOffset](#) (float offset)
Manually set offset.

Protected Attributes

- [RPI_PWM_pwm](#)
[RPI_PWM](#) object to generate PWM voltage for servo control.
- int [_freq](#) = 50
PWM frequency in Hz.

Protected Attributes inherited from [bioscara_hardware_drivers::BaseGripper](#)

- float [_reduction](#) = 1
gripper width to actuator reduction ratio
- float [_offset](#) = 0
gripper width position offset
- float [_min](#) = 0
gripper width lower limit
- float [_max](#) = 0
gripper width upper limit
- float [_backup_init_pos](#) = 0.0
Initial position assumed if none can be retrieved from the buffer file.
- float [_pos](#) = [_backup_init_pos](#)
stored position
- float [_pos_get](#) = [_pos](#)
reported position

Additional Inherited Members

Protected Member Functions inherited from bioscara_hardware_drivers::BaseGripper

- [err_type_t save_last_position](#) (float pos)
Stores the latest position to the buffer file.
- [err_type_t retrieve_last_position](#) (float &pos)
Retrieves the stored position from the buffer file.

8.6.1 Detailed Description

Child class implementing control of the hardware gripper.

Use this class if you wish to control the actual hardware by generating a PWM voltage on GPIO18 of the Raspberry Pi.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 Gripper()

```
bioscara_hardware_drivers::Gripper::Gripper (
    float reduction,
    float offset,
    float min,
    float max,
    float backup_init_pos )
```

Construct a new [Gripper](#) object.

see the [BaseGripper](#) constructor for a description of the parameters.

8.6.3 Member Function Documentation

8.6.3.1 disable()

```
err_type_t bioscara_hardware_drivers::Gripper::disable (
    void ) [override], [virtual]
```

Deactivates the hardware.

Invokes [BaseGripper::disable\(\)](#) and stops the servo by disabling the PWM generation.

Returns

[err_type_t::OK](#)

Reimplemented from [bioscara_hardware_drivers::BaseGripper](#).

8.6.3.2 enable()

```
err_type_t bioscara_hardware_drivers::Gripper::enable (
    void ) [override], [virtual]
```

Activates the hardware.

Invokes [BaseGripper::enable\(\)](#) and starts the PWM generation but does not set a position. Must be called before a position is set. The PWM pin is GPIO18 on the Raspberry Pi 4. PWM chip is 0, channel 0.

Returns

[err_type_t::OK](#) on success, [err_type_t::COMM_ERROR](#) if PWM can not be enabled

Reimplemented from [bioscara_hardware_drivers::BaseGripper](#).

8.6.3.3 setPosition()

```
err_type_t bioscara_hardware_drivers::Gripper::setPosition (
    float width ) [override], [virtual]
```

Sets the gripper width in m.

Invokes [BaseGripper::setPosition\(\)](#), transforms the desired gripper width to a servo angle using the [JOINT2ACTUATOR\(\)](#) macro and finally sets the servo position using [setServoPosition\(\)](#)

Parameters

<i>width</i>	width in m.
--------------	-------------

Returns

see [setServoPosition\(\)](#)

Reimplemented from [bioscara_hardware_drivers::BaseGripper](#).

8.6.3.4 setServoPosition()

```
err_type_t bioscara_hardware_drivers::Gripper::setServoPosition (
    float angle )
```

Sets the servo position of the gripper actuator in degrees.

Calculates a PWM dutycycle for the desired servo angle and invokes [RPI_PWM::setDutyCycle\(\)](#) to set it.

Parameters

<i>angle</i>	in degrees.
--------------	-------------

Returns

[err_type_t::OK](#) on success, [err_type_t::COMM_ERROR](#) if the dutycycle can not be set.

8.6.4 Member Data Documentation

8.6.4.1 _freq

```
int bioscara_hardware_drivers::Gripper::_freq = 50 [protected]
```

PWM frequency in Hz.

8.6.4.2 _pwm

```
RPI\_PWM bioscara_hardware_drivers::Gripper::_pwm [protected]
```

[RPI_PWM](#) object to generate PWM voltage for servo control.

The documentation for this class was generated from the following files:

- [lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mGripper.h](#)
- [lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mGripper.cpp](#)

8.7 bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t Struct Reference

configuration structure holding the passed paramters from the ros2_control urdf

Public Attributes

- float [reduction](#) = 1
- float [offset](#) = 0
- float [min](#)
- float [max](#)
- float [init_pos](#)

8.7.1 Detailed Description

configuration structure holding the passed paramters from the ros2_control urdf

Saving all parameters on initialization in a structure allows for quick access during runtime.

8.7.2 Member Data Documentation

8.7.2.1 init_pos

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t::init↵
_pos
```

8.7.2.2 max

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t::max
```

8.7.2.3 min

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t::min
```

8.7.2.4 offset

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t::offset
= 0
```

8.7.2.5 reduction

```
float bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t::reduction
= 1
```

The documentation for this struct was generated from the following file:

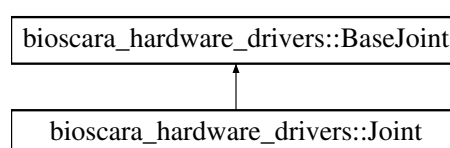
- `lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_interface/include/bioscara_gripper_↵`
hardware_interface/[gripper_hardware.hpp](#)

8.8 bioscara_hardware_drivers::Joint Class Reference

Representing a single hardware joint connected via I2C.

```
#include <mJoint.h>
```

Inheritance diagram for bioscara_hardware_drivers::Joint:



Public Member Functions

- **Joint** (const std::string [name](#), const int [address](#), const float [reduction](#), const float [min](#), const float [max](#))
Create a [Joint](#) object.
- **~Joint** (void)
Destroy the [Joint](#) object.
- **err_type_t init** (void) override
Initialize connection to a joint via I2C.
- **err_type_t deinit** (void) override
Disconnects from a joint.
- **err_type_t enable** (u_int8_t driveCurrent, u_int8_t holdCurrent) override
Engages the joint.
- **err_type_t postHoming** (void) override
perform tasks after a non-blocking homing
- **err_type_t getPosition** (float &pos) override
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- **err_type_t setPosition** (float pos) override
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- **err_type_t moveSteps** (int32_t steps) override
Move full steps.
- **err_type_t getVelocity** (float &vel) override
get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- **err_type_t setVelocity** (float vel) override
Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- **err_type_t checkOrientation** (float angle=2.0) override
Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.
- **err_type_t stop** (void) override
Stops the motor.
- **err_type_t disableCL** (void) override
- **err_type_t setDriveCurrent** (u_int8_t current) override
Set the Drive Current.
- **err_type_t setHoldCurrent** (u_int8_t current) override
Set the Hold Current.
- **err_type_t setBrakeMode** (u_int8_t mode) override
Set Brake Mode.
- **err_type_t setMaxAcceleration** (float maxAccel) override
Set the maximum permitted joint acceleration (and deceleration) in rad/s^2 or m/s^2 for cylindrical and prismatic joints respectively.
- **err_type_t setMaxVelocity** (float maxVel) override
Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.
- **err_type_t enableStallguard** (u_int8_t sensitivity) override
Enable encoder stall detection of the joint.
- **err_type_t getFlags** (void) override
Overload of [getFlags\(u_int8_t &flags\)](#)

Public Member Functions inherited from bioscara_hardware_drivers::BaseJoint

- [BaseJoint](#) (const std::string [name](#))
Create a [Joint](#) object.
- [~BaseJoint](#) (void)
Destroy the [BaseJoint](#) object.
- virtual [err_type_t](#) [disable](#) (void)
disengages the joint
- virtual [err_type_t](#) [home](#) (float velocity, u_int8_t sensitivity, u_int8_t current)
Blocking implementation to home the joint.
- virtual [err_type_t](#) [startHoming](#) (float velocity, u_int8_t sensitivity, u_int8_t current)
non-blocking implementation to home the joint
- virtual bool [isHomed](#) (void)
Checks the state if the motor is homed.
- virtual bool [isEnabled](#) (void)
Checks the state if the motor is enabled.
- virtual bool [isStalled](#) (void)
Checks if the motor is stalled.
- virtual bool [isBusy](#) (void)
Checks if the joint controller is busy processing a blocking command.
- virtual [err_type_t](#) [getFlags](#) (u_int8_t &[flags](#))
get the latest driver state [flags](#) from the joint
- virtual [stp_reg_t](#) [getCurrentBCmd](#) (void)
get the currently active blocking command

Protected Member Functions

- [err_type_t](#) [getHomingOffset](#) (float &[offset](#))
Retrieves the homing position from the last homing.
- [err_type_t](#) [setHomingOffset](#) (const float [offset](#))
Stores the homing position on the joint.
- [err_type_t](#) [_home](#) (float velocity, u_int8_t sensitivity, u_int8_t current) override
Call to start the homing sequence of a joint.
- [err_type_t](#) [checkCom](#) (void)
Check if communication to the joint is established.

Protected Member Functions inherited from bioscara_hardware_drivers::BaseJoint

- virtual void [wait_while_busy](#) (const float period_ms)
Blocking loop waiting for BUSY flag to reset.

Protected Attributes

- float [reduction](#) = 1
[Joint](#) to actuator reduction ratio.
- float [offset](#) = 0
[Joint](#) position offset.
- float [min](#) = 0
[Joint](#) lower limit.
- float [max](#) = 0
[Joint](#) upper limit.

Protected Attributes inherited from bioscara_hardware_drivers::BaseJoint

- `u_int8_t flags` = 0b00001100
State `flags` transmitted with every I2C transaction.
- `stp_reg_t current_b_cmd` = NONE
Keeps track if a blocking command is being executed.

Private Member Functions

- `template<typename T >`
`int read` (const `stp_reg_t` reg, T &data, `u_int8_t` &flags)
Wrapper function to request data from the I2C slave.
- `template<typename T >`
`int write` (const `stp_reg_t` reg, T data, `u_int8_t` &flags)
Wrapper function to send command to the I2C slave.

Private Attributes

- `int address`
I2C adress.
- `int handle` = -1
I2C bus handle.

Additional Inherited Members

Public Types inherited from bioscara_hardware_drivers::BaseJoint

- `enum stp_reg_t` {
`NONE` = 0x00 , `PING` = 0x0f , `SETUP` = 0x10 , `SETRPM` = 0x11 ,
`GETDRIVERRPM` = 0x12 , `MOVESTEPS` = 0x13 , `MOVEANGLE` = 0x14 , `MOVETOANGLE` = 0x15 ,
`GETMOTORSTATE` = 0x16 , `RUNCOTINOUS` = 0x17 , `ANGLEMOVED` = 0x18 , `SETCURRENT` = 0x19 ,
`SETHOLDCURRENT` = 0x1A , `SETMAXACCELERATION` = 0x1B , `SETMAXDECELERATION` = 0x1C ,
`SETMAXVELOCITY` = 0x1D ,
`ENABLESTALLGUARD` = 0x1E , `DISABLESTALLGUARD` = 0x1F , `CLEARSTALL` = 0x20 , `SETBRAKEMODE`
= 0x22 ,
`ENABLEPID` = 0x23 , `DISABLEPID` = 0x24 , `ENABLECLOSEDLOOP` = 0x25 , `DISABLECLOSEDLOOP` =
0x26 ,
`SETCONTROLTHRESHOLD` = 0x27 , `MOVETOEND` = 0x28 , `STOP` = 0x29 , `GETPIDERROR` = 0x2A ,
`CHECKORIENTATION` = 0x2B , `GETENCODERRPM` = 0x2C , `HOME` = 0x2D , `HOMEOFFSET` = 0x2E }
register and command definitions

Public Attributes inherited from bioscara_hardware_drivers::BaseJoint

- `std::string name`
Joint name for logging.

8.8.1 Detailed Description

Representing a single hardware joint connected via I2C.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 Joint()

```
bioscara_hardware_drivers::Joint::Joint (
    const std::string name,
    const int address,
    const float reduction,
    const float min,
    const float max )
```

Create a [Joint](#) object.

The [Joint](#) object represents a single joint and its actuator. Each [Joint](#) has a transmission with the following relationship:

$$\begin{aligned} \text{actuator position} &= (\text{joint position} - \text{offset}) * \text{reduction} \\ \text{joint position} &= \text{actuator position} / \text{reduction} + \text{offset} \end{aligned}$$

Parameters

<i>name</i>	string device name for identification
<i>address</i>	1-byte I2C device address (0x11 ... 0x14) for J1 ... J4
<i>reduction</i>	gear reduction of the joint. This is used to transform position and velocity values between in joint units and actuator (stepper) units. The sign depends on the direction the motor is mounted and is turning. Adjust such that the joint moves in the positive direction on on positive joint commands. Cable polarity has no effect since the motors automatically adjust to always run in the 'right' direction from their point of view. J1: 35 J2: -2*pi/0.004 (4 mm linear movement per stepper revolution) J3: 24 J4: 12
<i>min</i>	lower joint limit in joint units. J1: -3.04647 J2: -0.0016 J3: -2.62672 J4: -3.01069
<i>max</i>	upper joint limit in joint units. J1: 3.04647 J2: 0.3380 J3: 2.62672 J4: 3.01069

8.8.2.2 ~Joint()

```
bioscara_hardware_drivers::Joint::~~Joint (
    void )
```

Destroy the [Joint](#) object.

Create a [Joint](#) object.

8.8.3 Member Function Documentation

8.8.3.1 _home()

```
err_type_t bioscara_hardware_drivers::Joint::_home (
    float velocity,
    u_int8_t sensitivity,
    u_int8_t current ) [override], [protected], [virtual]
```

Call to start the homing sequence of a joint.

The joint will start rotating with the specified speed until a resistance which drives the PID error above the specified threshold is encountered. At this point the stepper stops and zeros the encoder.

Parameters

<i>velocity</i>	signed velocity in rad/s or m/s. Must be between $1.0 < \text{RAD2DEG}(\text{JOINT2ACTUATOR}(\text{velocity}, \text{reduction}, 0)) / 6 < 250.0$
<i>sensitivity</i>	Encoder pid error threshold 0 to 255.
<i>current</i>	homing current, determines how easy it is to stop the motor and thereby provoke a stall

Returns

err_type_t

Implements [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.2 checkCom()

```
err_type_t bioscara_hardware_drivers::Joint::checkCom (
    void ) [protected]
```

Check if communication to the joint is established.

Sends a PING to and expects a ACK from the joint.

Returns

err_type_t

8.8.3.3 checkOrientation()

```
err_type_t bioscara_hardware_drivers::Joint::checkOrientation (
    float angle = 2.0 ) [override], [virtual]
```

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

<i>angle</i>	degrees how much the motor should turn. A few degrees is sufficient.
--------------	--

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.4 deinit()

```
err_type_t bioscara_hardware_drivers::Joint::deinit (
    void ) [override], [virtual]
```

Disconnects from a joint.

Removes the joint from the I2C bus by invoking [closeI2CDevHandle\(\)](#).

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.5 disableCL()

```
err_type_t bioscara_hardware_drivers::Joint::disableCL (
    void ) [override], [virtual]
```

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.6 enable()

```
err_type_t bioscara_hardware_drivers::Joint::enable (
    u_int8_t driveCurrent,
    u_int8_t holdCurrent ) [override], [virtual]
```

Engages the joint.

This function prepares the motor for movement. After successfull execution the joint is ready to accept [setPosition\(\)](#) and [setVelocity\(\)](#) commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

<i>driveCurrent</i>	drive current in 0-100 % of 2.5A output (check uStepper doc.)
<i>holdCurrent</i>	hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.7 enableStallguard()

```
err_type_t bioscara_hardware_drivers::Joint::enableStallguard (
    u_int8_t sensitivity ) [override], [virtual]
```

Enable encoder stall detection of the joint.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. A detected stall can be reset by homing or reenabling the joint using [enable\(\)](#).

Note

If stall detection shall be enabled, invoke this method AFTER enabling the joint with [enable\(\)](#).

Parameters

<i>sensitivity</i>	value of threshold. 0 - 255 where lower is more sensitive.
--------------------	--

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.8 getFlags()

```
err_type_t bioscara_hardware_drivers::Joint::getFlags (
    void ) [override], [virtual]
```

Overload of [getFlags\(u_int8_t &flags\)](#)

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.9 getHomingOffset()

```
err_type_t bioscara_hardware_drivers::Joint::getHomingOffset (
    float & offset ) [protected]
```

Retrieves the homing position from the last homing.

The homing position is stored on the joint to make it persistent as long as the joint is powered up.

Returns

`err_type_t`

8.8.3.10 getPosition()

```
err_type_t bioscara_hardware_drivers::Joint::getPosition (
    float & pos ) [override], [virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Warning

If the joint is not homed this method does not return an error. Instead `pos` will be 0.0.

Parameters

<code>pos</code>	the current joint position in rad or m.
------------------	---

Returns

`err_type_t`

Implements [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.11 getVelocity()

```
err_type_t bioscara_hardware_drivers::Joint::getVelocity (
    float & vel ) [override], [virtual]
```

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<code>vel</code>	the current joint velocity in rad/s or m/s.
------------------	---

Returns

`err_type_t`

Implements [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.12 init()

```
err_type_t bioscara_hardware_drivers::Joint::init (
    void ) [override], [virtual]
```

Initialize connection to a joint via I2C.

Adds the joint to the I2C bus and tests if is responsive by invoking [checkCom\(\)](#). If the joint is homed, retrieve the homing position stored on the joint by invoking [getHomingOffset\(\)](#).

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.13 moveSteps()

```
err_type_t bioscara_hardware_drivers::Joint::moveSteps (
    int32_t steps ) [override], [virtual]
```

Move full steps.

This function can be called even when not homed.

Parameters

<i>steps</i>	number of full steps
--------------	----------------------

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.14 postHoming()

```
err_type_t bioscara_hardware_drivers::Joint::postHoming (
    void ) [override], [virtual]
```

perform tasks after a non-blocking homing

Invokes [BaseJoint::postHoming\(\)](#) and additionally saves the homing offset to the joint controller [setHomingOffset\(\)](#)

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.15 read()

```
template<typename T >
int bioscara_hardware_drivers::Joint::read (
    const stp_reg_t reg,
    T & data,
    u_int8_t & flags ) [private]
```

Wrapper function to request data from the I2C slave.

Allocates a buffer of size sizeof(T) + RFLAGS_SIZE. Invokes [readFromI2CDev\(\)](#), and copies the received payload to *data* and the transmission flags to *flags*. See [BaseJoint::flags](#) for details.

Template Parameters

<i>T</i>	Datatype of value to be transmitted
----------	-------------------------------------

Parameters

<i>reg</i>	stp_reg_t register to read
<i>data</i>	reference to store payload.
<i>flags</i>	reference to a byte which stores the return flags

Returns

0 on OK, negative on error

8.8.3.16 setBrakeMode()

```
err_type_t bioscara_hardware_drivers::Joint::setBrakeMode (
    u_int8_t mode ) [override], [virtual]
```

Set Brake Mode.

Parameters

<i>mode</i>	Freewheel: 0, Coolbrake: 1, Hardbrake: 2
-------------	--

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.17 setDriveCurrent()

```
err_type_t bioscara_hardware_drivers::Joint::setDriveCurrent (
    u_int8_t current ) [override], [virtual]
```

Set the Drive Current.

Parameters

<i>current</i>	0% - 100% of driver current
----------------	-----------------------------

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.18 setHoldCurrent()

```
err_type_t bioscara_hardware_drivers::Joint::setHoldCurrent (
    u_int8_t current ) [override], [virtual]
```

Set the Hold Current.

Parameters

<i>current</i>	0% - 100% of driver current
----------------	-----------------------------

Returns

`err_type_t` -1 on communication error,

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.19 setHomingOffset()

```
err_type_t bioscara_hardware_drivers::Joint::setHomingOffset (
    const float offset ) [protected]
```

Stores the homing position on the joint.

The homing position is stored on the joint to make it persistent as long as the joint is powered up.

Returns

`err_type_t`

8.8.3.20 setMaxAcceleration()

```
err_type_t bioscara_hardware_drivers::Joint::setMaxAcceleration (
    float maxAccel ) [override], [virtual]
```

Set the maximum permitted joint acceleration (and deceleration) in rad/s² or m/s² for cylindrical and prismatic joints respectively.

Parameters

<i>maxAccel</i>	maximum joint acceleration.
-----------------	-----------------------------

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.21 setMaxVelocity()

```
err_type_t bioscara_hardware_drivers::Joint::setMaxVelocity (
    float maxVel ) [override], [virtual]
```

Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<i>maxVel</i>	maximum joint velocity.
---------------	-------------------------

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.22 setPosition()

```
err_type_t bioscara_hardware_drivers::Joint::setPosition (
    float pos ) [override], [virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters

<i>pos</i>	the commanded joint position in rad or m.
------------	---

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.23 setVelocity()

```
err_type_t bioscara_hardware_drivers::Joint::setVelocity (
    float vel ) [override], [virtual]
```

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<i>vel</i>	the commanded joint velocity in rad/s or m/s.
------------	---

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.24 stop()

```
err_type_t bioscara_hardware_drivers::Joint::stop (
    void ) [override], [virtual]
```

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.8.3.25 write()

```
template<typename T >
int bioscara_hardware_drivers::Joint::write (
    const stp_reg_t reg,
    T data,
    u_int8_t & flags ) [private]
```

Wrapper function to send command to the I2C slave.

Allocates a buffer of size `sizeof(T) + RFLAGS_SIZE`. Copies *data* to the buffer and invokes [writeToI2CDev\(\)](#). The flags received from the transaction are copied to *flags*. The flags are described in [Joint::read\(\)](#).

Template Parameters

<i>T</i>	Datatype of value to be transmitted
----------	-------------------------------------

Parameters

<i>reg</i>	stp_reg_t command to execute
<i>data</i>	payload to transmit. It is the users responsibility to populate the right amount of data for the relevant register
<i>flags</i>	reference to a byte which stores the return flags

Returns

0 on OK, negative on error

8.8.4 Member Data Documentation

8.8.4.1 address

```
int bioscara_hardware_drivers::Joint::address [private]
```

I2C adress.

8.8.4.2 handle

```
int bioscara_hardware_drivers::Joint::handle = -1 [private]
```

I2C bus handle.

8.8.4.3 max

```
float bioscara_hardware_drivers::Joint::max = 0 [protected]
```

[Joint](#) upper limit.

8.8.4.4 min

```
float bioscara_hardware_drivers::Joint::min = 0 [protected]
```

[Joint](#) lower limit.

8.8.4.5 offset

```
float bioscara_hardware_drivers::Joint::offset = 0 [protected]
```

[Joint](#) position offset.

8.8.4.6 reduction

```
float bioscara_hardware_drivers::Joint::reduction = 1 [protected]
```

[Joint](#) to actuator reduction ratio.

The documentation for this class was generated from the following files:

- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔
driver/[mJoint.h](#)
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_↔
driver/[mJoint.hpp](#)
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/[mJoint.cpp](#)

8.9 bioscara_hardware_interfaces::BioscaraArmHardwareInterface↔ ::joint_config_t Struct Reference

configuration structure holding the passed paramters from the ros2_control urdf

Public Attributes

- `int i2c_address`
[int, HEX] I2C device adress
- `float reduction = 1`
[float] Gear reduction of the joint
- `float min`
[float] Lower joint limit in joint units
- `float max`
[float] Upper joint limit in joint units
- `u_int8_t drive_current`
[int, DEC] Drive current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)
- `u_int8_t hold_current`
[int, DEC] Hold current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)
- `u_int8_t stall_threshold`
[int, DEC] Stall protection threshold in 0-255, where lower is more sensitive.
- `float max_velocity`
[float] Maximum permitted joint velocity.
- `float max_acceleration`
[float] Maximum permitted joint acceleration (and deceleration).
- `joint_homing_config_t homing`
Holding the `joint_homing_config_t` configuration values.

8.9.1 Detailed Description

configuration structure holding the passed paramters from the ros2_control urdf

Saving all parameters on initialization in a structure allows for quick access during runtime.

8.9.2 Member Data Documentation

8.9.2.1 drive_current

```
u_int8_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::drive↔
current
```

[int, DEC] Drive current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)

Set when joint is enabled, see [bioscara_hardware_drivers::BaseJoint::enable\(\)](#).

8.9.2.2 hold_current

```
u_int8_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::hold↔
current
```

[int, DEC] Hold current of stepper driver in 0-100 % of 2.5A output (check uStepper doc.)

Set when joint is enabled, see [bioscara_hardware_drivers::BaseJoint::enable\(\)](#).

8.9.2.3 homing

```
joint_homing_config_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_↵
config_t::homing
```

Holding the [joint_homing_config_t](#) configuration values.

8.9.2.4 i2c_address

```
int bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::i2c_address
```

[int, HEX] I2C device address

1-byte I2C device address (0x11 ... 0x14) for J1 ... J4

8.9.2.5 max

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::max
```

[float] Upper joint limit in joint units

Initial values are (subject to tuning)

J1: 3.04647

J2: 0.3380

J3: 2.62672

J4: 3.01069

8.9.2.6 max_acceleration

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::max_acceleration
```

[float] Maximum permitted joint acceleration (and deceleration).

In rad/s^2 or m/s^2 for cylindrical and prismatic joints respectively. Set when joint is enabled, see [bioscara_hardware_drivers::BaseJoint::setMaxAcceleration\(\)](#).

Note

This is the "last" limit and shall ALWAYS be greater than all limits set in the controller and application (MoveIt) configuration.

8.9.2.7 max_velocity

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::max_velocity
```

[float] Maximum permitted joint velocity.

In rad/s or m/s for cylindrical and prismatic joints respectively. Set when joint is enabled, see [bioscara_hardware_drivers::BaseJoint::se](#)

Note

This is the "last" limit and shall ALWAYS be greater than all limits set in the controller and application (MoveIt) configuration.

8.9.2.8 min

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::min
```

[float] Lower joint limit in joint units

Initial values are (subject to tuning)

J1: -3.04647

J2: -0.0016

J3: -2.62672

J4: -3.01069

8.9.2.9 reduction

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::reduction =
1
```

[float] Gear reduction of the joint

This is used to transform position and velocity values between in joint units and actuator (stepper) units. The sign depends on the direction the motor is mounted and is turning. Adjust such that the joint moves in the positive direction on on positive joint commands. Cable polarity has no effect since the motors automatically adjust to always run in the 'right' direction from their point of view.

J1: 35

J2: $-2\pi/0.004$ (4 mm linear movement per stepper revolution, positive rotation makes the joint go down (negative), hence the minus sign for correction)

J3: 24

J4: 12

8.9.2.10 stall_threshold

```
u_int8_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t::stall_↵
threshold
```

[int, DEC] Stall protection threshold in 0-255, where lower is more sensitive.

If the PID error exceeds the set threshold a stall is triggered and the motor disabled. Set when joint is enabled, see [bioscara_hardware_drivers::BaseJoint::enableStallguard\(\)](#).

The documentation for this struct was generated from the following file:

- [lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_hardware_↵
interface/arm_hardware.hpp](#)

8.10 bioscara_hardware_interfaces::BioscaraArmHardwareInterface↵ ::joint_homing_config_t Struct Reference

configuration structure holding the passed homing paramters from the ros2_control urdf

Public Attributes

- float [speed](#)
[float] Signed homing velocity.
- u_int8_t [threshold](#)
[int, DEC] Encoder error threshold 0 to 255.
- u_int8_t [current](#)
[int, DEC] Homing current between 0 and 100.
- float [acceleration](#) = 0.01
[float] Joint acceleration during homing.

8.10.1 Detailed Description

configuration structure holding the passed homing paramters from the ros2_control urdf

Saving all parameters on initialization in a structure allows for quick access during runtime. See [bioscara_hardware_drivers::BaseJoint](#) for the description of the homing procedure.

8.10.2 Member Data Documentation

8.10.2.1 acceleration

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t↔  
::acceleration = 0.01
```

[float] Joint acceleration during homing.

In rad/s² or m/s² for cylindrical and prismatic joints respectively. Accelerating too fast can trigger a stall and hence false positive homing. See [joint_config_t::max_acceleration](#).

8.10.2.2 current

```
u_int8_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t↔  
::current
```

[int, DEC] Homing current between 0 and 100.

Determines how easy it is to stop the motor and thereby provoke a stall. See [joint_config_t::drive_current](#).

8.10.2.3 speed

```
float bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t::speed
```

[float] Signed homing velocity.

In rad/s or m/s for cylindrical and prismatic joints respectively. Must satisfy: $1.0 < \text{RAD2DEG}(\text{JOINT2} \leftrightarrow \text{ACTUATOR}(<\text{speed}>, \text{reduction}, 0)) / 6 < 250.0$

8.10.2.4 threshold

```
u_int8_t bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t↵
::threshold
```

[int, DEC] Encoder error threshold 0 to 255.

Lower values make the homing more sensitive but also more susceptible to false positives.

The documentation for this struct was generated from the following file:

- `lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_hardware_↵
interface/arm_hardware.hpp`

8.11 bioscara_joint_firmware::Lowpass Class Reference

Simple discrete IIR lowpass filter.

```
#include <filters.h>
```

Public Member Functions

- [Lowpass](#) (float gain=1, float sampleTime=0.1, float timeconstant=1.0)
Constructs and initializes the [Lowpass](#) filter.
- float [updateState](#) (float u)
Update the filter state. Must be called with the constant period [Ts](#).
- void [resetState](#) (void)
Resets the filter state [x](#) to 0.0.

Protected Attributes

- float [K](#)
Filter gain.
- float [Ts](#)
Filter sampling time.
- float [tau](#)
Filter timeconstant.
- float [x](#)
Filter state.

8.11.1 Detailed Description

Simple discrete IIR lowpass filter.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 Lowpass()

```
bioscara_joint_firmware::Lowpass::Lowpass (
    float gain = 1,
    float sampleTime = 0.1,
    float timeconstant = 1.0 ) [inline]
```

Constructs and initializes the [Lowpass](#) filter.

Parameters

<i>gain</i>	
<i>sampleTime</i>	
<i>timeconstant</i>	

8.11.3 Member Function Documentation

8.11.3.1 resetState()

```
void bioscara_joint_firmware::Lowpass::resetState (
    void ) [inline]
```

Resets the filter state x to 0.0.

8.11.3.2 updateState()

```
float bioscara_joint_firmware::Lowpass::updateState (
    float u ) [inline]
```

Update the filter state. Must be called with the constant period T_s .

Returns

The updated filter state

8.11.4 Member Data Documentation

8.11.4.1 K

```
float bioscara_joint_firmware::Lowpass::K [protected]
```

Filter gain.

8.11.4.2 τ

```
float bioscara_joint_firmware::Lowpass::tau [protected]
```

Filter timeconstant.

8.11.4.3 T_s

```
float bioscara_joint_firmware::Lowpass::Ts [protected]
```

Filter sampling time.

8.11.4.4 x

```
float bioscara_joint_firmware::Lowpass::x [protected]
```

Filter state.

The documentation for this class was generated from the following file:

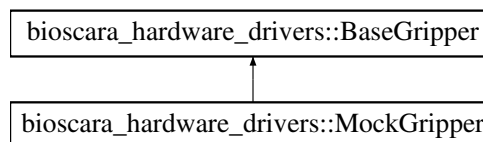
- [lib/joint_firmware/joint/filters.h](#)

8.12 bioscara_hardware_drivers::MockGripper Class Reference

Child class for to mock the gripper hardware.

```
#include <mMockGripper.h>
```

Inheritance diagram for bioscara_hardware_drivers::MockGripper:



Public Member Functions

- [MockGripper](#) (float reduction, float offset, float min, float max, float backup_init_pos)
Construct a new [MockGripper](#) object.

Public Member Functions inherited from [bioscara_hardware_drivers::BaseGripper](#)

- [BaseGripper](#) (float reduction, float offset, float min, float max, float backup_init_pos)
Construct a new [BaseGripper](#) object.
- [~BaseGripper](#) (void)
Destroy the [BaseGripper](#) object.
- virtual [err_type_t init](#) (void)
Initializes the gripper.
- virtual [err_type_t deinit](#) (void)
Deinitializes the gripper.
- virtual [err_type_t enable](#) (void)
Enables the gripper.
- virtual [err_type_t disable](#) (void)
Disables the gripper.
- virtual [err_type_t setPosition](#) (float width)
Sets the gripper width in m.
- virtual [err_type_t getPosition](#) (float &width)
Gets the gripper width.
- virtual void [setReduction](#) (float reduction)
Manually set reduction.
- virtual void [setOffset](#) (float offset)
Manually set offset.

Additional Inherited Members

Protected Member Functions inherited from [bioscara_hardware_drivers::BaseGripper](#)

- [err_type_t save_last_position](#) (float pos)
Stores the latest position to the buffer file.
- [err_type_t retrieve_last_position](#) (float &pos)
Retrieves the stored position from the buffer file.

Protected Attributes inherited from [bioscara_hardware_drivers::BaseGripper](#)

- float [_reduction](#) = 1
gripper width to actuator reduction ratio
- float [_offset](#) = 0
gripper width position offset
- float [_min](#) = 0
gripper width lower limit
- float [_max](#) = 0
gripper width upper limit
- float [_backup_init_pos](#) = 0.0
Initial position assumed if none can be retrieved from the buffer file.
- float [_pos](#) = [_backup_init_pos](#)
stored position
- float [_pos_get](#) = [_pos](#)
reported position

8.12.1 Detailed Description

Child class for to mock the gripper hardware.

Use this class if you dont wish to use the gripper hardware. Commands are simply mirrored.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 MockGripper()

```
bioscara_hardware_drivers::MockGripper::MockGripper (
    float reduction,
    float offset,
    float min,
    float max,
    float backup_init_pos )
```

Construct a new [MockGripper](#) object.

see the [BaseGripper](#) constructor for a description of the parameters.

The documentation for this class was generated from the following files:

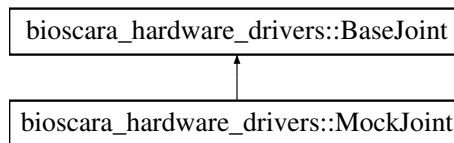
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↔ hardware_driver/[mMockGripper.h](#)
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/[mMockGripper.cpp](#)

8.13 bioscara_hardware_drivers::MockJoint Class Reference

Representing a single joint mocking the hardware joint.

```
#include <mMockJoint.h>
```

Inheritance diagram for bioscara_hardware_drivers::MockJoint:



Public Member Functions

- [MockJoint](#) (const std::string [name](#))
- [err_type_t enable](#) (u_int8_t driveCurrent, u_int8_t holdCurrent) override
Engages the joint.
- [err_type_t disable](#) (void) override
disengages the joint
- [err_type_t getPosition](#) (float &pos) override
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- [err_type_t setPosition](#) (float pos) override
get the current joint position in radians or m for cylindrical and prismatic joints respectively.
- [err_type_t getVelocity](#) (float &vel) override
get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- [err_type_t setVelocity](#) (float vel) override
Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.
- [err_type_t checkOrientation](#) (float angle=2.0) override
Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.
- [err_type_t stop](#) (void) override
Stops the motor.
- [err_type_t getFlags](#) (void) override
Overload of [getFlags\(u_int8_t &flags\)](#)
- bool [isHomed](#) (void) override
Checks the state if the motor is homed.

Public Member Functions inherited from [bioscara_hardware_drivers::BaseJoint](#)

- [BaseJoint](#) (const std::string [name](#))
Create a [Joint](#) object.
- [~BaseJoint](#) (void)
Destroy the [BaseJoint](#) object.
- virtual [err_type_t init](#) (void)
Initialize the joint communication.
- virtual [err_type_t deinit](#) (void)
Denitalize the joint communication.
- virtual [err_type_t home](#) (float velocity, u_int8_t sensitivity, u_int8_t current)
Blocking implementation to home the joint.

- virtual [err_type_t startHoming](#) (float velocity, u_int8_t sensitivity, u_int8_t current)
non-blocking implementation to home the joint
- virtual [err_type_t postHoming](#) (void)
perform tasks after a non-blocking homing
- virtual [err_type_t moveSteps](#) (int32_t steps)
Move full steps.
- virtual [err_type_t disableCL](#) (void)
- virtual [err_type_t setDriveCurrent](#) (u_int8_t current)
Set the Drive Current.
- virtual [err_type_t setHoldCurrent](#) (u_int8_t current)
Set the Hold Current.
- virtual [err_type_t setBrakeMode](#) (u_int8_t mode)
Set Brake Mode.
- virtual [err_type_t setMaxAcceleration](#) (float maxAccel)
Set the maximum permitted joint acceleration (and deceleration) in rad/s^2 or m/s^2 for cylindrical and prismatic joints respectively.
- virtual [err_type_t setMaxVelocity](#) (float maxVel)
Set the maximum permitted joint velocity in rad/s or m/s for cylindrical and prismatic joints respectively.
- virtual [err_type_t enableStallguard](#) (u_int8_t sensitivity)
Enable encoder stall detection of the joint.
- virtual bool [isEnabled](#) (void)
Checks the state if the motor is enabled.
- virtual bool [isStalled](#) (void)
Checks if the motor is stalled.
- virtual bool [isBusy](#) (void)
Checks if the joint controller is busy processing a blocking command.
- virtual [err_type_t getFlags](#) (u_int8_t &flags)
*get the latest driver state *flags* from the joint*
- virtual [stp_reg_t getCurrentBCmd](#) (void)
get the currently active blocking command

Protected Member Functions

- [err_type_t _home](#) (float velocity, u_int8_t sensitivity, u_int8_t current)
Call to start the homing sequence of a joint.

Protected Member Functions inherited from [bioscara_hardware_drivers::BaseJoint](#)

- virtual void [wait_while_busy](#) (const float period_ms)
Blocking loop waiting for BUSY flag to reset.

Private Member Functions

- float [getDeltaT](#) (std::chrono::_V2::system_clock::time_point &last_call, bool update=true)
Compute the time since the given time point.

Private Attributes

- float `q` = 0.0
position
- float `qd` = 0.0
velocity
- std::chrono::_V2::system_clock::time_point `last_set_position` = std::chrono::high_resolution_clock::now()
- std::chrono::_V2::system_clock::time_point `last_set_velocity` = `last_set_position`
- std::chrono::_V2::system_clock::time_point `async_start_time` = `last_set_position`
- `stp_reg_t op_mode` = NONE

Additional Inherited Members

Public Types inherited from `bioscara_hardware_drivers::BaseJoint`

- enum `stp_reg_t` {
`NONE` = 0x00 , `PING` = 0x0f , `SETUP` = 0x10 , `SETRPM` = 0x11 ,
`GETDRIVERRPM` = 0x12 , `MOVESTEPS` = 0x13 , `MOVEANGLE` = 0x14 , `MOVETOANGLE` = 0x15 ,
`GETMOTORSTATE` = 0x16 , `RUNCOTINOUS` = 0x17 , `ANGLEMOVED` = 0x18 , `SETCURRENT` = 0x19 ,
`SETHOLDCURRENT` = 0x1A , `SETMAXACCELERATION` = 0x1B , `SETMAXDECELERATION` = 0x1C ,
`SETMAXVELOCITY` = 0x1D ,
`ENABLESTALLGUARD` = 0x1E , `DISABLESTALLGUARD` = 0x1F , `CLEARSTALL` = 0x20 , `SETBRAKEMODE`
= 0x22 ,
`ENABLEPID` = 0x23 , `DISABLEPID` = 0x24 , `ENABLECLOSEDLOOP` = 0x25 , `DISABLECLOSEDLOOP` =
0x26 ,
`SETCONTROLTHRESHOLD` = 0x27 , `MOVETOEND` = 0x28 , `STOP` = 0x29 , `GETPIDERROR` = 0x2A ,
`CHECKORIENTATION` = 0x2B , `GETENCODERRPM` = 0x2C , `HOME` = 0x2D , `HOMEOFFSET` = 0x2E }
register and command definitions

Public Attributes inherited from `bioscara_hardware_drivers::BaseJoint`

- std::string `name`
Joint name for logging.

Protected Attributes inherited from `bioscara_hardware_drivers::BaseJoint`

- u_int8_t `flags` = 0b00001100
State `flags` transmitted with every I2C transaction.
- `stp_reg_t current_b_cmd` = NONE
Keeps track if a blocking command is being executed.

8.13.1 Detailed Description

Representing a single joint mocking the hardware joint.

This mock hardware implementation simply mirrors (potentially delayed) commands.

Note

Much of this implementation is very basic not well written.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 MockJoint()

```
bioscara_hardware_drivers::MockJoint::MockJoint (
    const std::string name )
```

8.13.3 Member Function Documentation

8.13.3.1 _home()

```
err_type_t bioscara_hardware_drivers::MockJoint::_home (
    float velocity,
    u_int8_t sensitivity,
    u_int8_t current ) [protected], [virtual]
```

Call to start the homing sequence of a joint.

The joint will start rotating with the specified speed until a resistance which drives the PID error above the specified threshold is encountered. At this point the stepper stops and zeros the encoder.

Parameters

<i>velocity</i>	signed velocity in rad/s or m/s. Must be between $1.0 < \text{RAD2DEG}(\text{JOINT2ACTUATOR}(\text{velocity}, \text{reduction}, 0)) / 6 < 250.0$
<i>sensitivity</i>	Encoder pid error threshold 0 to 255.
<i>current</i>	homing current, determines how easy it is to stop the motor and thereby provoke a stall

Returns

err_type_t

Implements [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.2 checkOrientation()

```
err_type_t bioscara_hardware_drivers::MockJoint::checkOrientation (
    float angle = 2.0 ) [override], [virtual]
```

Calls the checkOrientation method of the motor. Checks in which direction the motor is turning.

Note

This is a blocking function.

Parameters

<i>angle</i>	degrees how much the motor should turn. A few degrees is sufficient.
--------------	--

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.3 disable()

```
err_type_t bioscara_hardware_drivers::MockJoint::disable (
    void ) [override], [virtual]
```

disenganges the joint

invokes [stop\(\)](#), sets hold and drive current to 0 and sets the the joint brake mode to freewheeling

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.4 enable()

```
err_type_t bioscara_hardware_drivers::MockJoint::enable (
    u_int8_t driveCurrent,
    u_int8_t holdCurrent ) [override], [virtual]
```

Engages the joint.

This function prepares the motor for movement. After successfull execution the joint is ready to accept [setPosition\(\)](#) and [setVelocity\(\)](#) commands.

The function sets the drive and hold current for the specified joint and engages the motor. The currents are in percent of driver max. output (2.5A, check with TMC5130 datasheet or Ustepper documentation)

Parameters

<i>driveCurrent</i>	drive current in 0-100 % of 2.5A output (check uStepper doc.)
<i>holdCurrent</i>	hold current in 0-100 % of 2.5A output (check uStepper doc.)

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.5 getDeltaT()

```
float bioscara_hardware_drivers::MockJoint::getDeltaT (
    std::chrono::_V2::system_clock::time_point & last_call,
    bool update = true ) [private]
```

Compute the time since the given time point.

Parameters

<i>last_call</i>	the time point to calculate the duration
<i>update</i>	If true compute the duration and also reset <i>last_call</i> to current time

Returns

elapsed time in seconds since *last_call*

8.13.3.6 `getFlags()`

```
err_type_t bioscara_hardware_drivers::MockJoint::getFlags (
    void ) [override], [virtual]
```

Overload of `getFlags(u_int8_t &flags)`

Returns

`err_type_t`

Reimplemented from `bioscara_hardware_drivers::BaseJoint`.

8.13.3.7 `getPosition()`

```
err_type_t bioscara_hardware_drivers::MockJoint::getPosition (
    float & pos ) [override], [virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Warning

If the joint is not homed this method does not return an error. Instead `pos` will be 0.0.

Parameters

<i>pos</i>	the current joint position in rad or m.
------------	---

Returns

`err_type_t`

Implements `bioscara_hardware_drivers::BaseJoint`.

8.13.3.8 `getVelocity()`

```
err_type_t bioscara_hardware_drivers::MockJoint::getVelocity (
    float & vel ) [override], [virtual]
```

get the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<i>vel</i>	the current joint velocity in rad/s or m/s.
------------	---

Returns

err_type_t

Implements [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.9 isHomed()

```
bool bioscara_hardware_drivers::MockJoint::isHomed (
    void ) [override], [virtual]
```

Checks the state if the motor is homed.

Reads the internal state [flags](#) from the last transmission. If an update is necessary call [getFlags\(\)](#) before invoking this function.

Returns

true if the motor is homed, false if not.

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.10 setPosition()

```
err_type_t bioscara_hardware_drivers::MockJoint::setPosition (
    float pos ) [override], [virtual]
```

get the current joint position in radians or m for cylindrical and prismatic joints respectively.

Parameters

<i>pos</i>	the commanded joint position in rad or m.
------------	---

Returns

err_type_t

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.11 setVelocity()

```
err_type_t bioscara_hardware_drivers::MockJoint::setVelocity (
    float vel ) [override], [virtual]
```

Set the current joint velocity in radians/s or m/s for cylindrical and prismatic joints respectively.

Parameters

<code>vel</code>	the commanded joint velocity in rad/s or m/s.
------------------	---

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.3.12 stop()

```
err_type_t bioscara_hardware_drivers::MockJoint::stop (
    void ) [override], [virtual]
```

Stops the motor.

Stops the motor by setting the maximum velocity to zero and the position setpoint to the current position

Returns

`err_type_t`

Reimplemented from [bioscara_hardware_drivers::BaseJoint](#).

8.13.4 Member Data Documentation**8.13.4.1 async_start_time**

```
std::chrono::_V2::system_clock::time_point bioscara_hardware_drivers::MockJoint::async_start←
_time = last_set_position [private]
```

8.13.4.2 last_set_position

```
std::chrono::_V2::system_clock::time_point bioscara_hardware_drivers::MockJoint::last_set←
position = std::chrono::high_resolution_clock::now() [private]
```

8.13.4.3 last_set_velocity

```
std::chrono::_V2::system_clock::time_point bioscara_hardware_drivers::MockJoint::last_set←
velocity = last_set_position [private]
```

8.13.4.4 op_mode

```
stp_reg_t bioscara_hardware_drivers::MockJoint::op_mode = NONE [private]
```


8.13.4.5 q

```
float bioscara_hardware_drivers::MockJoint::q = 0.0 [private]
```

position

8.13.4.6 qd

```
float bioscara_hardware_drivers::MockJoint::qd = 0.0 [private]
```

velocity

The documentation for this class was generated from the following files:

- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mMockJoint.h
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mMockJoint.cpp

8.14 bioscara_joint_firmware::MovMax Class Reference

Simple FIR moving maximum filter.

```
#include <filters.h>
```

Public Member Functions

- [MovMax](#) (float windowSize)
Constructs and initializes the [MovMax](#) filter.
- float [updateState](#) (float u)
Update and return the filter state.

Protected Attributes

- unsigned int [M](#) = 200
Window Size.
- float * [cb_data](#)
Pointer to ring buffer holding the data.
- unsigned int [cb_index](#)
Current ringbuffer index.

8.14.1 Detailed Description

Simple FIR moving maximum filter.

Implements a circular ringbuffer holding the sampled data over the window size [M](#)

8.14.2 Constructor & Destructor Documentation

8.14.2.1 MovMax()

```
bioscara_joint_firmware::MovMax::MovMax (
    float windowSize ) [inline]
```

Constructs and initializes the [MovMax](#) filter.

Allocates a ring buffer [cb_data](#) of size [M](#).

Parameters

<i>windowSize</i>	the window size
-------------------	-----------------

8.14.3 Member Function Documentation

8.14.3.1 `updateState()`

```
float bioscara_joint_firmware::MovMax::updateState (
    float u ) [inline]
```

Update and return the filter state.

Returns

The updated filter state

8.14.4 Member Data Documentation

8.14.4.1 `cb_data`

```
float* bioscara_joint_firmware::MovMax::cb_data [protected]
```

Pointer to ring buffer holding the data.

8.14.4.2 `cb_index`

```
unsigned int bioscara_joint_firmware::MovMax::cb_index [protected]
```

Current ringbuffer index.

8.14.4.3 `M`

```
unsigned int bioscara_joint_firmware::MovMax::M = 200 [protected]
```

Window Size.

The documentation for this class was generated from the following file:

- `lib/joint_firmware/joint/filters.h`

8.15 RPI_PWM Class Reference

Class to create a Pulse Width Modulated (PWM) signal on the Raspberry PI 4 and 5.

```
#include <uPWM.h>
```

Public Member Functions

- int [start](#) (int channel, int frequency, float duty_cycle=0, int chip=2)
Starts the PWM.
- void [stop](#) ()
Stops the PWM.
- [~RPI_PWM](#) ()
Destroy the [RPI_PWM](#) object.
- int [setDutyCycle](#) (float v) const
Sets the duty cycle in percent 0 - 100.

Private Member Functions

- void [setPeriod](#) (int ns) const
- int [setDutyCycleNS](#) (int ns) const
- void [enable](#) () const
- void [disable](#) () const
- int [writeSYS](#) (std::string filename, int value) const

Private Attributes

- int [per](#) = 0
- std::string [chippath](#)
- std::string [pwmpath](#)

8.15.1 Detailed Description

Class to create a Pulse Width Modulated (PWM) signal on the Raspberry PI 4 and 5.

Based on https://github.com/berndporr/rpi_pwm/blob/main/rpi_pwm.h and slightly modified.

For servo control it is important to use a hardware generated PWM, since software PWM, like it is created by the lgpio library, is subject to timing jitter which can cause the servo to figet which might lead to overheating and wear.

8.15.2 Constructor & Destructor Documentation

8.15.2.1 [~RPI_PWM\(\)](#)

```
RPI_PWM::~~RPI_PWM ( ) [inline]
```

Destroy the [RPI_PWM](#) object.

Invokes [disable\(\)](#)

8.15.3 Member Function Documentation

8.15.3.1 [disable\(\)](#)

```
void RPI_PWM::disable ( ) const [inline], [private]
```

8.15.3.2 enable()

```
void RPI_PWM::enable ( ) const [inline], [private]
```

8.15.3.3 setDutyCycle()

```
int RPI_PWM::setDutyCycle (
    float v ) const [inline]
```

Sets the duty cycle in percent 0 - 100.

Parameters

<i>v</i>	The duty cycle in percent.
----------	----------------------------

Returns

>0 on success and -1 after an error.

8.15.3.4 setDutyCycleNS()

```
int RPI_PWM::setDutyCycleNS (
    int ns ) const [inline], [private]
```

8.15.3.5 setPeriod()

```
void RPI_PWM::setPeriod (
    int ns ) const [inline], [private]
```

8.15.3.6 start()

```
int RPI_PWM::start (
    int channel,
    int frequency,
    float duty_cycle = 0,
    int chip = 2 ) [inline]
```

Starts the PWM.

Parameters

<i>channel</i>	The GPIO channel
<i>frequency</i>	The PWM frequency
<i>duty_cycle</i>	The initial duty cycle of the PWM (default 0)
<i>chip</i>	The chip number

Returns

>0 on success and -1 if an error has happened.

8.15.3.7 stop()

```
void RPI_PWM::stop ( ) [inline]
```

Stops the PWM.

8.15.3.8 writeSYS()

```
int RPI_PWM::writeSYS (
    std::string filename,
    int value ) const [inline], [private]
```

8.15.4 Member Data Documentation**8.15.4.1 chippath**

```
std::string RPI_PWM::chippath [private]
```

8.15.4.2 per

```
int RPI_PWM::per = 0 [private]
```

8.15.4.3 pwmpath

```
std::string RPI_PWM::pwmpath [private]
```

The documentation for this class was generated from the following file:

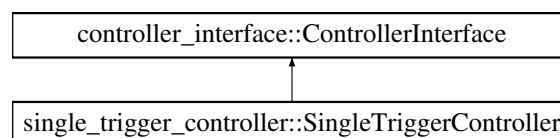
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_↵ hardware_driver/[uPWM.h](#)

8.16 single_trigger_controller::SingleTriggerController Class Reference

A general purpose controller which can be used to trigger command interfaces.

```
#include <single_trigger_controller.hpp>
```

Inheritance diagram for single_trigger_controller::SingleTriggerController:



Public Member Functions

- [SingleTriggerController](#) ()
- [controller_interface::InterfaceConfiguration](#) [command_interface_configuration](#) () const override
Sets the command interface configuration to INDIVIDUAL.
- [controller_interface::InterfaceConfiguration](#) [state_interface_configuration](#) () const override
Sets the state interface configuration to INDIVIDUAL.
- [CallbackReturn](#) [on_init](#) () override
*Called on initialization to the *unconfigured* state.*
- [CallbackReturn](#) [on_configure](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *unconfigured* to the *inactive* state.*
- [CallbackReturn](#) [on_activate](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *inactive* to the *active* state.*
- [CallbackReturn](#) [on_deactivate](#) (const rclcpp_lifecycle::State &previous_state) override
*Called on the transition from the *active* to the *inactive* state.*
- [controller_interface::return_type](#) [update](#) (const rclcpp::Time &time, const rclcpp::Duration &period) override
*Realtime update method called when in *active* state.*

Protected Attributes

- [InterfacesNames](#) [command_interface_types_](#)
vector of string holding all command interface names as configured from the parameters.
- [InterfacesNames](#) [state_interface_types_](#)
vector of string holding all state interface names as configured from the parameters.
- [MapOfReferencesToCommandInterfaces](#) [command_interfaces_map_](#)
map containing a reference to the actual hardware's command interfaces
- [MapOfReferencesToStateInterfaces](#) [state_interfaces_map_](#)
map containing a reference to the actual hardware's state interfaces
- rclcpp::Subscription< [CmdType](#) >::SharedPtr [command_subscriber_](#) {}
Subscriber object for the '~/'commands' topic.
- std::shared_ptr< rclcpp::Publisher< [StateType](#) > > [state_publisher_](#) {}
Publisher object to the '~/'states' topic.
- std::shared_ptr< realtime_tools::RealtimePublisher< [StateType](#) > > [realtime_state_publisher_](#) {}
Realtime publisher wrapper of the [state_publisher_](#).
- [StateType](#) [state_msg_](#)
Message object holding the states to be published.
- std::shared_ptr< single_trigger_controller_parameters::ParamListener > [param_listener_](#) {}
parameter listener object.
- single_trigger_controller_parameters::Params [params_](#)
Parameter structure holding controller configuration parameters.

Private Member Functions

- void [store_command_interface_types](#) ()
Stores all configured command interfaces in [command_interface_types_](#).
- void [store_state_interface_types](#) ()
Stores all configured state interfaces in [state_interface_types_](#).
- void [initialize_state_msg](#) ()
Initializes the state message [state_msg_](#) with NaN values for each configured state interface.
- void [update_states](#) ()

- Publishes current state values.*
- controller_interface::return_type [update_commands](#) (const [CmdType](#) &commands)
Callback method for received command msg. Calls [apply_command\(\)](#) for each command interface.
- template<typename T >
std::unordered_map< std::string, std::reference_wrapper< T > > [create_map_of_references_to_interfaces](#)
(const [InterfacesNames](#) &interfaces_from_params, std::vector< T > &configured_interfaces)
Create a map of references to the hardware components interfaces.
- template<typename T >
bool [check_if_configured_interfaces_matches_received](#) (const [InterfacesNames](#) &interfaces_from_params,
const T &configured_interfaces)
performs checks on the configured interfaces vs received interfaces.
- void [apply_state_value](#) ([StateType](#) &state_msg, std::size_t index, std::size_t interface_index) const
retrieves the state values from the hardware and populates the state_msg with the result.
- void [apply_command](#) (const [CmdType](#) &commands, std::size_t index, std::size_t command_interface_index)
const
calls the [set_value\(\)](#) method for the hardware's command interface references from the [command_interfaces_map_](#)
- [InterfacesNames](#) [get_state_interfaces_names](#) (const std::string &name) const
*Gets the all qualified state interface names from the [state_interface_types_](#) for the joint matching *name*.*
- bool [update_dynamic_map_parameters](#) ()
Update the controller parameters.

8.16.1 Detailed Description

A general purpose controller which can be used to trigger command interfaces.

This controller is a derivative of the `ros2_control` [GpioCommandController](#). The key differences between this controller and the `GpioCommandController` is two-fold:

1. Besides GPIO interfaces, this controller allows Joint interfaces to be used, enabling the user to utilize the controller in broader applications and simplifying the URDF description.
2. The `GpioCommandController` writes the latest received command to the configured state interfaces on every update cycle. On the contrary, the [SingleTriggerController](#) only writes to the command interfaces once in the subscription callback when a new command has been received. The user can for example leverage this by clearing the command interface (writing NaN to it) in the hardware interface once the latest trigger has been handled.

The controller subscribes and publishes to the same topics as the `GpioCommandController`.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 SingleTriggerController()

```
single_trigger_controller::SingleTriggerController::SingleTriggerController ( )
```

8.16.3 Member Function Documentation

8.16.3.1 apply_command()

```
void single_trigger_controller::SingleTriggerController::apply_command (
    const CmdType & commands,
    std::size_t index,
    std::size_t command_interface_index ) const [private]
```

calls the `set_value()` method for the hardware's command interface references from the [command_interfaces_map_](#)

Parameters

<i>commands</i>	
<i>index</i>	
<i>command_interface_index</i>	

8.16.3.2 apply_state_value()

```
void single_trigger_controller::SingleTriggerController::apply_state_value (
    StateType & state_msg,
    std::size_t index,
    std::size_t interface_index ) const [private]
```

retrieves the state values from the hardware and populates the state_msg with the result.

calls the get_optional() method of each state interface from the [state_interfaces_map_](#) for the state referenced by the index and interface_index.

Parameters

<i>state_msg</i>	
<i>index</i>	
<i>interface_index</i>	

8.16.3.3 check_if_configured_interfaces_matches_received()

```
template<typename T >
bool single_trigger_controller::SingleTriggerController::check_if_configured_interfaces_↔
matches_received (
    const InterfacesNames & interfaces_from_params,
    const T & configured_interfaces ) [private]
```

performs checks on the configured interfaces vs received interfaces.

Template Parameters

<i>T</i>	
----------	--

Parameters

<i>interfaces_from_params</i>	
<i>configured_interfaces</i>	

Returns

std::unordered_map<std::string, std::reference_wrapper<T>>>

8.16.3.4 command_interface_configuration()

```
controller_interface::InterfaceConfiguration single_trigger_controller::SingleTriggerController↵
::command_interface_configuration ( ) const [override]
```

Sets the command interface configuration to INDIVIDUAL.

See ControllerInterfaceBase::command_interface_configuration()

Returns

controller_interface::InterfaceConfiguration

8.16.3.5 create_map_of_references_to_interfaces()

```
template<typename T >
std::unordered_map< std::string, std::reference_wrapper< T > > single_trigger_controller::↵
SingleTriggerController::create_map_of_references_to_interfaces (
    const InterfacesNames & interfaces_from_params,
    std::vector< T > & configured_interfaces ) [private]
```

Create a map of references to the hardware components interfaces.

Template Parameters

<i>T</i>	
----------	--

Parameters

<i>interfaces_from_params</i>	
<i>configured_interfaces</i>	

Returns

std::unordered_map<std::string, std::reference_wrapper<T>>

8.16.3.6 get_state_interfaces_names()

```
InterfacesNames single_trigger_controller::SingleTriggerController::get_state_interfaces_names
(
    const std::string & name ) const [private]
```

Gets the all qualified state interface names from the [state_interface_types_](#) for the joint matching name.

Parameters

<i>name</i>	
-------------	--

Returns

InterfacesNames

8.16.3.7 initialize_state_msg()

```
void single_trigger_controller::SingleTriggerController::initialize_state_msg ( ) [private]
```

Initializes the state message [state_msg_](#) with NaN values for each configured state interface.

For example the state message could look like this:

```
{
  header:
  {
    stamp: 12345678
  },
  interface_groups: ['j1', 'gpio1'],
  interface_values:
  [
    {
      interface_names: ['home'],
      values: [NaN]
    },
    {
      interface_names: ['ai0', 'ai1'],
      values: [NaN, NaN]
    }
  ]
}
```

8.16.3.8 on_activate()

```
CallbackReturn single_trigger_controller::SingleTriggerController::on_activate (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transistion from the `inactive` to the `active` state.

Populates `command_interfaces_map_` and `state_interfaces_map_` by invoking [create_map_of_references_to_interfaces\(\)](#) for both command and state interfaces.

Initializes an empty state message by calling [initialize_state_msg\(\)](#).

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

CallbackReturn

8.16.3.9 on_configure()

```
CallbackReturn single_trigger_controller::SingleTriggerController::on_configure (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transistion from the `unconfigured` to the `inactive` state.

Stores all configured command and state interfaces using the [store_command_interface_types\(\)](#) and [store_state_interface_types\(\)](#) methods.

Creates a subscription to the '~/commands' topic if command interfaces are configured. Creates a realtime publisher to the '~/states' topic if state interfaces are configured.

8.16.3.10 on_deactivate()

```
CallbackReturn single_trigger_controller::SingleTriggerController::on_deactivate (
    const rclcpp_lifecycle::State & previous_state ) [override]
```

Called on the transition from the active to the inactive state.

Parameters

<i>previous_state</i>	
-----------------------	--

Returns

CallbackReturn

8.16.3.11 on_init()

```
CallbackReturn single_trigger_controller::SingleTriggerController::on_init ( ) [override]
```

Called on initialization to the unconfigured state.

Retrieves and stores the parameters supplied to the controller in [params_](#)

Returns

hardware_interface::CallbackReturn

8.16.3.12 state_interface_configuration()

```
controller_interface::InterfaceConfiguration single_trigger_controller::SingleTriggerController↔
::state_interface_configuration ( ) const [override]
```

Sets the state interface configuration to INDIVIDUAL.

See ControllerInterfaceBase::state_interface_configuration()

Returns

controller_interface::InterfaceConfiguration

8.16.3.13 store_command_interface_types()

```
void single_trigger_controller::SingleTriggerController::store_command_interface_types ( )
[private]
```

Stores all configured command interfaces in [command_interface_types_](#).

8.16.3.14 store_state_interface_types()

```
void single_trigger_controller::SingleTriggerController::store_state_interface_types ( ) [private]
```

Stores all configured state interfaces in [state_interface_types_](#).

8.16.3.15 update()

```
controller_interface::return_type single_trigger_controller::SingleTriggerController::update (
    const rclcpp::Time & time,
    const rclcpp::Duration & period ) [override]
```

Realtime update method called when in active state.

Only publishes the states by invoking [update_states\(\)](#).

Parameters

<i>time</i>	
<i>period</i>	

Returns

controller_interface::return_type

8.16.3.16 update_commands()

```
controller_interface::return_type single_trigger_controller::SingleTriggerController::update←
_commands (
    const CmdType & commands ) [private]
```

Callback method for received command msg. Calls [apply_command\(\)](#) for each command interface.

Sets all command interfaces to 0.0 if an empty command message is received.

Parameters

<i>commands</i>	
-----------------	--

Returns

controller_interface::return_type

8.16.3.17 update_dynamic_map_parameters()

```
bool single_trigger_controller::SingleTriggerController::update_dynamic_map_parameters ( )
[private]
```

Update the controller parameters.

Only called in [on_configure\(\)](#) to update the parameters in case they have been changed.

Returns

true

8.16.3.18 update_states()

```
void single_trigger_controller::SingleTriggerController::update_states ( ) [private]
```

Publishes current state values.

Loops over all interface groups and command interfaces and calls the [apply_state_value\(\)](#). Then publishes the result using the [realtime_state_publisher_](#).

8.16.4 Member Data Documentation**8.16.4.1 command_interface_types_**

```
InterfacesNames single_trigger_controller::SingleTriggerController::command_interface_types_
[protected]
```

vector of string holding all command interface names as configured from the parameters.

Looks like this for example:

```
['j1/home', 'gpio1/ao0', 'gpio1/ao1']
```

8.16.4.2 command_interfaces_map_

```
MapOfReferencesToCommandInterfaces single_trigger_controller::SingleTriggerController::command↵
_interfaces_map_ [protected]
```

map containing a reference to the actual hardware's command interfaces

8.16.4.3 command_subscriber_

```
roscpp::Subscription<CmdType>::SharedPtr single_trigger_controller::SingleTriggerController↔
::command_subscriber_ {} [protected]
```

Subscriber object for the '~/commands' topic.

8.16.4.4 param_listener_

```
std::shared_ptr<single_trigger_controller_parameters::ParamListener> single_trigger_controller↔
::SingleTriggerController::param_listener_ {} [protected]
```

parameter listerner object.

8.16.4.5 params_

```
single_trigger_controller_parameters::Params single_trigger_controller::SingleTriggerController↔
::params_ [protected]
```

Parameter structure holding controller configuration parameters.

8.16.4.6 realtime_state_publisher_

```
std::shared_ptr<realtime_tools::RealtimePublisher<StateType> > single_trigger_controller::↔
SingleTriggerController::realtime_state_publisher_ {} [protected]
```

Realtime publisher wrapper of the [state_publisher_](#).

8.16.4.7 state_interface_types_

```
InterfacesNames single_trigger_controller::SingleTriggerController::state_interface_types_↔
[protected]
```

vector of string holding all state interface names as configured from the parameters.

Looks like this for example:

```
['j1/home', 'gpio1/ai0', 'gpio1/ai1']
```

8.16.4.8 state_interfaces_map_

```
MapOfReferencesToStateInterfaces single_trigger_controller::SingleTriggerController::state↔
interfaces_map_ [protected]
```

map containing a reference to the actual hardware's state interfaces

8.16.4.9 state_msg_

`StateType` single_trigger_controller::SingleTriggerController::state_msg_ [protected]

Message object holding the states to be published.

8.16.4.10 state_publisher_

```
std::shared_ptr<rclcpp::Publisher<StateType> > single_trigger_controller::SingleTriggerController::state_publisher_ {} [protected]
```

Publisher object to the '~/states' topic.

The documentation for this class was generated from the following files:

- lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/include/single_trigger_controller/single_trigger_controller.hpp
- lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/src/single_trigger_controller.cpp

Chapter 9

File Documentation

9.1 docs/doxygen/index.md File Reference

9.2 lib/joint_firmware/joint/configuration.h File Reference

Configuration definitions for Joint 1 to Joint 4.

Macros

- #define `ADR` 0x11
I2C adress of joint n is $0x1n$.
- #define `MAXACCEL` 100
Maximum acceleration in steps/s². Can be set for each joint depending on inertia. If set to high stalls might trigger since PID error grows too large.
- #define `MAXVEL` 10
Maximum velocity in steps/s. Can be set for each joint. If set to high stalls might trigger since PID error grows too large.

9.2.1 Detailed Description

Configuration definitions for Joint 1 to Joint 4.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

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This file shall be included AFTER one of J1, J2, J3 or J4 have been defined.

9.2.2 Macro Definition Documentation

9.2.2.1 ADR

```
#define ADR 0x11
```

I2C adress of joint n is 0x1n.

9.2.2.2 MAXACCEL

```
#define MAXACCEL 100
```

Maximum acceleration in steps/s². Can be set for each joint depending on inertia. If set to high stalls might trigger since PID error grows too large.

9.2.2.3 MAXVEL

```
#define MAXVEL 10
```

Maximum velocity in steps/s. Can be set for each joint. If set to high stalls might trigger since PID error grows too large.

9.3 configuration.h

[Go to the documentation of this file.](#)

```
00001
00014 #ifndef CONFIGURATION_H
00015 #define CONFIGURATION_H
00016
00017 #if defined(J1)
00019 #define ADR 0x11
00020 #define MAXACCEL 100
00021 #define MAXVEL 10
00022 #define STALL_WINDOW_B1 12
00023 #define STALL_WINDOW_B2 450
00024 #define STALL_WINDOW_OFFSET 90.0
00025 #define STALL_SLOPE 0.0
00026
00027 #elif defined(J2)
00028 #define ADR 0x12
00029 #define MAXACCEL 100
00030 #define MAXVEL 10
00031 #define STALL_WINDOW_B1 12
00032 #define STALL_WINDOW_B2 450
00033 #define STALL_WINDOW_OFFSET 90.0
00034 #define STALL_SLOPE 0.0
00035
00036 #elif defined(J3)
00037 #define ADR 0x13
00038 #define MAXACCEL 100
00039 #define MAXVEL 10
00040 #define STALL_WINDOW_B1 12
00041 #define STALL_WINDOW_B2 450
00042 #define STALL_WINDOW_OFFSET 90.0
00043 #define STALL_SLOPE 0.0
00044
00045 #elif defined(J4)
00046 #define ADR 0x14
00047 #define MAXACCEL 100
00048 #define MAXVEL 10
00049 #define STALL_WINDOW_B1 12
00050 #define STALL_WINDOW_B2 450
00051 #define STALL_WINDOW_OFFSET 90.0
00052 #define STALL_SLOPE 0.0
```

```
00053 #else
00054
00055 /* Below only defined for documentation */
00059 #define ADR 0x11
00060
00065 #define MAXACCEL 100
00066
00071 #define MAXVEL 10
00072 #error "No Joint has been defined. Define one of 'JX' where X 1,2,3,4"
00073 #endif
00074
00075 #endif
```

9.4 lib/joint_firmware/joint/filters.h File Reference

Helper classes for FIR and IIR filters.

```
#include <stdlib.h>
```

Classes

- class [bioscara_joint_firmware::Lowpass](#)
Simple discrete IIR lowpass filter.
- class [bioscara_joint_firmware::MovMax](#)
Simple FIR moving maximum filter.

Namespaces

- namespace [bioscara_joint_firmware](#)
Joint firmware.

9.4.1 Detailed Description

Helper classes for FIR and IIR filters.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

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This file contains time series filters that can be used to filter data.

9.5 filters.h

[Go to the documentation of this file.](#)

```

00001
00013 #include <stdlib.h>
00014
00015 namespace bioscara_joint_firmware {
00016
00021 class Lowpass {
00022
00023 protected:
00024     float K;
00025     float Ts;
00026     float tau;
00027     float x;
00028
00029 public:
00030
00038     Lowpass(float gain = 1, float sampleTime = 0.1, float timeconstant = 1.0) {
00039         this->K = gain;
00040         this->Ts = sampleTime;
00041         this->tau = timeconstant;
00042         x = 0.0;
00043     }
00044
00049     float updateState(float u) {
00050         x = (1 - (Ts / tau)) * x + K * (Ts / tau) * u;
00051         return x;
00052     }
00053
00058     void resetState(void) {
00059         x = 0.0;
00060     }
00061 };
00062
00068 class MovMax {
00069
00070 protected:
00071     unsigned int M = 200;
00072
00073     float *cb_data;
00074     unsigned int cb_index;
00075
00076 public:
00077     MovMax(float windowSize)
00085         : M(windowSize), cb_index(0), cb_data(0) {
00086
00087         cb_data = (float *)malloc(windowSize * sizeof(float)); // allocate memory for buffer
00088     }
00089
00095     float updateState(float u) {
00096
00097         cb_data[cb_index] = u;
00098         cb_index = (cb_index + 1) % M;
00099
00100
00101         float max = 0;
00102         for (size_t i = 0; i < M; i++) {
00103             if (cb_data[i] > max) {
00104                 max = cb_data[i];
00105             }
00106         }
00107
00108         return max;
00109     }
00110 };
00111 }

```

9.6 lib/joint_firmware/joint/joint.h File Reference

joint firmware header

```
#include <Arduino.h>
```

Namespaces

- namespace [bioscara_joint_firmware](#)
Joint firmware.

Macros

- #define [ACK](#) 'O'
- #define [NACK](#) 'N'
- #define [MAX_BUFFER](#) 4
Maximum size of I2C Payload in bytes.
- #define [RFLAGS_SIZE](#) 1
Size of the return flags in bytes.
- #define [DUMP_BUFFER](#)(buffer, size)
Macro to dump a buffer to the serial console.

Enumerations

- enum [bioscara_joint_firmware::stp_reg_t](#) {
[bioscara_joint_firmware::PING](#) = 0x0f , [bioscara_joint_firmware::SETUP](#) = 0x10 , [bioscara_joint_firmware::SETRPM](#) = 0x11 , [bioscara_joint_firmware::GETDRIVERRPM](#) = 0x12 ,
[bioscara_joint_firmware::MOVESTEPS](#) = 0x13 , [bioscara_joint_firmware::MOVEANGLE](#) = 0x14 ,
[bioscara_joint_firmware::MOVETOANGLE](#) = 0x15 , [bioscara_joint_firmware::GETMOTORSTATE](#) = 0x16
, [bioscara_joint_firmware::RUNCOTINOUS](#) = 0x17 , [bioscara_joint_firmware::ANGLEMOVED](#) = 0x18 ,
[bioscara_joint_firmware::SETCURRENT](#) = 0x19 , [bioscara_joint_firmware::SETHOLDCURRENT](#) = 0x1A
, [bioscara_joint_firmware::SETMAXACCELERATION](#) = 0x1B , [bioscara_joint_firmware::SETMAXDECELERATION](#) = 0x1C , [bioscara_joint_firmware::SETMAXVELOCITY](#) = 0x1D , [bioscara_joint_firmware::ENABLESTALLGUARD](#) = 0x1E ,
[bioscara_joint_firmware::DISABLESTALLGUARD](#) = 0x1F , [bioscara_joint_firmware::CLEARSTALL](#) = 0x20 ,
[bioscara_joint_firmware::SETBRAKEMODE](#) = 0x22 , [bioscara_joint_firmware::ENABLEPID](#) = 0x23 ,
[bioscara_joint_firmware::DISABLEPID](#) = 0x24 , [bioscara_joint_firmware::ENABLECLOSEDLOOP](#) = 0x25 ,
[bioscara_joint_firmware::DISABLECLOSEDLOOP](#) = 0x26 , [bioscara_joint_firmware::SETCONTROLTHRESHOLD](#) = 0x27 ,
[bioscara_joint_firmware::MOVETOEND](#) = 0x28 , [bioscara_joint_firmware::STOP](#) = 0x29 , [bioscara_joint_firmware::GETPIDERR](#) = 0x2A , [bioscara_joint_firmware::CHECKORIENTATION](#) = 0x2B ,
[bioscara_joint_firmware::GETENCODERRPM](#) = 0x2C , [bioscara_joint_firmware::HOME](#) = 0x2D ,
[bioscara_joint_firmware::HOMEOFFSET](#) = 0x2E }
register and command definitions

Functions

- template<typename T >
void [bioscara_joint_firmware::readValue](#) (T &val, uint8_t *rxBuf, size_t rx_length)
Reads a value from a buffer to a value of the specified type.
- template<typename T >
int [bioscara_joint_firmware::writeValue](#) (const T val, uint8_t *txBuf, size_t &tx_length)
Writes a value of the specified type to a buffer.

9.6.1 Detailed Description

joint firmware header

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

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This file contains definitions and macros for the joint firmware.

9.6.2 Macro Definition Documentation

9.6.2.1 ACK

```
#define ACK 'O'
```

9.6.2.2 DUMP_BUFFER

```
#define DUMP_BUFFER(  
    buffer,  
    size )
```

Value:

```
{  
    Serial.print("Buffer dump: ");  
    for (size_t i = 0; i < size; i++)  
    {  
        Serial.print(buffer[i], HEX);  
        Serial.print(" ");  
    }  
    Serial.println();  
}
```

Macro to dump a buffer to the serial console.

Parameters

<i>buffer</i>	pointer to a buffer to dump to the console
<i>size</i>	number of bytes to dump

9.6.2.3 MAX_BUFFER

```
#define MAX_BUFFER 4
```

Maximum size of I2C Payload in bytes.

4 bytes used to transmit floats and int32_t

9.6.2.4 NACK

```
#define NACK 'N'
```

9.6.2.5 RFLAGS_SIZE

```
#define RFLAGS_SIZE 1
```

Size of the return flags in bytes.

Only one byte used and hence set to 1.

9.7 joint.h

[Go to the documentation of this file.](#)

```
00001
00014 #ifndef JOINT_H
00015 #define JOINT_H
00016 #include <Arduino.h>
00017
00018 #define ACK 'O'
00019 #define NACK 'N'
00020
00026 #define MAX_BUFFER 4 // Bytes
00027
00033 #define RFLAGS_SIZE 1
00034
00041 #define DUMP_BUFFER(buffer, size) \
00042 { \
00043     Serial.print("Buffer dump: "); \
00044     for (size_t i = 0; i < size; i++) \
00045     { \
00046         Serial.print(buffer[i], HEX); \
00047         Serial.print(" "); \
00048     } \
00049     Serial.println(); \
00050 }
00051
00052 namespace bioscara_joint_firmware {
00061 enum stp_reg_t
00062 {
00063     PING = 0x0f,
00064     SETUP = 0x10,
00065     SETRPM = 0x11,
00066     GETDRIVERRPM = 0x12,
00067     MOVESTEPS = 0x13,
00068     MOVEANGLE = 0x14,
00069     MOVETOANGLE = 0x15,
00070     GETMOTORSTATE = 0x16,
00071     RUNCOTINOUS = 0x17,
00072     ANGLEMOVED = 0x18,
00073     SETCURRENT = 0x19,
00074     SETHOLDCURRENT = 0x1A,
00075     SETMAXACCELERATION = 0x1B,
00076     SETMAXDECELERATION = 0x1C,
00077     SETMAXVELOCITY = 0x1D,
00078     ENABLESTALLGUARD = 0x1E,
00079     DISABLESTALLGUARD = 0x1F,
```

```

00080 CLEARSTALL = 0x20,
00081 SETBRAKEMODE = 0x22,
00082 ENABLEPID = 0x23,
00083 DISABLEPID = 0x24,
00084 ENABLECLOSEDLOOP = 0x25,
00085 DISABLECLOSEDLOOP = 0x26,
00086 SETCONTROLTHRESHOLD = 0x27,
00087 MOVETOEND = 0x28,
00088 STOP = 0x29,
00089 GETPIDERROR = 0x2A,
00090 CHECKORIENTATION = 0x2B,
00091 GETENCODERRPM = 0x2C,
00092 HOME = 0x2D,
00093 HOMEOFFSET = 0x2E,
00094 };
00095
00102 template <typename T>
00103 void readValue(T &val, uint8_t *rxBuf, size_t rx_length)
00104 {
00105     memcpy(&val, rxBuf, rx_length);
00106 }
00107
00115 template <typename T>
00116 int writeValue(const T val, uint8_t *txBuf, size_t &tx_length)
00117 {
00118     tx_length = sizeof(T);
00119     memcpy(txBuf, &val, tx_length);
00120     return 0;
00121 }
00122 }
00123 #endif

```

9.8 lib/joint_firmware/joint/joint.ino File Reference

joint firmware

```

#include "configuration.h"
#include <UstepperS32.h>
#include <Wire.h>
#include "joint.h"
#include "filters.h"
#include "stall.h"

```

Namespaces

- namespace `bioscara_joint_firmware`
Joint firmware.

Macros

- `#define J1`
Define either joint that is to be flashed.

Functions

- void `bioscara_joint_firmware::blocking_handler` (uint8_t reg)
Handles commands received via I2C.
- void `bioscara_joint_firmware::non_blocking_handler` (uint8_t reg)
Handles read request received via I2C.
- void `bioscara_joint_firmware::set_flags_for_blocking_handler` (uint8_t reg)

- prepare flags to initiate the blocking handling of the received comman.*
- void `bioscara_joint_firmware::receiveEvent` (int n)
I2C receive event Handler.
- void `bioscara_joint_firmware::requestEvent` ()
I2C request event Handler.
- void `bioscara_joint_firmware::setup` (void)
Setup Peripherals.
- void `bioscara_joint_firmware::loop` (void)
Main loop.
- void `setup` (void)
- void `loop` (void)

Variables

- UstepperS32 `bioscara_joint_firmware::stepper`
The core UstepperS32 stepper object to control the motor.
- uint8_t `bioscara_joint_firmware::reg` = 0
- uint8_t `bioscara_joint_firmware::blk_reg` = 0
- uint8_t `bioscara_joint_firmware::rx_buf` [MAX_BUFFER] = {0}
- uint8_t `bioscara_joint_firmware::tx_buf` [MAX_BUFFER+RFLAGS_SIZE] = {0}
- bool `bioscara_joint_firmware::rx_data_ready` = 0
- size_t `bioscara_joint_firmware::tx_length` = 0
- size_t `bioscara_joint_firmware::rx_length` = 0

9.8.1 Detailed Description

joint firmware

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

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This file contains the joint firmware.

9.8.2 Macro Definition Documentation

9.8.2.1 J1

```
#define J1
```

Define either joint that is to be flashed.

Define either J1, J2, J3 or J4 and subsequently include [configuration.h](#)

9.8.3 Function Documentation

9.8.3.1 loop()

```
void loop (
    void )
```

9.8.3.2 setup()

```
void setup (
    void )
```

9.9 lib/joint_firmware/joint/stall.h File Reference

Helper functions for improved stall detection.

```
#include <stdlib.h>
```

Namespaces

- namespace [bioscara_joint_firmware](#)
Joint firmware.

Functions

- float [bioscara_joint_firmware::stall_threshold](#) (float qd_rad, float offset)
computes the speed adaptive threshold.

9.9.1 Detailed Description

Helper functions for improved stall detection.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

Copyright (c) 2025

9.10 stall.h

[Go to the documentation of this file.](#)

```
00001
00012 #ifndef STALL_H
00013 #define STALL_H
00014 #include <stdlib.h>
00015
00016 namespace bioscara_joint_firmware {
00023 float stall_threshold(float qd_rad, float offset) {
00024     /* y = ax + b */
00025     float a = STALL_SLOPE;
00026     float b = offset;
00027     if ((abs(qd_rad) >= STALL_WINDOW_B1) && (abs(qd_rad) <= STALL_WINDOW_B2)) {
00028         b += STALL_WINDOW_OFFSET;
00029     }
00030     return a * qd_rad + b;
00031 }
00032 }
00033 #endif
```

9.11 lib/ros2_ws/src/bioscara_rviz_plugin/include/bioscara_rviz_↵ plugin/bioscara_panel.hpp File Reference

```
#include <QLabel>
#include <QPushButton>
#include <unordered_map>
#include <vector>
#include <rviz_common/panel.hpp>
#include <rviz_common/ros_integration/ros_node_abstraction_iface.hpp>
#include "controller_manager_msgs/msg/controller_manager_activity.hpp"
#include "controller_manager_msgs/msg/named_lifecycle_state.hpp"
#include "controller_manager_msgs/srv/list_controllers.hpp"
#include "controller_manager_msgs/srv/set_hardware_component_state.hpp"
#include "controller_manager_msgs/srv/switch_controller.hpp"
#include "controller_manager_msgs/srv/configure_controller.hpp"
#include "control_msgs/msg/dynamic_joint_state.hpp"
#include "control_msgs/msg/dynamic_interface_group_values.hpp"
```

Classes

- class [bioscara_rviz_plugin::BioscaraPanel](#)
RViz Panel to control the hardware specific functions of the Bioscara robot.

Namespaces

- namespace [bioscara_rviz_plugin](#)

9.12 bioscara_panel.hpp

[Go to the documentation of this file.](#)

```
00001  /* Author: sbstorz */
00002
00003  #ifndef BIOSCARA_RVIZ_PANEL_HPP_
00004  #define BIOSCARA_RVIZ_PANEL_HPP_
00005
00006  #include <QLabel>
00007  #include <QPushButton>
00008  #include <unordered_map>
00009  #include <vector>
00010
00011  #include <rviz_common/panel.hpp>
00012  #include <rviz_common/ros_integration/ros_node_abstraction_iface.hpp>
00013  #include "controller_manager_msgs/msg/controller_manager_activity.hpp"
00014  #include "controller_manager_msgs/msg/named_lifecycle_state.hpp"
00015  #include "controller_manager_msgs/srv/list_controllers.hpp"
00016  #include "controller_manager_msgs/srv/set_hardware_component_state.hpp"
00017  #include "controller_manager_msgs/srv/switch_controller.hpp"
00018  #include "controller_manager_msgs/srv/configure_controller.hpp"
00019  #include "control_msgs/msg/dynamic_joint_state.hpp"
00020  #include "control_msgs/msg/dynamic_interface_group_values.hpp"
00021
00022  namespace Ui
00023  {
00024      class BioscaraUI;
00025  }
00026
00027  namespace bioscara_rviz_plugin
00028  {
00029      using namespace controller_manager_msgs::msg;
00030      using namespace controller_manager_msgs::srv;
00031      using namespace control_msgs::msg;
00032
00033      class BioscaraPanel : public rviz_common::Panel
00034      {
00035      Q_OBJECT
00036      public:
00037          explicit BioscaraPanel(QWidget *parent = 0);
00038
00039          ~BioscaraPanel() override;
00040
00041          void onInitialize() override;
00042
00043      protected:
00044          rclcpp::Node::SharedPtr node_;
00045
00046          rclcpp::Subscription<ControllerManagerActivity>::SharedPtr cm_state_subscription_;
00047
00048          rclcpp::Subscription<DynamicJointState>::SharedPtr joint_state_subscription_;
00049
00050          rclcpp::Publisher<DynamicInterfaceGroupValues>::SharedPtr homing_publisher_;
00051
00052          rclcpp::Client<SwitchController>::SharedPtr switch_controller_client_;
00053
00054          rclcpp::Client<ConfigureController>::SharedPtr configure_controller_client_;
00055
00056          rclcpp::Client<SetHardwareComponentState>::SharedPtr hardware_state_client_;
00057
00058          rclcpp::TimerBase::SharedPtr prune_timer_;
00059
00060          std::unordered_map<std::string, InterfaceValue> joint_states_;
00061
00062          std::unordered_map<std::string, NamedLifecycleState> hardware_states_;
00063
00064          std::unordered_map<std::string, NamedLifecycleState> controller_states_;
00065      };
00066  }
```

```

00151
00158     Ui::BioscaraUI *ui_;
00159
00170     void cm_state_callback(const ControllerManagerActivity &msg);
00171
00180     void joint_state_callback(const DynamicJointState &msg);
00181
00190     void ensure_jsb_is_active(void);
00191
00201     void set_hardware_component_state(const std::string component, const lifecycle_msgs::msg::State
target_state);
00202
00211     void configure_controller(const std::string controller);
00212
00223     void switch_controllers(const std::vector<std::string> &activate_controllers,
00224                             const std::vector<std::string> &deactivate_controllers,
00225                             const int32_t = SwitchController::Request::BEST_EFFORT,
00226                             const bool activate_asap = false,
00227                             const builtin_interfaces::msg::Duration timeout =
builtin_interfaces::msg::Duration());
00228
00238     void set_controller_state(const std::string controller,
00239                             const lifecycle_msgs::msg::State target_state);
00240
00247     void dynamic_joint_state_msg_to_map(const DynamicJointState &dynamic_joint_state_in,
00248                                         std::unordered_map<std::string, InterfaceValue> &map_out);
00249
00256     void named_lcs_msg_to_map(const std::vector<NamedLifecycleState> &named_lcs_in,
00257                             std::unordered_map<std::string, NamedLifecycleState> &map_out);
00258
00265     void print_cm_map(const std::string prefix,
00266                     const std::unordered_map<std::string, NamedLifecycleState> &map_in);
00267
00288     void prepopulate_joint_state_map(std::unordered_map<std::string, InterfaceValue> &state_map,
00289                                     std::vector<std::string> augment_vec);
00290
00303     void prepopulate_state_map(std::unordered_map<std::string, NamedLifecycleState> &state_map,
00304                               std::vector<std::string> augment_vec);
00305
00310     void update_homing_grp_state(void);
00311
00318     void update_homing_state_labels(void);
00319
00327     void set_homing_state_label(QLabel *label, const InterfaceValue &state);
00328
00334     void update_state_labels_and_btns(void);
00335
00346     void update_state_label_and_btn(
00347         const std::unordered_map<std::string, NamedLifecycleState> &state_map,
00348         QLabel *state_label,
00349         QPushButton *en_button,
00350         const std::string &state_key);
00351
00360     void set_state_label(QLabel *label, const NamedLifecycleState &state);
00361
00370     void set_en_btn(QPushButton *button, const NamedLifecycleState &state);
00371
00382     bool check_activation_conditions(const std::string component_key);
00383
00391     lifecycle_msgs::msg::State target_state_from_current(lifecycle_msgs::msg::State current_state);
00392
00401     void homing_cmd(const std::string joint, const int cmd);
00402
00408     void ctrl_en_btn_cb(const std::string controller);
00409
00410 private Q_SLOTS:
00411
00417     void arm_en_btn_cb(void);
00418
00424     void gripper_en_btn_cb(void);
00425 };
00426
00427 } // namespace bioscara_rviz_plugin
00428
00429 #endif // BIOSCARA_RVIZ_PANEL_HPP_

```

9.13 lib/ros2_ws/src/bioscara_rviz_plugin/src/bioscara_panel.cpp File Reference

```

#include <bioscara_rviz_plugin/bioscara_panel.hpp>
#include <rviz_common/display_context.hpp>

```

```
#include <chrono>
#include "ui_bioscara_rviz_plugin_frame.h"
#include <pluginlib/class_list_macros.hpp>
```

Namespaces

- namespace [bioscara_rviz_plugin](#)

9.14 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/common.h File Reference

A file containing utility macros and functions.

Macros

- #define [DUMP_BUFFER](#)(buffer, size)
Macro to dump a buffer to cout.

9.14.1 Detailed Description

A file containing utility macros and functions.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

Copyright (c) 2025

9.14.2 Macro Definition Documentation

9.14.2.1 DUMP_BUFFER

```
#define DUMP_BUFFER(
    buffer,
    size )
```

Value:

```
{
    std::cout << "Buffer dump: ";
    for (size_t i = 0; i < size; i++)
    {
        printf("%#x ", buffer[i]);
    }
    std::cout << std::endl;
}
```

Macro to dump a buffer to cout.

Parameters

<i>buffer</i>	pointer to a buffer to dump to the console
<i>size</i>	number of bytes to dump

9.15 common.h

[Go to the documentation of this file.](#)

```

00001
00011 #ifndef COMMON_H
00012 #define COMMON_H
00013
00020 #define DUMP_BUFFER(buffer, size) \
00021 { \
00022     std::cout << "Buffer dump: "; \
00023     for (size_t i = 0; i < size; i++) \
00024     { \
00025         printf("%#x ", buffer[i]); \
00026     } \
00027     std::cout << std::endl; \
00028 }
00029
00030 #endif // COMMON_H

```

9.16 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mBaseJoint.h File Reference

File including the BaseJoint class.

```

#include <iostream>
#include "bioscara_arm_hardware_driver/uErr.h"

```

Classes

- class [bioscara_hardware_drivers::BaseJoint](#)
Generic base class to control a single joint.

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.16.1 Detailed Description

File including the BaseJoint class.

Author

sbstorz

Version

0.1

Date

2025-05-29

Copyright

Copyright (c) 2025

9.17 mBaseJoint.h

[Go to the documentation of this file.](#)

```

00001
00012 #ifndef MBASEJOINT_H
00013 #define MBASEJOINT_H
00014
00015 #include <iostream>
00016 #include "bioscara_arm_hardware_driver/uErr.h"
00017
00018 namespace bioscara_hardware_drivers
00019 {
00020     class BaseJoint
00021     {
00022     public:
00023         enum stp_reg_t
00024         {
00025             NONE = 0x00,
00026             PING = 0x0f,
00027             SETUP = 0x10,
00028             SETRPM = 0x11,
00029             GETDRIVERRPM = 0x12,
00030             MOVESTEPS = 0x13,
00031             MOVEANGLE = 0x14,
00032             MOVETOANGLE = 0x15,
00033             GETMOTORSTATE = 0x16,
00034             RUNCOTINOUS = 0x17,
00035             ANGLEMOVED = 0x18,
00036             SETCURRENT = 0x19,
00037             SETHOLDCURRENT = 0x1A,
00038             SETMAXACCELERATION = 0x1B,
00039             SETMAXDECELERATION = 0x1C,
00040             SETMAXVELOCITY = 0x1D,
00041             ENABLESTALLGUARD = 0x1E,
00042             DISABLESTALLGUARD = 0x1F,
00043             CLEARSTALL = 0x20,
00044             SETBRAKEMODE = 0x22,
00045             ENABLEPID = 0x23,
00046             DISABLEPID = 0x24,
00047             ENABLECLOSEDLOOP = 0x25,
00048             DISABLECLOSEDLOOP = 0x26,
00049             SETCONTROLTHRESHOLD = 0x27,
00050             MOVETOEND = 0x28,
00051             STOP = 0x29,
00052             GETPIDERROR = 0x2A,
00053             CHECKORIENTATION = 0x2B,
00054             GETENCODERRPM = 0x2C,
00055             HOME = 0x2D,

```



```

00071     HOMEOFFSET = 0x2E,
00072 };
00073
00081 BaseJoint(const std::string name);
00082
00088 ~BaseJoint(void);
00089
00096 virtual err_type_t init(void);
00097
00104 virtual err_type_t deinit(void);
00105
00118 virtual err_type_t enable(u_int8_t driveCurrent, u_int8_t holdCurrent);
00119
00127 virtual err_type_t disable(void);
00128
00142 virtual err_type_t home(float velocity, u_int8_t sensitivity, u_int8_t current);
00143
00160 virtual err_type_t startHoming(float velocity, u_int8_t sensitivity, u_int8_t current);
00161
00170 virtual err_type_t postHoming(void);
00171
00182 virtual err_type_t getPosition(float &pos) = 0;
00183
00191 virtual err_type_t setPosition(float pos);
00192
00201 virtual err_type_t moveSteps(int32_t steps);
00202
00210 virtual err_type_t getVelocity(float &vel) = 0;
00211
00219 virtual err_type_t setVelocity(float vel);
00220
00228 virtual err_type_t checkOrientation(float angle = 2.0);
00229
00238 virtual err_type_t stop(void);
00239
00240 virtual err_type_t disableCL(void);
00241
00247 virtual err_type_t setDriveCurrent(u_int8_t current);
00248
00255 virtual err_type_t setHoldCurrent(u_int8_t current);
00256
00262 virtual err_type_t setBrakeMode(u_int8_t mode);
00263
00271 virtual err_type_t setMaxAcceleration(float maxAccel);
00272
00280 virtual err_type_t setMaxVelocity(float maxVel);
00281
00291 virtual err_type_t enableStallguard(u_int8_t sensitivity);
00292
00301 virtual bool isHomed(void);
00302
00312 virtual bool isEnabled(void);
00313
00321 virtual bool isStalled(void);
00322
00330 virtual bool isBusy(void);
00331
00337 virtual err_type_t getFlags(u_int8_t &flags);
00338
00343 virtual err_type_t getFlags(void);
00344
00350 virtual stp_reg_t getCurrentBCmd(void);
00351
00352     std::string name;
00353
00354 protected:
00360     virtual void wait_while_busy(const float period_ms);
00361
00374     virtual err_type_t _home(float velocity, u_int8_t sensitivity, u_int8_t current) = 0;
00375
00394     u_int8_t flags = 0b00001100;
00395
00396     stp_reg_t current_b_cmd = NONE;
00397
00398 private:
00399 };
00400 }
00401 #endif

```

9.18 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mJoint.h File Reference

File including the Joint class.

```
#include <iostream>
#include "bioscara_arm_hardware_driver/mBaseJoint.h"
#include "bioscara_arm_hardware_driver/uTransmission.h"
#include "bioscara_arm_hardware_driver/mJoint.hpp"
```

Classes

- class [bioscara_hardware_drivers::Joint](#)
Representing a single hardware joint connected via I2C.

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.18.1 Detailed Description

File including the Joint class.

Author

sbstorz

Version

0.1

Date

2025-05-29

Copyright

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9.19 mJoint.h

[Go to the documentation of this file.](#)

```

00001
00012 #ifndef MJOINT_H
00013 #define MJOINT_H
00014
00015 #include <iostream>
00016 #include "bioscara_arm_hardware_driver/mBaseJoint.h"
00017 #include "bioscara_arm_hardware_driver/uTransmission.h"
00018
00019 namespace bioscara_hardware_drivers
00020 {
00021     class Joint : public BaseJoint
00022     {
00023     public:
00024         Joint(const std::string name, const int address, const float reduction, const float min, const
00060 float max);
00061
00062         ~Joint(void);
00063
00064         err_type_t init(void) override;
00065
00066         err_type_t deinit(void) override;
00067
00068         err_type_t enable(u_int8_t driveCurrent, u_int8_t holdCurrent) override;
00069
00070         err_type_t postHoming(void) override;
00071
00072         err_type_t getPosition(float &pos) override;
00073
00074         err_type_t setPosition(float pos) override;
00075
00076         err_type_t moveSteps(int32_t steps) override;
00077
00078         err_type_t getVelocity(float &vel) override;
00079
00080         err_type_t setVelocity(float vel) override;
00081
00082         err_type_t checkOrientation(float angle = 2.0) override;
00083
00084         err_type_t stop(void) override;
00085
00086         err_type_t disableCL(void) override;
00087
00088         err_type_t setDriveCurrent(u_int8_t current) override;
00089
00090         err_type_t setHoldCurrent(u_int8_t current) override;
00091
00092         err_type_t setBrakeMode(u_int8_t mode) override;
00093
00094         err_type_t setMaxAcceleration(float maxAccel) override;
00095
00096         err_type_t setMaxVelocity(float maxVel) override;
00097
00098         err_type_t enableStallguard(u_int8_t sensitivity) override;
00099
00100         err_type_t getFlags(void) override;
00101
00102     protected:
00103
00104         err_type_t getHomingOffset(float &offset);
00105
00106         err_type_t setHomingOffset(const float offset);
00107
00108         err_type_t _home(float velocity, u_int8_t sensitivity, u_int8_t current) override;
00109
00110         err_type_t checkCom(void);
00111
00112         float reduction = 1;
00113         float offset = 0;
00114         float min = 0;
00115         float max = 0;
00116
00117     private:
00118         template <typename T>
00119         int read(const stp_reg_t reg, T &data, u_int8_t &flags);
00120
00121         template <typename T>
00122         int write(const stp_reg_t reg, T data, u_int8_t &flags);
00123
00124         int address;
00125         int handle = -1;
00126     };
00127 }

```

```
00178 #include "bioscara_arm_hardware_driver/mJoint.hpp"
00179
00180 #endif
```

9.20 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mJoint.hpp File Reference

Templated functions for the Joint class.

```
#include "bioscara_arm_hardware_driver/mJoint.h"
#include "bioscara_arm_hardware_driver/uI2C.h"
#include "bioscara_arm_hardware_driver/common.h"
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.20.1 Detailed Description

Templated functions for the Joint class.

Author

sbstorz

Version

0.1

Date

2025-05-29

Copyright

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This header must be included at the END of the [mJoint.h](#) file.

9.21 mJoint.hpp

[Go to the documentation of this file.](#)

```

00001
00012 #include "bioscara_arm_hardware_driver/mJoint.h"
00013 #include "bioscara_arm_hardware_driver/uI2C.h"
00014 #include "bioscara_arm_hardware_driver/common.h"
00015
00016 namespace bioscara_hardware_drivers
00017 {
00031     template <typename T>
00032     int Joint::read(const stp_reg_t reg, T &data, u_int8_t &flags)
00033     {
00034         size_t size = sizeof(T) + RFLAGS_SIZE;
00035         char buf[MAX_BUFFER + RFLAGS_SIZE];
00036         int n = readFromI2CDev(this->handle, reg, buf, size);
00037         if (n != static_cast<int>(size))
00038         {
00039             return -1;
00040         }
00041         memcpy(&data, buf, size - RFLAGS_SIZE);
00042         memcpy(&flags, buf + size - RFLAGS_SIZE, RFLAGS_SIZE);
00043         return 0;
00044     }
00045
00061     template <typename T>
00062     int Joint::write(const stp_reg_t reg, T data, u_int8_t &flags)
00063     {
00064         size_t size = sizeof(T) + RFLAGS_SIZE;
00065         char buf[MAX_BUFFER + RFLAGS_SIZE];
00066         memcpy(buf, &data, size - RFLAGS_SIZE);
00067         int rc = writeToI2CDev(this->handle, reg, buf, size - RFLAGS_SIZE, buf + size - RFLAGS_SIZE);
00068         rc = rc > 0 ? 0 : rc;
00069         memcpy(&flags, buf + size - RFLAGS_SIZE, RFLAGS_SIZE);
00070         return rc;
00071     }
00072 }

```

9.22 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mMockJoint.h File Reference

File including the MockJoint class.

```

#include <iostream>
#include "bioscara_arm_hardware_driver/mBaseJoint.h"
#include <chrono>

```

Classes

- class `bioscara_hardware_drivers::MockJoint`
Representing a single joint mocking the hardware joint.

Namespaces

- namespace `bioscara_hardware_drivers`

9.22.1 Detailed Description

File including the MockJoint class.

Author

sbstorz

Version

0.1

Date

2025-05-29

Copyright

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9.23 mMockJoint.h

[Go to the documentation of this file.](#)

```

00001
00012 #ifndef MMOCKJOINT_H
00013 #define MMOCKJOINT_H
00014
00015 #include <iostream>
00016 #include "bioscara_arm_hardware_driver/mBaseJoint.h"
00017 #include <chrono>
00018
00019 namespace bioscara_hardware_drivers
00020 {
00021     class MockJoint : public BaseJoint
00022     {
00023     public:
00024         MockJoint(const std::string name);
00025
00026         err_type_t enable(u_int8_t driveCurrent, u_int8_t holdCurrent) override;
00027
00028         err_type_t disable(void) override;
00029
00030         err_type_t getPosition(float &pos) override;
00031
00032         err_type_t setPosition(float pos) override;
00033
00034         err_type_t getVelocity(float &vel) override;
00035
00036         err_type_t setVelocity(float vel) override;
00037
00038         err_type_t checkOrientation(float angle = 2.0) override;
00039
00040         err_type_t stop(void) override;
00041
00042         err_type_t getFlags(void) override;
00043
00044         bool isHomed(void) override;
00045
00046     protected:
00047         err_type_t _home(float velocity, u_int8_t sensitivity, u_int8_t current);
00048
00049     private:
00050         float q = 0.0;
00051         float qd = 0.0;
00052
00053         std::chrono::_V2::system_clock::time_point last_set_position =
00054             std::chrono::high_resolution_clock::now();
00055         std::chrono::_V2::system_clock::time_point last_set_velocity = last_set_position;
00056         std::chrono::_V2::system_clock::time_point async_start_time = last_set_position;
00057
00058         float getDeltaT(std::chrono::_V2::system_clock::time_point &last_call, bool update = true);
00059
00060         stp_reg_t op_mode = NONE;
00061     };
00062 }
00063 #endif

```

9.24 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/uErr.h File Reference

Defining common return types.

```
#include <string>
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

Macros

- #define [RETURN_ON_ERROR](#)(x)
Macro which executes a function and returns from the calling function with the error code if the called function fails.
- #define [RETURN_ON_FALSE](#)(a, err_code)
Macro which returns the calling function with specified error_code if the given condition is false.
- #define [RETURN_ON_NEGATIVE](#)(a, err_code)
Macro which returns the calling function with specified error_code if the given condition is negative.

Enumerations

- enum class [bioscara_hardware_drivers::err_type_t](#) {
[bioscara_hardware_drivers::OK](#) = 0 , [bioscara_hardware_drivers::ERROR](#) = -1 , [bioscara_hardware_drivers::NOT_HOMED](#) = -2 , [bioscara_hardware_drivers::NOT_ENABLED](#) = -3 ,
[bioscara_hardware_drivers::STALLED](#) = -4 , [bioscara_hardware_drivers::NOT_INIT](#) = -5 , [bioscara_hardware_drivers::COMM_I](#) = -6 , [bioscara_hardware_drivers::INVALID_ARGUMENT](#) = -101 ,
[bioscara_hardware_drivers::INCORRECT_STATE](#) = -109 }
Enum defining common error types.

Functions

- std::string [bioscara_hardware_drivers::error_to_string](#) (err_type_t err)
Converts an error code to a string and returns it.

9.24.1 Detailed Description

Defining common return types.

Author

sbstorz

Version

0.1

Date

2025-11-05

Copyright

Copyright (c) 2025

9.24.2 Macro Definition Documentation

9.24.2.1 RETURN_ON_ERROR

```
#define RETURN_ON_ERROR(  
    x )
```

Value:

```
do  
{  
    bioscara_hardware_drivers::err_type_t err_rc_ = (x);  
    if (err_rc_ != bioscara_hardware_drivers::err_type_t::OK)  
    {  
        return err_rc_;  
    }  
} while (0);
```

Macro which executes a function and returns from the calling function with the error code if the called function fails.

Adapted from the [ESP-IDF](#)

Parameters

<i>x</i>	function to call
----------	------------------

9.24.2.2 RETURN_ON_FALSE

```
#define RETURN_ON_FALSE(  
    a,  
    err_code )
```

Value:

```
do  
{  
    if (!a)  
    {  
        return err_code;  
    }  
} while (0);
```

Macro which returns the calling function with specified `err_code` if the given condition is false.

Adapted from the [ESP-IDF](#)

Parameters

<i>a</i>	expression that evaluates to true or false
<i>err_code</i>	return code to return on false

9.24.2.3 RETURN_ON_NEGATIVE

```
#define RETURN_ON_NEGATIVE(  
    a,  
    err_code )
```

Value:


```

do
{
    if ((a) < 0)
    {
        return err_code;
    }
} while (0);

```

Macro which returns the calling function with specified error_code if the given condition is negative.

Adapted from the [ESP-IDF](#)

Parameters

<i>a</i>	expression that evaluates to a signed number
<i>err_code</i>	return code to return on false

9.25 uErr.h

[Go to the documentation of this file.](#)

```

00001
00011 #ifndef UERR_H
00012 #define UERR_H
00013
00014 #include <string>
00015
00016 namespace bioscara_hardware_drivers
00017 {
00022     enum class err_type_t
00023     {
00024         OK = 0,
00025         ERROR = -1,
00026         NOT_HOMED = -2,
00027         NOT_ENABLED = -3,
00028         STALLED = -4,
00029         NOT_INIT = -5,
00030         COMM_ERROR = -6,
00031         INVALID_ARGUMENT = -101,
00032         INCORRECT_STATE = -109,
00033     };
00034
00035     std::string error_to_string(err_type_t err);
00043 }
00044
00052 #define RETURN_ON_ERROR(x)
00053 do
00054 {
00055     bioscara_hardware_drivers::err_type_t err_rc_ = (x);
00056     if (err_rc_ != bioscara_hardware_drivers::err_type_t::OK)
00057     {
00058         return err_rc_;
00059     }
00060 } while (0);
00061
00069 #define RETURN_ON_FALSE(a, err_code) \
00070 do
00071 {
00072     if (!(a))
00073     {
00074         return err_code;
00075     }
00076 } while (0);
00077
00085 #define RETURN_ON_NEGATIVE(a, err_code) \
00086 do
00087 {
00088     if ((a) < 0)
00089     {
00090         return err_code;
00091     }
00092 } while (0);
00093
00094 #endif // UERR_H

```

9.26 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/ul2C.h File Reference

Low level utility for I2C communication on Raspberry Pi using I2C library.

```
#include <cstring>
#include <errno.h>
#include <fcntl.h>
#include <iostream>
#include <termios.h>
#include <unistd.h>
```

Macros

- #define `ACK` 'O'
- #define `NACK` 'N'
- #define `RFLAGS_SIZE` 1
Size of the return flags in bytes.
- #define `MAX_BUFFER` 4
Maximum size of I2C Payload in bytes.

Functions

- int `openI2CDevHandle` (const int dev_addr)
Initiates an I2C device on the bus.
- int `readFromI2CDev` (const int dev_handle, const int reg, char *buffer, const int data_length)
reads block of bytes from device to buffer
- int `writeToI2CDev` (const int dev_handle, const int reg, char *tx_buffer, const int data_length, char *RFLAGS_buffer)
writes block of bytes from buffer to device
- int `closeI2CDevHandle` (int &dev_handle)
close an I2C device on the bus

9.26.1 Detailed Description

Low level utility for I2C communication on Raspberry Pi using I2C library.

Author

sbstorz

Version

0.1

Date

2025-05-28

Copyright

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lgpio needs to be installed and linked! Installation:

```
cd ~
sudo apt update
sudo apt install -y swig
wget https://github.com/joan2937/lg/archive/master.zip
unzip master.zip
cd lg-master
make
sudo make install
cd ..
sudo rm -rf lg-master
rm master.zip
```

bash

9.26.2 Macro Definition Documentation

9.26.2.1 ACK

```
#define ACK 'O'
```

9.26.2.2 MAX_BUFFER

```
#define MAX_BUFFER 4
```

Maximum size of I2C Payload in bytes.

4 bytes used to transmit floats and int32_t

9.26.2.3 NACK

```
#define NACK 'N'
```

9.26.2.4 RFLAGS_SIZE

```
#define RFLAGS_SIZE 1
```

Size of the return flags in bytes.

Only one byte used and hence set to 1.

9.26.3 Function Documentation

9.26.3.1 closeI2CDevHandle()

```
int closeI2CDevHandle (
    int & dev_handle )
```

close an I2C device on the bus

Parameters

<i>dev_handle</i>	device handle obtained from <code>openI2CDevHandle</code>
-------------------	---

Returns

0 on OK, negative on error.

9.26.3.2 openI2CDevHandle()

```
int openI2CDevHandle (
    const int dev_addr )
```

Initiates an I2C device on the bus.

Parameters

<i>dev_addr</i>	7-bit device address [0 - 0x7F]
-----------------	---------------------------------

Returns

the device handle, negative on error.

9.26.3.3 readFromI2CDev()

```
int readFromI2CDev (
    const int dev_handle,
    const int reg,
    char * buffer,
    const int data_length )
```

reads block of bytes from device to buffer

Parameters

<i>dev_handle</i>	device handle obtained from <code>openI2CDevHandle</code>
<i>reg</i>	the command/data register
<i>buffer</i>	pointer to data buffer to hold received values
<i>data_length</i>	number of bytes to read

Returns

number of bytes read, negative on error.

9.26.3.4 writeToI2CDev()

```
int writeToI2CDev (
    const int dev_handle,
```

```

    const int reg,
    char * tx_buffer,
    const int data_length,
    char * RFLAGS_buffer )

```

writes block of bytes from buffer to device

Parameters

<i>dev_handle</i>	device handle obtained from openI2CDevHandle
<i>reg</i>	the command/data register
<i>tx_buffer</i>	pointer to data buffer holding the data to send
<i>data_length</i>	number of bytes to send
<i>RFLAGS_buffer</i>	buffer to hold returned flags

Returns

0 on OK, negative on error.

9.27 ul2C.h

[Go to the documentation of this file.](#)

```

00001
00028 #ifndef SERIAL_H
00029 #define SERIAL_H
00030 #include <cstring>
00031 #include <errno.h>
00032 #include <fcntl.h>
00033 #include <iostream>
00034 #include <termios.h>
00035 #include <unistd.h>
00036
00037 #define ACK 'O'
00038 #define NACK 'N'
00039
00043 #define RFLAGS_SIZE 1
00044
00048 #define MAX_BUFFER 4 // Bytes
00049
00055 int openI2CDevHandle(const int dev_addr);
00056
00065 int readFromI2CDev(const int dev_handle, const int reg, char *buffer, const int data_length);
00066
00076 int writeToI2CDev(const int dev_handle, const int reg, char *tx_buffer, const int data_length, char
    *RFLAGS_buffer);
00077
00083 int closeI2CDevHandle(int &dev_handle);
00084
00085
00086 #endif

```

9.28 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_↵ driver/include/bioscara_arm_hardware_driver/uTransmission.h File Reference

Macros

- `#define JOINT2ACTUATOR(in, reduction, offset) (reduction * (in - offset))`
Macro for a simple transmission from joint units to actuator units.

- `#define ACTUATOR2JOINT(in, reduction, offset) (in / reduction + offset)`
Macro for a simple transmission from actuator units to joint units.
- `#define M_PI 3.14159265358979323846`
pi
- `#define RAD2DEG(rad) (rad / M_PI * 180)`
Macro to convert radians to degree.
- `#define DEG2RAD(deg) (deg * M_PI / 180)`
Macro to convert degree to radians.

9.28.1 Macro Definition Documentation

9.28.1.1 ACTUATOR2JOINT

```
#define ACTUATOR2JOINT(
    in,
    reduction,
    offset ) (in / reduction + offset)
```

Macro for a simple transmission from actuator units to joint units.

The translation is based on the `ros2_control` transmission interface, simple transmission. For position reduction and offset need to be used.

For velocity and acceleration only use reduction and NO offset

For effort/torque use 1/reduction and NO offset

9.28.1.2 DEG2RAD

```
#define DEG2RAD(
    deg ) (deg * M_PI / 180)
```

Macro to convert degree to radians.

9.28.1.3 JOINT2ACTUATOR

```
#define JOINT2ACTUATOR(
    in,
    reduction,
    offset ) (reduction * (in - offset))
```

Macro for a simple transmission from joint units to actuator units.

The translation is based on the `ros2_control` transmission interface, simple transmission. For position reduction and offset need to be used.

For velocity and acceleration only use reduction and NO offset

For effort/torque use 1/reduction and NO offset

9.28.1.4 M_PI

```
#define M_PI 3.14159265358979323846
```

pi

9.28.1.5 RAD2DEG

```
#define RAD2DEG(  
    rad ) (rad / M_PI * 180)
```

Macro to convert radians to degree.

9.29 uTransmission.h

[Go to the documentation of this file.](#)

```
00001 #ifndef UTRANSMISSION_H
00002 #define UTRANSMISSION_H
00003
00012 #define JOINT2ACTUATOR(in, reduction, offset) (reduction * (in - offset))
00013
00022 #define ACTUATOR2JOINT(in, reduction, offset) (in / reduction + offset)
00023
00028 #define M_PI 3.14159265358979323846
00029
00034 #define RAD2DEG(rad) (rad / M_PI * 180)
00035
00040 #define DEG2RAD(deg) (deg * M_PI / 180)
00041 #endif //UTRANSMISSION_H
```

9.30 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_↵ driver/src/joint_comm_node.cpp File Reference

```
#include <signal.h>
#include <unistd.h>
#include "bioscara_arm_hardware_driver/mJoint.h"
#include <vector>
#include <cmath>
```

Functions

- void [INT_handler](#) (int s)
- int [main](#) (int argc, char **argv)

Variables

- [Joint J1](#) ("j1", 0x11, 35, -3.04647, 3.04647)
- [Joint J2](#) ("j2", 0x12, -2 * [M_PI](#) / 0.004, 0.338, 0.0)
- [Joint J3](#) ("j3", 0x13, 24, -2.62672, 2.62672)
- [Joint J4](#) ("j4", 0x14, 12, -3.01069, 3.01069)

9.30.1 Function Documentation

9.30.1.1 INT_handler()

```
void INT_handler (
    int s )
```

9.30.1.2 main()

```
int main (
    int argc,
    char ** argv )
```

9.30.2 Variable Documentation

9.30.2.1 J1

```
Joint J1("j1", 0x11, 35, -3.04647, 3.04647) (
    "j1" ,
    0x11 ,
    35 ,
    -3. 04647,
    3. 04647 )
```

9.30.2.2 J2

```
Joint J2("j2", 0x12, -2 *M_PI/0.004, 0.338, 0.0) (
    "j2" ,
    0x12 ,
    -2 *M_PI/0. 004,
    0. 338,
    0. 0 )
```

9.30.2.3 J3

```
Joint J3("j3", 0x13, 24, -2.62672, 2.62672) (
    "j3" ,
    0x13 ,
    24 ,
    -2. 62672,
    2. 62672 )
```

9.30.2.4 J4

```
Joint J4("j4", 0x14, 12, -3.01069, 3.01069) (
    "j4" ,
    0x14 ,
    12 ,
    -3. 01069,
    3. 01069 )
```


9.31 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mBaseJoint.cpp File Reference

```
#include "bioscara_arm_hardware_driver/mBaseJoint.h"
#include <unistd.h>
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.32 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mJoint.cpp File Reference

```
#include "bioscara_arm_hardware_driver/uI2C.h"
#include "bioscara_arm_hardware_driver/mJoint.h"
#include <cmath>
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.33 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/mMockJoint.cpp File Reference

```
#include "bioscara_arm_hardware_driver/mMockJoint.h"
#include <unistd.h>
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.34 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/uErr.cpp File Reference

```
#include "bioscara_arm_hardware_driver/uErr.h"
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

Functions

- `std::string bioscara_hardware_drivers::error_to_string (err_type_t err)`
Converts an error code to a string and returns it.

9.35 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/uI2C.cpp File Reference

```
#include "bioscara_arm_hardware_driver/uI2C.h"
#include "bioscara_arm_hardware_driver/common.h"
#include <lgpio.h>
```

Functions

- `int openI2CDevHandle (const int dev_addr)`
Initiates an I2C device on the bus.
- `int readFromI2CDev (const int dev_handle, const int reg, char *buffer, const int data_length)`
reads block of bytes from device to buffer
- `int writeToI2CDev (const int dev_handle, const int reg, char *tx_buffer, const int data_length, char *RFLAGS_buffer)`
writes block of bytes from buffer to device
- `int closeI2CDevHandle (int &dev_handle)`
close an I2C device on the bus

9.35.1 Function Documentation

9.35.1.1 closeI2CDevHandle()

```
int closeI2CDevHandle (
    int & dev_handle )
```

close an I2C device on the bus

Parameters

<i>dev_handle</i>	device handle obtained from <code>openI2CDevHandle</code>
-------------------	---

Returns

0 on OK, negative on error.

9.35.1.2 openI2CDevHandle()

```
int openI2CDevHandle (
    const int dev_addr )
```

Initiates an I2C device on the bus.

Parameters

<i>dev_addr</i>	7-bit device address [0 - 0x7F]
-----------------	---------------------------------

Returns

the device handle, negative on error.

9.35.1.3 readFromI2CDev()

```
int readFromI2CDev (
    const int dev_handle,
    const int reg,
    char * buffer,
    const int data_length )
```

reads block of bytes from device to buffer

Parameters

<i>dev_handle</i>	device handle obtained from openI2CDevHandle
<i>reg</i>	the command/data register
<i>buffer</i>	pointer to data buffer to hold received values
<i>data_length</i>	number of bytes to read

Returns

number of bytes read, negative on error.

9.35.1.4 writeToI2CDev()

```
int writeToI2CDev (
    const int dev_handle,
    const int reg,
    char * tx_buffer,
    const int data_length,
    char * RFLAGS_buffer )
```

writes block of bytes from buffer to device

Parameters

<i>dev_handle</i>	device handle obtained from openI2CDevHandle
<i>reg</i>	the command/data register
<i>tx_buffer</i>	pointer to data buffer holding the data to send
<i>data_length</i>	number of bytes to send
<i>RFLAGS_buffer</i>	buffer to hold returned flags

Returns

0 on OK, negative on error.

9.36 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_interface/include/bioscara_arm_hardware_interface/arm_hardware.hpp File Reference

```
#include <memory>
#include <string>
#include <vector>
#include <set>
#include <unordered_map>
#include <mutex>
#include "bioscara_arm_hardware_driver/mJoint.h"
#include "bioscara_arm_hardware_driver/mMockJoint.h"
#include "hardware_interface/handle.hpp"
#include "hardware_interface/hardware_info.hpp"
#include "hardware_interface/system_interface.hpp"
#include "hardware_interface/types/hardware_interface_return_values.hpp"
#include "rclcpp/macros.hpp"
#include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
#include "rclcpp_lifecycle/state.hpp"
```

Classes

- class [bioscara_hardware_interfaces::BioscaraArmHardwareInterface](#)
The bioscara arm hardware interface class.
- struct [bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_homing_config_t](#)
configuration structure holding the passed homing paramters from the ros2_control urdf
- struct [bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_config_t](#)
configuration structure holding the passed paramters from the ros2_control urdf

Namespaces

- namespace [bioscara_hardware_interfaces](#)

Variables

- constexpr char [bioscara_hardware_interfaces::HW_IF_HOME](#) [] = "home"

9.37 arm_hardware.hpp

[Go to the documentation of this file.](#)

```

00001 // Copyright 2023 ros2_control Development Team
00002 //
00003 // Licensed under the Apache License, Version 2.0 (the "License");
00004 // you may not use this file except in compliance with the License.
00005 // You may obtain a copy of the License at
00006 //
00007 //     http://www.apache.org/licenses/LICENSE-2.0
00008 //
00009 // Unless required by applicable law or agreed to in writing, software
00010 // distributed under the License is distributed on an "AS IS" BASIS,
00011 // WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00012 // See the License for the specific language governing permissions and
00013 // limitations under the License.
00014
00015 #ifndef ARM_HARDWARE_HPP_
00016 #define ARM_HARDWARE_HPP_
00017
00018 #include <memory>
00019 #include <string>
00020 #include <vector>
00021 #include <set>
00022 #include <unordered_map>
00023 #include <memory>
00024 #include <mutex>
00025
00026 #include "bioscara_arm_hardware_driver/mJoint.h"
00027 #include "bioscara_arm_hardware_driver/mMockJoint.h"
00028
00029 #include "hardware_interface/handle.hpp"
00030 #include "hardware_interface/hardware_info.hpp"
00031 #include "hardware_interface/system_interface.hpp"
00032 #include "hardware_interface/types/hardware_interface_return_values.hpp"
00033 #include "rclcpp/macros.hpp"
00034 #include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
00035 #include "rclcpp_lifecycle/state.hpp"
00036
00037 namespace bioscara_hardware_interfaces
00038 {
00039     constexpr char HW_IF_HOME[] = "home";
00040
00041     class BioscaraArmHardwareInterface : public hardware_interface::SystemInterface
00042     {
00043     public:
00044         RCLCPP_SHARED_PTR_DEFINITIONS(BioscaraArmHardwareInterface)
00045
00046         hardware_interface::CallbackReturn on_init(
00047             const hardware_interface::HardwareComponentInterfaceParams &params) override;
00048
00049         hardware_interface::CallbackReturn on_shutdown(
00050             const rclcpp_lifecycle::State &previous_state) override;
00051
00052         hardware_interface::CallbackReturn on_configure(
00053             const rclcpp_lifecycle::State &previous_state) override;
00054
00055         hardware_interface::CallbackReturn on_cleanup(
00056             const rclcpp_lifecycle::State &previous_state) override;
00057
00058         hardware_interface::CallbackReturn on_activate(
00059             const rclcpp_lifecycle::State &previous_state) override;
00060
00061         hardware_interface::CallbackReturn on_deactivate(
00062             const rclcpp_lifecycle::State &previous_state) override;
00063
00064         hardware_interface::return_type read(
00065             const rclcpp::Time &time,
00066             const rclcpp::Duration &period) override;
00067
00068         hardware_interface::return_type write(
00069             const rclcpp::Time &time,
00070             const rclcpp::Duration &period) override;
00071
00072         hardware_interface::return_type prepare_command_mode_switch(
00073             const std::vector<std::string> &start_interfaces,
00074             const std::vector<std::string> &stop_interfaces) override;
00075
00076         hardware_interface::return_type perform_command_mode_switch(
00077             const std::vector<std::string> &start_interfaces,
00078             const std::vector<std::string> &stop_interfaces) override;
00079
00080         hardware_interface::CallbackReturn on_error(
00081             const rclcpp_lifecycle::State &previous_state) override;
00082     };
00083 }

```

```

00288
00289     private:
00296         struct joint_homing_config_t
00297         {
00304             float speed;
00305
00311             u_int8_t threshold;
00312
00319             u_int8_t current;
00320
00328             float acceleration = 0.01;
00329         };
00330
00337         struct joint_config_t
00338         {
00344             int i2c_address;
00345
00360             float reduction = 1;
00361
00371             float min;
00372
00382             float max;
00383
00389             u_int8_t drive_current;
00390
00396             u_int8_t hold_current;
00397
00404             u_int8_t stall_threshold;
00405
00414             float max_velocity;
00415
00424             float max_acceleration;
00425
00430             joint_homing_config_t homing;
00431         };
00432
00443         std::unordered_map<std::string, std::unique_ptr<bioscara_hardware_drivers::BaseJoint> _joints;
00444
00452         std::unordered_map<std::string, joint_config_t> _joint_cfg;
00453
00467         std::unordered_map<std::string, std::set<std::string> _joint_command_modes;
00468
00476         std::unordered_map<std::string, std::set<std::string> _new_joint_command_modes;
00477
00493         std::vector<std::pair<std::string, hardware_interface::InterfaceDescription *>
00494         _ordered_joint_state_interfaces_ptr;
00508
00509         std::mutex mtx;
00518
00519         bioscara_hardware_drivers::err_type_t start_homing(const std::string name, float velocity);
00527
00528         bioscara_hardware_drivers::err_type_t stop_homing(const std::string name);
00536
00537         void split_interface_string_to_joint_and_name(std::string interface, std::string &joint_name,
00544         std::string &interface_name);
00545
00552         bioscara_hardware_drivers::err_type_t activate_joint(const std::string name);
00553         bioscara_hardware_drivers::err_type_t deactivate_joint(const std::string name);
00554     };
00555 } // namespace bioscara_hardware_interfaces
00556
00557 #endif // ARM_HARDWARE_HPP_

```

9.38 lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_↵ interface/src/arm_hardware.cpp File Reference

```

#include "bioscara_arm_hardware_interface/arm_hardware.hpp"
#include <chrono>
#include <cmath>
#include <limits>
#include <memory>
#include <vector>
#include "hardware_interface/types/hardware_interface_type_values.hpp"
#include "rclcpp/rclcpp.hpp"

```

```
#include "pluginlib/class_list_macros.hpp"
```

Namespaces

- namespace [bioscara_hardware_interfaces](#)

9.39 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mBaseGripper.h File Reference

File containing the BaseGripper class.

```
#include "bioscara_arm_hardware_driver/uErr.h"
#include <chrono>
```

Classes

- class [bioscara_hardware_drivers::BaseGripper](#)
Generic base class for gripper control implementations.

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.39.1 Detailed Description

File containing the BaseGripper class.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

Copyright (c) 2025

Dont include this file directly, instead use one of the derived classes.

9.40 mBaseGripper.h

[Go to the documentation of this file.](#)

```

00001
00012 #ifndef MBASEGRIPPER_H
00013 #define MBASEGRIPPER_H
00014 #include "bioscara_arm_hardware_driver/uErr.h"
00015 #include <chrono>
00016
00017 namespace bioscara_hardware_drivers
00018 {
00019
00026     class BaseGripper
00027     {
00028     public:
00058         BaseGripper(float reduction, float offset, float min, float max, float backup_init_pos);
00059
00065         ~BaseGripper(void);
00066
00072         virtual err_type_t init(void);
00073
00079         virtual err_type_t deinit(void);
00080
00090         virtual err_type_t enable(void);
00091
00098         virtual err_type_t disable(void);
00099
00107         virtual err_type_t setPosition(float width);
00108
00117         virtual err_type_t getPosition(float &width);
00118
00124         virtual void setReduction(float reduction);
00125
00129         virtual void setOffset(float offset);
00130
00131     protected:
00138         err_type_t save_last_position(float pos);
00139
00146         err_type_t retrieve_last_position(float &pos);
00147
00148         float _reduction = 1;
00149         float _offset = 0;
00150         float _min = 0;
00151         float _max = 0;
00152         float _backup_init_pos = 0.0;
00153         float _pos = _backup_init_pos;
00154         float _pos_get = _pos;
00155     private:
00156         std::chrono::V2::system_clock::time_point _new_cmd_time =
std::chrono::V2::system_clock::time_point();
00157     };
00158 }
00159 #endif // MBASEGRIPPER_H

```

9.41 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_↵ hardware_driver/include/bioscara_gripper_hardware_driver/m↵ Gripper.h File Reference

File containing the Gripper class.

```

#include "bioscara_gripper_hardware_driver/mBaseGripper.h"
#include "bioscara_gripper_hardware_driver/uPWM.h"
#include "bioscara_arm_hardware_driver/uErr.h"

```

Classes

- class [bioscara_hardware_drivers::Gripper](#)
Child class implementing control of the hardware gripper.

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.41.1 Detailed Description

File containing the Gripper class.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

Copyright (c) 2025

Include this file for API functions to interact with the hardware gripper.

9.42 mGripper.h

[Go to the documentation of this file.](#)

```

00001
00013 #ifndef MGRIPPER_H
00014 #define MGRIPPER_H
00015 #include "bioscara_gripper_hardware_driver/mBaseGripper.h"
00016 #include "bioscara_gripper_hardware_driver/uPWM.h"
00017 #include "bioscara_arm_hardware_driver/uErr.h"
00018
00019 namespace bioscara_hardware_drivers
00020 {
00021     class Gripper : public BaseGripper
00022     {
00023     public:
00035         Gripper(float reduction, float offset, float min, float max, float backup_init_pos);
00036
00046         err_type_t enable(void) override;
00047
00056         err_type_t disable(void) override;
00057
00068         err_type_t setPosition(float width) override;
00069
00078         err_type_t setServoPosition(float angle);
00079
00080     protected:
00081
00082         RPI_PWM _pwm;
00083         int _freq = 50;
00084
00085     private:
00086
00087     };
00088 }
00089 #endif // MGRIPPER_H

```

9.43 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_↵ hardware_driver/include/bioscara_gripper_hardware_driver/m↵ MockGripper.h File Reference

File containing the MockGripper class.

```
#include "bioscara_gripper_hardware_driver/mBaseGripper.h"  
#include "bioscara_arm_hardware_driver/uErr.h"
```

Classes

- class [bioscara_hardware_drivers::MockGripper](#)
Child class for to mock the gripper hardware.

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.43.1 Detailed Description

File containing the MockGripper class.

Author

sbstorz

Version

0.1

Date

2025-05-27

Copyright

Copyright (c) 2025

Include this file for API functions to interact with the MockGripper.

9.44 mMockGripper.h

[Go to the documentation of this file.](#)

```
00001
00013 #ifndef MMOCKGRIPPER_H
00014 #define MMOCKGRIPPER_H
00015 #include "bioscara_gripper_hardware_driver/mBaseGripper.h"
00016 #include "bioscara_arm_hardware_driver/uErr.h"
00017
00018 namespace bioscara_hardware_drivers
00019 {
00026     class MockGripper : public BaseGripper
00027     {
00028     public:
00034         MockGripper(float reduction, float offset, float min, float max, float backup_init_pos);
00035
00036     protected:
00037     private:
00038     };
00039 }
00040 #endif // MMOCKGRIPPER_H
```

9.45 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_↵ hardware_driver/include/bioscara_gripper_hardware_driver/u↵ PWM.h File Reference

Includes source code for Hardware PWM generation on Raspberry Pi 4.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <string>
#include <iostream>
#include <math.h>
```

Classes

- class [RPI_PWM](#)

Class to create a Pulse Width Modulated (PWM) signal on the Raspberry PI 4 and 5.

9.45.1 Detailed Description

Includes source code for Hardware PWM generation on Raspberry Pi 4.

Author

sbstorz and Bernd Porr, bernd.porr@glasgow.ac.uk

Version

0.1

Date

2025-05-27

Copyright

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9.46 uPWM.h

[Go to the documentation of this file.](#)

```

00001
00012 #ifndef __RPIPWM
00013 #define __RPIPWM
00014
00015 #include <stdio.h>
00016 #include <stdlib.h>
00017 #include <string.h>
00018 #include <unistd.h>
00019 #include <string>
00020 #include <iostream>
00021 #include <math.h>
00022
00023 class RPI_PWM
00024 {
00025 public:
00026     int start(int channel, int frequency, float duty_cycle = 0, int chip = 2)
00027     {
00028         chippath = "/sys/class/pwm/pwmchip" + std::to_string(chip);
00029         pwmpath = chippath + "/pwm" + std::to_string(channel);
00030         std::string p = chippath + "/export";
00031         FILE *const fp = fopen(p.c_str(), "w");
00032         if (NULL == fp)
00033         {
00034             std::cerr << "PWM device does not exist. Make sure to add 'dtoverlay=pwm-2chan' to
/boot/firmware/config.txt.\n";
00035             return -1;
00036         }
00037         const int r = fprintf(fp, "%d", channel);
00038         fclose(fp);
00039         if (r < 0)
00040             return r;
00041         usleep(100000); // it takes a while till the PWM subdir is created
00042         per = (int)1E9 / frequency;
00043         setPeriod(per);
00044         setDutyCycle(duty_cycle);
00045         enable();
00046         return r;
00047     }
00048
00049     void stop()
00050     {
00051         disable();
00052     }
00053
00054 ~RPI_PWM()
00055 {
00056     disable();
00057 }
00058
00059 inline int setDutyCycle(float v) const
00060 {
00061     const int dc = (int)round((float)per * (v / 100.0));
00062     const int r = setDutyCycleNS(dc);
00063     return r;
00064 }
00065
00066 private:
00067
00068     void setPeriod(int ns) const
00069     {
00070         writeSYS(pwmpath + "/" + "period", ns);
00071     }
00072
00073     inline int setDutyCycleNS(int ns) const
00074     {
00075         const int r = writeSYS(pwmpath + "/" + "duty_cycle", ns);
00076         return r;
00077     }
00078
00079     void enable() const
00080     {
00081         writeSYS(pwmpath + "/" + "enable", 1);
00082     }
00083
00084     void disable() const
00085     {
00086         writeSYS(pwmpath + "/" + "enable", 0);
00087     }
00088
00089     int per = 0;
00090
00091     std::string chippath;

```

```
00125     std::string pwmpath;
00126
00127     inline int writeSYS(std::string filename, int value) const
00128     {
00129         FILE *const fp = fopen(filename.c_str(), "w");
00130         if (NULL == fp)
00131         {
00132             return -1;
00133         }
00134         const int r = fprintf(fp, "%d", value);
00135         fclose(fp);
00136         return r;
00137     }
00138 };
00139
00140 #endif
```

9.47 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mBaseGripper.cpp File Reference

```
#include "bioscara_gripper_hardware_driver/mBaseGripper.h"
#include <fstream>
#include <iostream>
#include <cmath>
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.48 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mGripper.cpp File Reference

```
#include "bioscara_gripper_hardware_driver/mGripper.h"
#include "bioscara_arm_hardware_driver/uTransmission.h"
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.49 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/src/mMockGripper.cpp File Reference

```
#include "bioscara_gripper_hardware_driver/mMockGripper.h"
```

Namespaces

- namespace [bioscara_hardware_drivers](#)

9.50 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_↵ hardware_interface/include/bioscara_gripper_hardware_↵ interface/gripper_hardware.hpp File Reference

```
#include <memory>
#include <string>
#include <vector>
#include <set>
#include <unordered_map>
#include <limits>
#include "bioscara_gripper_hardware_driver/mGripper.h"
#include "bioscara_gripper_hardware_driver/mMockGripper.h"
#include "hardware_interface/handle.hpp"
#include "hardware_interface/hardware_info.hpp"
#include "hardware_interface/system_interface.hpp"
#include "hardware_interface/types/hardware_interface_return_values.hpp"
#include "rclcpp/macros.hpp"
#include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
#include "rclcpp_lifecycle/state.hpp"
```

Classes

- class [bioscara_hardware_interfaces::BioscaraGripperHardwareInterface](#)
The bioscara gripper hardware interface class.
- struct [bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::gripper_config_t](#)
configuration structure holding the passed paramters from the ros2_control urdf

Namespaces

- namespace [bioscara_hardware_interfaces](#)

9.51 gripper_hardware.hpp

[Go to the documentation of this file.](#)

```
00001 // Copyright 2023 ros2_control Development Team
00002 //
00003 // Licensed under the Apache License, Version 2.0 (the "License");
00004 // you may not use this file except in compliance with the License.
00005 // You may obtain a copy of the License at
00006 //
00007 //     http://www.apache.org/licenses/LICENSE-2.0
00008 //
00009 // Unless required by applicable law or agreed to in writing, software
00010 // distributed under the License is distributed on an "AS IS" BASIS,
00011 // WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00012 // See the License for the specific language governing permissions and
00013 // limitations under the License.
00014
00015 #ifndef GRIPPER_HARDWARE_HPP_
00016 #define GRIPPER_HARDWARE_HPP_
00017
00018 #include <memory>
00019 #include <string>
00020 #include <vector>
00021 #include <set>
00022 #include <unordered_map>
00023 #include <memory>
00024 #include <limits>
```

```

00025
00026 #include "bioscara_gripper_hardware_driver/mGripper.h"
00027 #include "bioscara_gripper_hardware_driver/mMockGripper.h"
00028
00029 #include "hardware_interface/handle.hpp"
00030 #include "hardware_interface/hardware_info.hpp"
00031 #include "hardware_interface/system_interface.hpp"
00032 #include "hardware_interface/types/hardware_interface_return_values.hpp"
00033 #include "rclcpp/macros.hpp"
00034 #include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
00035 #include "rclcpp_lifecycle/state.hpp"
00036
00037 namespace bioscara_hardware_interfaces
00038 {
00039
00046     class BioscaraGripperHardwareInterface : public hardware_interface::SystemInterface
00047     {
00048     public:
00049         RCLCPP_SHARED_PTR_DEFINITIONS(BioscaraGripperHardwareInterface)
00050
00051         hardware_interface::CallbackReturn on_init(
00052             const hardware_interface::HardwareComponentInterfaceParams &params) override;
00053
00062         hardware_interface::CallbackReturn on_shutdown(
00063             const rclcpp_lifecycle::State &previous_state) override;
00064
00071         hardware_interface::CallbackReturn on_configure(
00072             const rclcpp_lifecycle::State &previous_state) override;
00073
00080         hardware_interface::CallbackReturn on_cleanup(
00081             const rclcpp_lifecycle::State &previous_state) override;
00082
00091         hardware_interface::CallbackReturn on_activate(
00092             const rclcpp_lifecycle::State &previous_state) override;
00093
00102         hardware_interface::CallbackReturn on_deactivate(
00103             const rclcpp_lifecycle::State &previous_state) override;
00104
00114         hardware_interface::return_type read(
00115             const rclcpp::Time &time,
00116             const rclcpp::Duration &period) override;
00117
00127         hardware_interface::return_type write(
00128             const rclcpp::Time &time,
00129             const rclcpp::Duration &period) override;
00130
00153         hardware_interface::CallbackReturn on_error(
00154             const rclcpp_lifecycle::State &previous_state) override;
00155
00156     private:
00163         struct gripper_config_t
00164         {
00165             float reduction = 1;
00166             float offset = 0;
00167             float min;
00168             float max;
00169             float init_pos;
00170         };
00171
00178         std::unique_ptr<bioscara_hardware_drivers::BaseGripper> _gripper;
00179
00184         gripper_config_t _gripper_cfg;
00185
00186         float _last_pos = std::numeric_limits<double>::quiet_NaN();
00187         float _vel = std::numeric_limits<double>::quiet_NaN();
00188     };
00189
00190 } // namespace bioscara_hardware_interfaces
00191
00192 #endif // GRIPPER_HARDWARE_HPP_

```

9.52 lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_↵ hardware_interface/src/gripper_hardware.cpp File Reference

```

#include "bioscara_gripper_hardware_interface/gripper_hardware.hpp"
#include <chrono>
#include <cmath>
#include <memory>

```

```
#include <vector>
#include "hardware_interface/types/hardware_interface_type_values.hpp"
#include "rclcpp/rclcpp.hpp"
#include "pluginlib/class_list_macros.hpp"
```

Namespaces

- namespace [bioscara_hardware_interfaces](#)

9.53 lib/ros2_ws/src/dalsa_controllers/single_trigger_↵ controller/include/single_trigger_controller/single_trigger_↵ controller.hpp File Reference

```
#include <memory>
#include <string>
#include <unordered_map>
#include <vector>
#include "control_msgs/msg/dynamic_interface_group_values.hpp"
#include "controller_interface/controller_interface.hpp"
#include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
#include "rclcpp_lifecycle/state.hpp"
#include "realtime_tools/realtime_publisher.hpp"
#include "realtime_tools/realtime_thread_safe_box.hpp"
#include "single_trigger_controller/single_trigger_controller_parameters.↵  
hpp"
```

Classes

- class [single_trigger_controller::SingleTriggerController](#)
A general purpose controller which can be used to trigger command interfaces.

Namespaces

- namespace [single_trigger_controller](#)

Typedefs

- using [single_trigger_controller::CmdType](#) = control_msgs::msg::DynamicInterfaceGroupValues
- using [single_trigger_controller::StateType](#) = control_msgs::msg::DynamicInterfaceGroupValues
- using [single_trigger_controller::CallbackReturn](#) = controller_interface::CallbackReturn
- using [single_trigger_controller::InterfacesNames](#) = std::vector< std::string >
- using [single_trigger_controller::MapOfReferencesToCommandInterfaces](#) = std::unordered_map< std::string, std::reference_wrapper< hardware_interface::LoanedCommandInterface > >
- using [single_trigger_controller::MapOfReferencesToStateInterfaces](#) = std::unordered_map< std::string, std::reference_wrapper< hardware_interface::LoanedStateInterface > >
- using [single_trigger_controller::StateInterfaces](#) = std::vector< std::reference_wrapper< hardware_↵
interface::LoanedStateInterface > >

9.54 single_trigger_controller.hpp

[Go to the documentation of this file.](#)

```
00001 // Copyright 2022 ICUBE Laboratory, University of Strasbourg
00002 //
00003 // Licensed under the Apache License, Version 2.0 (the "License");
00004 // you may not use this file except in compliance with the License.
00005 // You may obtain a copy of the License at
00006 //
00007 //     http://www.apache.org/licenses/LICENSE-2.0
00008 //
00009 // Unless required by applicable law or agreed to in writing, software
00010 // distributed under the License is distributed on an "AS IS" BASIS,
00011 // WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
00012 // See the License for the specific language governing permissions and
00013 // limitations under the License.
00014
00015 #ifndef SINGLE_TRIGGER_CONTROLLER__GPIO_COMMAND_CONTROLLER_HPP_
00016 #define SINGLE_TRIGGER_CONTROLLER__GPIO_COMMAND_CONTROLLER_HPP_
00017
00018 #include <memory>
00019 #include <string>
00020 #include <unordered_map>
00021 #include <vector>
00022
00023 #include "control_msgs/msg/dynamic_interface_group_values.hpp"
00024 #include "controller_interface/controller_interface.hpp"
00025 #include "rclcpp_lifecycle/node_interfaces/lifecycle_node_interface.hpp"
00026 #include "rclcpp_lifecycle/state.hpp"
00027 #include "realtime_tools/realtime_publisher.hpp"
00028 #include "realtime_tools/realtime_thread_safe_box.hpp"
00029
00030 #include "single_trigger_controller/single_trigger_controller_parameters.hpp"
00031
00032 namespace single_trigger_controller
00033 {
00034     using CmdType = control_msgs::msg::DynamicInterfaceGroupValues;
00035     using StateType = control_msgs::msg::DynamicInterfaceGroupValues;
00036     using CallbackReturn = controller_interface::CallbackReturn;
00037     using InterfacesNames = std::vector<std::string>;
00038     using MapOfReferencesToCommandInterfaces = std::unordered_map<
00039         std::string, std::reference_wrapper<hardware_interface::LoanedCommandInterface>;
00040     using MapOfReferencesToStateInterfaces =
00041         std::unordered_map<std::string,
00042             std::reference_wrapper<hardware_interface::LoanedStateInterface>;
00043     using StateInterfaces =
00044         std::vector<std::reference_wrapper<hardware_interface::LoanedStateInterface>;
00045
00046     class SingleTriggerController : public controller_interface::ControllerInterface
00047     {
00048     public:
00049         SingleTriggerController();
00050
00051         controller_interface::InterfaceConfiguration command_interface_configuration() const override;
00052
00053         controller_interface::InterfaceConfiguration state_interface_configuration() const override;
00054
00055         CallbackReturn on_init() override;
00056
00057         CallbackReturn on_configure(const rclcpp_lifecycle::State &previous_state) override;
00058
00059         CallbackReturn on_activate(const rclcpp_lifecycle::State &previous_state) override;
00060
00061         CallbackReturn on_deactivate(const rclcpp_lifecycle::State &previous_state) override;
00062
00063         controller_interface::return_type update(
00064             const rclcpp::Time &time, const rclcpp::Duration &period) override;
00065
00066     private:
00067         void store_command_interface_types();
00068
00069         void store_state_interface_types();
00070
00071         void initialize_state_msg();
00072
00073         void update_states();
00074
00075         controller_interface::return_type update_commands(const CmdType &commands);
00076
00077     template <typename T>
00078         std::unordered_map<std::string, std::reference_wrapper<T> > create_map_of_references_to_interfaces(
00079             const InterfacesNames &interfaces_from_params, std::vector<T> &configured_interfaces);
00080
00081 }
```

```

00212     template <typename T>
00213     bool check_if_configured_interfaces_matches_received(
00214         const InterfacesNames &interfaces_from_params, const T &configured_interfaces);
00215
00226     void apply_state_value(
00227         StateType &state_msg, std::size_t index, std::size_t interface_index) const;
00228
00237     void apply_command(
00238         const CmdType &commands, std::size_t index,
00239         std::size_t command_interface_index) const;
00240
00241
00248     InterfacesNames get_state_interfaces_names(const std::string &name) const;
00249
00257     bool update_dynamic_map_parameters();
00258
00259 protected:
00268     InterfacesNames command_interface_types_;
00269
00278     InterfacesNames state_interface_types_;
00279
00284     MapOfReferencesToCommandInterfaces command_interfaces_map_;
00285
00290     MapOfReferencesToStateInterfaces state_interfaces_map_;
00291
00296     rclcpp::Subscription<CmdType>::SharedPtr command_subscriber_{};
00297
00302     std::shared_ptr<rclcpp::Publisher<StateType>> state_publisher_{};
00303
00307     std::shared_ptr<realtime_tools::RealtimePublisher<StateType>> realtime_state_publisher_{};
00308
00313     StateType state_msg_;
00314
00319     std::shared_ptr<single_trigger_controller_parameters::ParamListener> param_listener_{};
00320
00325     single_trigger_controller_parameters::Params params_;
00326
00327 };
00328
00329 } // namespace single_trigger_controller
00330
00331 #endif // SINGLE_TRIGGER_CONTROLLER__GPIO_COMMAND_CONTROLLER_HPP_

```

9.55 lib/ros2_ws/src/dalsa_controllers/single_trigger_↔ controller/src/single_trigger_controller.cpp File Reference

```

#include "single_trigger_controller/single_trigger_controller.hpp"
#include <algorithm>
#include "controller_interface/helpers.hpp"
#include "hardware_interface/component_parser.hpp"
#include "rclcpp/logging.hpp"
#include "rclcpp/qos.hpp"
#include "rclcpp/subscription.hpp"
#include "pluginlib/class_list_macros.hpp"

```

Namespaces

- namespace [single_trigger_controller](#)

Index

`_backup_init_pos`
 bioscara hardware drivers::BaseGripper, [28](#)

`_freq`
 bioscara hardware drivers::Gripper, [79](#)

`_gripper`
 bioscara hardware interfaces::BioscaraGripperHardwareInterface, [62](#)

`_gripper_cfg`
 bioscara hardware interfaces::BioscaraGripperHardwareInterface, [62](#)

`_home`
 bioscara hardware drivers::BaseJoint, [33](#)
 bioscara hardware drivers::Joint, [85](#)
 bioscara hardware drivers::MockJoint, [107](#)

`_joint_cfg`
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [55](#)

`_joint_command_modes`
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [55](#)

`_joints`
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [56](#)

`_last_pos`
 bioscara hardware interfaces::BioscaraGripperHardwareInterface, [62](#)

`_max`
 bioscara hardware drivers::BaseGripper, [28](#)

`_min`
 bioscara hardware drivers::BaseGripper, [28](#)

`_new_cmd_time`
 bioscara hardware drivers::BaseGripper, [28](#)

`_new_joint_command_modes`
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [56](#)

`_offset`
 bioscara hardware drivers::BaseGripper, [29](#)

`_ordered_joint_state_interfaces_ptr`
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [56](#)

`_pos`
 bioscara hardware drivers::BaseGripper, [29](#)

`_pos_get`
 bioscara hardware drivers::BaseGripper, [29](#)

`_pwm`
 bioscara hardware drivers::Gripper, [79](#)

`_reduction`
 bioscara hardware drivers::BaseGripper, [29](#)

`_vel`
 bioscara hardware interfaces::BioscaraGripperHardwareInterface, [62](#)
 ~BaseGripper
 bioscara hardware drivers::BaseGripper, [25](#)
 ~BaseJoint
 bioscara hardware drivers::BaseJoint, [33](#)
 ~BioscaraPanel
 bioscara rviz plugin::BioscaraPanel, [65](#)
 ~RPI_PWM
 RPI_PWM, [115](#)

acceleration
 bioscara hardware interfaces::BioscaraArmHardwareInterface::joint, [99](#)

ACK
 joint.h, [134](#)
 u2C.h, [155](#)

activate_joint
 bioscara hardware interfaces::BioscaraArmHardwareInterface, [47](#)

ACTUATOR2JOINT
 uTransmission.h, [158](#)

address
 bioscara hardware drivers::Joint, [94](#)

ADR
 configuration.h, [130](#)

ANGLEMOVED
 bioscara hardware drivers::BaseJoint, [32](#)
 bioscara joint firmware, [16](#)

apply_command
 single_trigger_controller::SingleTriggerController, [119](#)

apply_state_value
 single_trigger_controller::SingleTriggerController, [120](#)

arm_en_btn_cb
 bioscara rviz plugin::BioscaraPanel, [66](#)

async_start_time
 bioscara hardware drivers::MockJoint, [112](#)

BaseGripper
 bioscara hardware drivers::BaseGripper, [24](#)

BaseJoint
 bioscara hardware drivers::BaseJoint, [33](#)

Bioscara C++ API Documentation, [1](#)

bioscara hardware drivers, [13](#)

COMM_ERROR, [14](#)

err_type_t, [13](#)

- ERROR, 14
- error_to_string, 14
- INCORRECT_STATE, 14
- INVALID_ARGUMENT, 14
- NOT_ENABLED, 14
- NOT_HOMED, 14
- NOT_INIT, 14
- OK, 14
- STALLED, 14
- bioscara_hardware_drivers::BaseGripper, 23
 - _backup_init_pos, 28
 - _max, 28
 - _min, 28
 - _new_cmd_time, 28
 - _offset, 29
 - _pos, 29
 - _pos_get, 29
 - _reduction, 29
 - ~BaseGripper, 25
 - BaseGripper, 24
 - deinit, 25
 - disable, 25
 - enable, 26
 - getPosition, 26
 - init, 26
 - retrieve_last_position, 27
 - save_last_position, 27
 - setOffset, 27
 - setPosition, 27
 - setReduction, 28
- bioscara_hardware_drivers::BaseJoint, 29
 - _home, 33
 - ~BaseJoint, 33
 - ANGLEMOVED, 32
 - BaseJoint, 33
 - CHECKORIENTATION, 32
 - checkOrientation, 33
 - CLEARSTALL, 32
 - current_b_cmd, 43
 - deinit, 34
 - disable, 34
 - disableCL, 34
 - DISABLECLOSEDLOOP, 32
 - DISABLEPID, 32
 - DISABLESTALLGUARD, 32
 - enable, 35
 - ENABLECLOSEDLOOP, 32
 - ENABLEPID, 32
 - ENABLESTALLGUARD, 32
 - enableStallguard, 35
 - flags, 43
 - getCurrentBCmd, 36
 - GETDRIVERRPM, 32
 - GETENCODERRPM, 32
 - getFlags, 36
 - GETMOTORSTATE, 32
 - GETPIDERROR, 32
 - getPosition, 36
 - getVelocity, 37
 - HOME, 32
 - home, 37
 - HOMEOFFSET, 32
 - init, 38
 - isBusy, 38
 - isEnabled, 38
 - isHomed, 38
 - isStalled, 39
 - MOVEANGLE, 32
 - MOVESTEPS, 32
 - moveSteps, 39
 - MOVETOANGLE, 32
 - MOVETOEND, 32
 - name, 44
 - NONE, 32
 - PING, 32
 - postHoming, 39
 - RUNCOTINOUS, 32
 - SETBRAKEMODE, 32
 - setBrakeMode, 40
 - SETCONTROLTHRESHOLD, 32
 - SETCURRENT, 32
 - setDriveCurrent, 40
 - SETHOLDCURRENT, 32
 - setHoldCurrent, 40
 - SETMAXACCELERATION, 32
 - setMaxAcceleration, 41
 - SETMAXDECELERATION, 32
 - SETMAXVELOCITY, 32
 - setMaxVelocity, 41
 - setPosition, 42
 - SETRPM, 32
 - SETUP, 32
 - setVelocity, 42
 - startHoming, 42
 - STOP, 32
 - stop, 43
 - stp_reg_t, 32
 - wait_while_busy, 43
- bioscara_hardware_drivers::Gripper, 75
 - _freq, 79
 - _pwm, 79
 - disable, 77
 - enable, 77
 - Gripper, 77
 - setPosition, 78
 - setServoPosition, 78
- bioscara_hardware_drivers::Joint, 80
 - _home, 85
 - ~Joint, 84
 - address, 94
 - checkCom, 85
 - checkOrientation, 85
 - deinit, 86
 - disableCL, 86
 - enable, 86
 - enableStallguard, 87

- getFlags, 87
- getHomingOffset, 87
- getPosition, 87
- getVelocity, 88
- handle, 94
- init, 88
- Joint, 84
- max, 94
- min, 95
- moveSteps, 88
- offset, 95
- postHoming, 89
- read, 89
- reduction, 95
- setBrakeMode, 90
- setDriveCurrent, 90
- setHoldCurrent, 90
- setHomingOffset, 92
- setMaxAcceleration, 92
- setMaxVelocity, 92
- setPosition, 93
- setVelocity, 93
- stop, 93
- write, 94
- bioscara_hardware_drivers::MockGripper, 102
 - MockGripper, 103
- bioscara_hardware_drivers::MockJoint, 104
 - _home, 107
 - async_start_time, 112
 - checkOrientation, 107
 - disable, 108
 - enable, 108
 - getDeltaT, 108
 - getFlags, 110
 - getPosition, 110
 - getVelocity, 110
 - isHomed, 111
 - last_set_position, 112
 - last_set_velocity, 112
 - MockJoint, 107
 - op_mode, 112
 - q, 112
 - qd, 113
 - setPosition, 111
 - setVelocity, 111
 - stop, 112
- bioscara_hardware_interfaces, 14
 - HW_IF_HOME, 14
- bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 44
 - _joint_cfg, 55
 - _joint_command_modes, 55
 - _joints, 56
 - _new_joint_command_modes, 56
 - _ordered_joint_state_interfaces_ptr, 56
 - activate_joint, 47
 - deactivate_joint, 47
 - mtx, 56
 - on_activate, 47
 - on_cleanup, 48
 - on_configure, 48
 - on_deactivate, 48
 - on_error, 49
 - on_init, 50
 - on_shutdown, 51
 - perform_command_mode_switch, 51
 - prepare_command_mode_switch, 52
 - read, 53
 - split_interface_string_to_joint_and_name, 53
 - start_homing, 54
 - stop_homing, 54
 - write, 54
- bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_conf, 95
 - drive_current, 96
 - hold_current, 96
 - homing, 96
 - i2c_address, 97
 - max, 97
 - max_acceleration, 97
 - max_velocity, 97
 - min, 97
 - reduction, 98
 - stall_threshold, 98
- bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint_hom, 98
 - acceleration, 99
 - current, 99
 - speed, 99
 - threshold, 99
- bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 57
 - _gripper, 62
 - _gripper_cfg, 62
 - _last_pos, 62
 - _vel, 62
 - on_activate, 58
 - on_cleanup, 59
 - on_configure, 59
 - on_deactivate, 59
 - on_error, 60
 - on_init, 60
 - on_shutdown, 61
 - read, 61
 - write, 61
- bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::grippe, 79
 - init_pos, 80
 - max, 80
 - min, 80
 - offset, 80
 - reduction, 80
- bioscara_joint_firmware, 15
 - ANGLEMOVED, 16
 - blk_reg, 20
 - blocking_handler, 17

- CHECKORIENTATION, 17
- CLEARSTALL, 17
- DISABLECLOSEDLOOP, 17
- DISABLEPID, 17
- DISABLESTALLGUARD, 17
- ENABLECLOSEDLOOP, 17
- ENABLEPID, 17
- ENABLESTALLGUARD, 16
- GETDRIVERRPM, 16
- GETENCODERRPM, 17
- GETMOTORSTATE, 16
- GETPIDERROR, 17
- HOME, 17
- HOMEOFFSET, 17
- loop, 17
- MOVEANGLE, 16
- MOVESTEPS, 16
- MOVETOANGLE, 16
- MOVETOEND, 17
- non_blocking_handler, 18
- PING, 16
- readValue, 18
- receiveEvent, 18
- reg, 20
- requestEvent, 19
- RUNCOTINOUS, 16
- rx_buf, 20
- rx_data_ready, 20
- rx_length, 20
- set_flags_for_blocking_handler, 19
- SETBRAKEMODE, 17
- SETCONTROLTHRESHOLD, 17
- SETCURRENT, 16
- SETHOLDCURRENT, 16
- SETMAXACCELERATION, 16
- SETMAXDECELERATION, 16
- SETMAXVELOCITY, 16
- SETRPM, 16
- SETUP, 16
- setup, 19
- stall_threshold, 19
- stepper, 20
- STOP, 17
- stp_reg_t, 16
- tx_buf, 21
- tx_length, 21
- writeValue, 20
- bioscara_joint_firmware::Lowpass, 100
 - K, 101
 - Lowpass, 100
 - resetState, 101
 - tau, 101
 - Ts, 101
 - updateState, 101
 - x, 101
- bioscara_joint_firmware::MovMax, 113
 - cb_data, 114
 - cb_index, 114
- M, 114
 - MovMax, 113
 - updateState, 114
- bioscara_rviz_plugin, 21
- bioscara_rviz_plugin::BioscaraPanel, 63
 - ~BioscaraPanel, 65
 - arm_en_btn_cb, 66
 - BioscaraPanel, 65
 - check_activation_conditions, 66
 - cm_state_callback, 66
 - cm_state_subscription_, 73
 - configure_controller, 66
 - configure_controller_client_, 73
 - controller_states_, 73
 - ctrl_en_btn_cb, 67
 - dynamic_joint_state_msg_to_map, 67
 - ensure_jsb_is_active, 67
 - gripper_en_btn_cb, 67
 - hardware_state_client_, 74
 - hardware_states_, 74
 - homing_cmd, 67
 - homing_publisher_, 74
 - joint_state_callback, 68
 - joint_state_subscription_, 74
 - joint_states_, 74
 - named_lcs_msg_to_map, 68
 - node_, 74
 - onInitialize, 68
 - prepopulate_joint_state_map, 69
 - prepopulate_state_map, 69
 - print_cm_map, 69
 - prune_timer_, 75
 - set_controller_state, 70
 - set_en_btn, 70
 - set_hardware_component_state, 70
 - set_homing_state_label, 71
 - set_state_label, 71
 - switch_controller_client_, 75
 - switch_controllers, 71
 - target_state_from_current, 72
 - ui_, 75
 - update_homing_grp_state, 72
 - update_homing_state_labels, 72
 - update_state_label_and_btn, 72
 - update_state_labels_and_btns, 73
- BioscaraPanel
 - bioscara_rviz_plugin::BioscaraPanel, 65
- blk_reg
 - bioscara_joint_firmware, 20
- blocking_handler
 - bioscara_joint_firmware, 17
- CallbackReturn
 - single_trigger_controller, 21
- cb_data
 - bioscara_joint_firmware::MovMax, 114
- cb_index
 - bioscara_joint_firmware::MovMax, 114
- check_activation_conditions

- bioscara_rviz_plugin::BioscaraPanel, 66
- check_if_configured_interfaces_matches_received
 - single_trigger_controller::SingleTriggerController, 120
- checkCom
 - bioscara_hardware_drivers::Joint, 85
- CHECKORIENTATION
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- checkOrientation
 - bioscara_hardware_drivers::BaseJoint, 33
 - bioscara_hardware_drivers::Joint, 85
 - bioscara_hardware_drivers::MockJoint, 107
- chippath
 - RPI_PWM, 117
- CLEARSTALL
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- closeI2CDevHandle
 - ul2C.cpp, 162
 - ul2C.h, 155
- cm_state_callback
 - bioscara_rviz_plugin::BioscaraPanel, 66
- cm_state_subscription_
 - bioscara_rviz_plugin::BioscaraPanel, 73
- CmdType
 - single_trigger_controller, 21
- COMM_ERROR
 - bioscara_hardware_drivers, 14
- command_interface_configuration
 - single_trigger_controller::SingleTriggerController, 120
- command_interface_types_
 - single_trigger_controller::SingleTriggerController, 125
- command_interfaces_map_
 - single_trigger_controller::SingleTriggerController, 125
- command_subscriber_
 - single_trigger_controller::SingleTriggerController, 125
- common.h
 - DUMP_BUFFER, 142
- configuration.h
 - ADR, 130
 - MAXACCEL, 130
 - MAXVEL, 130
- configure_controller
 - bioscara_rviz_plugin::BioscaraPanel, 66
- configure_controller_client_
 - bioscara_rviz_plugin::BioscaraPanel, 73
- controller_states_
 - bioscara_rviz_plugin::BioscaraPanel, 73
- create_map_of_references_to_interfaces
 - single_trigger_controller::SingleTriggerController, 121
- ctrl_en_btn_cb
 - bioscara_rviz_plugin::BioscaraPanel, 67

- current
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, 99
- current_b_cmd
 - bioscara_hardware_drivers::BaseJoint, 43
- deactivate_joint
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 47
- DEG2RAD
 - uTransmission.h, 158
- deinit
 - bioscara_hardware_drivers::BaseGripper, 25
 - bioscara_hardware_drivers::BaseJoint, 34
 - bioscara_hardware_drivers::Joint, 86
- disable
 - bioscara_hardware_drivers::BaseGripper, 25
 - bioscara_hardware_drivers::BaseJoint, 34
 - bioscara_hardware_drivers::Gripper, 77
 - bioscara_hardware_drivers::MockJoint, 108
 - RPI_PWM, 115
- disableCL
 - bioscara_hardware_drivers::BaseJoint, 34
 - bioscara_hardware_drivers::Joint, 86
- DISABLECLOSEDLOOP
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- DISABLEPID
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- DISABLESTALLGUARD
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- docs/doxygen/index.md, 129
- drive_current
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, 96
- DUMP_BUFFER
 - common.h, 142
 - joint.h, 134
- dynamic_joint_state_msg_to_map
 - bioscara_rviz_plugin::BioscaraPanel, 67
- enable
 - bioscara_hardware_drivers::BaseGripper, 26
 - bioscara_hardware_drivers::BaseJoint, 35
 - bioscara_hardware_drivers::Gripper, 77
 - bioscara_hardware_drivers::Joint, 86
 - bioscara_hardware_drivers::MockJoint, 108
 - RPI_PWM, 115
- ENABLECLOSEDLOOP
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- ENABLEPID
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- ENABLESTALLGUARD
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16

- enableStallguard
 - bioscara_hardware_drivers::BaseJoint, [35](#)
 - bioscara_hardware_drivers::Joint, [87](#)
- ensure_jsb_is_active
 - bioscara_rviz_plugin::BioscaraPanel, [67](#)
- err_type_t
 - bioscara_hardware_drivers, [13](#)
- ERROR
 - bioscara_hardware_drivers, [14](#)
- error_to_string
 - bioscara_hardware_drivers, [14](#)
- flags
 - bioscara_hardware_drivers::BaseJoint, [43](#)
- get_state_interfaces_names
 - single_trigger_controller::SingleTriggerController, [121](#)
- getCurrentBCmd
 - bioscara_hardware_drivers::BaseJoint, [36](#)
- getDeltaT
 - bioscara_hardware_drivers::MockJoint, [108](#)
- GETDRIVERRPM
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [16](#)
- GETENCODERRPM
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [17](#)
- getFlags
 - bioscara_hardware_drivers::BaseJoint, [36](#)
 - bioscara_hardware_drivers::Joint, [87](#)
 - bioscara_hardware_drivers::MockJoint, [110](#)
- getHomingOffset
 - bioscara_hardware_drivers::Joint, [87](#)
- GETMOTORSTATE
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [16](#)
- GETPIDERROR
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [17](#)
- getPosition
 - bioscara_hardware_drivers::BaseGripper, [26](#)
 - bioscara_hardware_drivers::BaseJoint, [36](#)
 - bioscara_hardware_drivers::Joint, [87](#)
 - bioscara_hardware_drivers::MockJoint, [110](#)
- getVelocity
 - bioscara_hardware_drivers::BaseJoint, [37](#)
 - bioscara_hardware_drivers::Joint, [88](#)
 - bioscara_hardware_drivers::MockJoint, [110](#)
- Gripper
 - bioscara_hardware_drivers::Gripper, [77](#)
- gripper_en_btn_cb
 - bioscara_rviz_plugin::BioscaraPanel, [67](#)
- handle
 - bioscara_hardware_drivers::Joint, [94](#)
- hardware_state_client_
 - bioscara_rviz_plugin::BioscaraPanel, [74](#)
- hardware_states_
 - bioscara_rviz_plugin::BioscaraPanel, [74](#)
- hold_current
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, [96](#)
- HOME
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [17](#)
- home
 - bioscara_hardware_drivers::BaseJoint, [37](#)
- HOMEOFFSET
 - bioscara_hardware_drivers::BaseJoint, [32](#)
 - bioscara_joint_firmware, [17](#)
- homing
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, [96](#)
- homing_cmd
 - bioscara_rviz_plugin::BioscaraPanel, [67](#)
- homing_publisher_
 - bioscara_rviz_plugin::BioscaraPanel, [74](#)
- HW_IF_HOME
 - bioscara_hardware_interfaces, [14](#)
- i2c_address
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, [97](#)
- INCORRECT_STATE
 - bioscara_hardware_drivers, [14](#)
- init
 - bioscara_hardware_drivers::BaseGripper, [26](#)
 - bioscara_hardware_drivers::BaseJoint, [38](#)
 - bioscara_hardware_drivers::Joint, [88](#)
- init_pos
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::g, [80](#)
- initialize_state_msg
 - single_trigger_controller::SingleTriggerController, [122](#)
- INT_handler
 - joint_comm_node.cpp, [160](#)
- InterfacesNames
 - single_trigger_controller, [22](#)
- INVALID_ARGUMENT
 - bioscara_hardware_drivers, [14](#)
- isBusy
 - bioscara_hardware_drivers::BaseJoint, [38](#)
- isEnabled
 - bioscara_hardware_drivers::BaseJoint, [38](#)
- isHomed
 - bioscara_hardware_drivers::BaseJoint, [38](#)
 - bioscara_hardware_drivers::MockJoint, [111](#)
- isStalled
 - bioscara_hardware_drivers::BaseJoint, [39](#)
- J1
 - joint.ino, [138](#)
 - joint_comm_node.cpp, [160](#)
- J2
 - joint_comm_node.cpp, [160](#)
- J3

- joint_comm_node.cpp, 160
- J4
 - joint_comm_node.cpp, 160
- Joint
 - bioscara_hardware_drivers::Joint, 84
- joint.h
 - ACK, 134
 - DUMP_BUFFER, 134
 - MAX_BUFFER, 135
 - NACK, 135
 - RFLAGS_SIZE, 135
- joint.ino
 - J1, 138
 - loop, 138
 - setup, 138
- JOINT2ACTUATOR
 - uTransmission.h, 158
- joint_comm_node.cpp
 - INT_handler, 160
 - J1, 160
 - J2, 160
 - J3, 160
 - J4, 160
 - main, 160
- joint_state_callback
 - bioscara_rviz_plugin::BioscaraPanel, 68
- joint_state_subscription_
 - bioscara_rviz_plugin::BioscaraPanel, 74
- joint_states_
 - bioscara_rviz_plugin::BioscaraPanel, 74
- K
 - bioscara_joint_firmware::Lowpass, 101
- last_set_position
 - bioscara_hardware_drivers::MockJoint, 112
- last_set_velocity
 - bioscara_hardware_drivers::MockJoint, 112
- lib/joint_firmware/joint/configuration.h, 129, 130
- lib/joint_firmware/joint/filters.h, 131, 132
- lib/joint_firmware/joint/joint.h, 132, 135
- lib/joint_firmware/joint/joint.ino, 136
- lib/joint_firmware/joint/stall.h, 138, 139
- lib/ros2_ws/src/dalsa_bioscara_rviz_plugin/include/bioscara_rviz_plugin/bioscara_rviz_plugin.h, 139, 140
- lib/ros2_ws/src/dalsa_bioscara_rviz_plugin/src/bioscara_panel.cpp
 - 141
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/common.h, 142, 143
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mBaseJoint.h, 143, 144
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mJoint.h, 146, 147
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mMockJoint.h, 148, 149
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/mTransmission.h, 149, 150
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/uErr.h, 151, 153
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/uTransmission.h, 154, 157
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/include/bioscara_arm_hardware_driver/uTransmission.h, 157, 159
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/joint.ino, 159
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 161
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 161
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 161
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 161
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 162
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 164, 165
- lib/ros2_ws/src/dalsa_bioscara_arm/bioscara_arm_hardware_driver/src/main.cpp, 166
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/common.h, 167, 168
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mBaseGripper.h, 168, 169
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mGripper.h, 170, 171
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/mMockGripper.h, 171, 172
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uTransmission.h, 173
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uTransmission.h, 173
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uTransmission.h, 173
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uTransmission.h, 174
- lib/ros2_ws/src/dalsa_bioscara_grippers/bioscara_gripper_hardware_driver/include/bioscara_gripper_hardware_driver/uTransmission.h, 175
- lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/include/single_trigger_controller/single_trigger_controller.h, 176, 177
- lib/ros2_ws/src/dalsa_controllers/single_trigger_controller/src/single_trigger_controller.cpp, 178
- loop
 - bioscara_rviz_plugin/bioscara_panel.h, 17
 - joint.ino, 138
- Lowpass
 - bioscara_joint_firmware::Lowpass, 100
- M
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143
- M_PI
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143
- main
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143
- MapOfReferencesToCommandInterfaces
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143
- MapOfReferencesToStateInterfaces
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143
- max
 - bioscara_arm_hardware_driver/mBaseJoint.h, 143
 - bioscara_arm_hardware_driver/mMockJoint.h, 143
 - bioscara_arm_hardware_driver/mTransmission.h, 143

- bioscara_hardware_drivers::Joint, 94
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 97
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 80
- max_acceleration
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 97
- MAX_BUFFER
 - joint.h, 135
 - ul2C.h, 155
- max_velocity
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 97
- MAXACCEL
 - configuration.h, 130
- MAXVEL
 - configuration.h, 130
- min
 - bioscara_hardware_drivers::Joint, 95
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 97
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 80
- MockGripper
 - bioscara_hardware_drivers::MockGripper, 103
- MockJoint
 - bioscara_hardware_drivers::MockJoint, 107
- MOVEANGLE
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- MOVESTEPS
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- moveSteps
 - bioscara_hardware_drivers::BaseJoint, 39
 - bioscara_hardware_drivers::Joint, 88
- MOVETOANGLE
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- MOVETOEND
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- MovMax
 - bioscara_joint_firmware::MovMax, 113
- mtx
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 56
- NACK
 - joint.h, 135
 - ul2C.h, 155
- name
 - bioscara_hardware_drivers::BaseJoint, 44
- named_lcs_msg_to_map
 - bioscara_rviz_plugin::BioscaraPanel, 68
- node_
 - bioscara_rviz_plugin::BioscaraPanel, 74
- non_blocking_handler
 - bioscara_joint_firmware, 18
 - BaseJoint::joint_config_t, bioscara_hardware_drivers::BaseJoint, 32
 - BaseGripper::gripper_config_t, bioscara_hardware_drivers, 14
 - NOT_HOMED
 - bioscara_hardware_drivers, 14
 - NOT_INIT
 - bioscara_hardware_drivers, 14
 - offset
 - bioscara_hardware_drivers::Joint, 95
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 80
 - OK
 - bioscara_hardware_drivers, 14
 - on_activate
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 47
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 58
 - single_trigger_controller::SingleTriggerController, 122
 - on_cleanup
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 48
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 59
 - on_configure
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 48
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 59
 - single_trigger_controller::SingleTriggerController, 123
 - on_deactivate
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 48
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 59
 - single_trigger_controller::SingleTriggerController, 123
 - on_error
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 49
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 60
 - on_init
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 50
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 60
 - single_trigger_controller::SingleTriggerController, 123
 - on_shutdown
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 51
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 61

- onInitialize
 - bioscara_rviz_plugin::BioscaraPanel, 68
- op_mode
 - bioscara_hardware_drivers::MockJoint, 112
- openI2CDevHandle
 - ul2C.cpp, 162
 - ul2C.h, 156
- param_listener_
 - single_trigger_controller::SingleTriggerController, 126
- params_
 - single_trigger_controller::SingleTriggerController, 126
- per
 - RPI_PWM, 117
- perform_command_mode_switch
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 51
- PING
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- postHoming
 - bioscara_hardware_drivers::BaseJoint, 39
 - bioscara_hardware_drivers::Joint, 89
- prepare_command_mode_switch
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 52
- prepopulate_joint_state_map
 - bioscara_rviz_plugin::BioscaraPanel, 69
- prepopulate_state_map
 - bioscara_rviz_plugin::BioscaraPanel, 69
- print_cm_map
 - bioscara_rviz_plugin::BioscaraPanel, 69
- prune_timer_
 - bioscara_rviz_plugin::BioscaraPanel, 75
- pwmpath
 - RPI_PWM, 117
- q
 - bioscara_hardware_drivers::MockJoint, 112
- qd
 - bioscara_hardware_drivers::MockJoint, 113
- RAD2DEG
 - uTransmission.h, 159
- read
 - bioscara_hardware_drivers::Joint, 89
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface, 53
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface, 61
- readFromI2CDev
 - ul2C.cpp, 163
 - ul2C.h, 156
- readValue
 - bioscara_joint_firmware, 18
- realtime_state_publisher_
 - single_trigger_controller::SingleTriggerController, 126
- receiveEvent
 - bioscara_joint_firmware, 18
- reduction
 - bioscara_hardware_drivers::Joint, 95
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::joint, 98
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface::joint, 80
- reg
 - bioscara_joint_firmware, 20
- requestEvent
 - bioscara_joint_firmware, 19
- resetState
 - bioscara_joint_firmware::Lowpass, 101
- retrieve_last_position
 - bioscara_hardware_drivers::BaseGripper, 27
- RETURN_ON_ERROR
 - uErr.h, 152
- RETURN_ON_FALSE
 - uErr.h, 152
- RETURN_ON_NEGATIVE
 - uErr.h, 152
- RFLAGS_SIZE
 - joint.h, 135
 - uErr.h, 155
- RPI_PWM, 114
 - ~RPI_PWM, 115
- chippath, 117
- disable, 115
- enable, 115
- per, 117
- pwmpath, 117
- setDutyCycle, 116
- setDutyCycleNS, 116
- setPeriod, 116
- start, 116
- stop, 117
- writeSYS, 117
- RUNCOTINOUS
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- rx_buf
 - bioscara_joint_firmware, 20
- rx_data_ready
 - bioscara_joint_firmware, 20
- rx_length
 - bioscara_joint_firmware, 20
- save_last_position
 - bioscara_hardware_drivers::BaseGripper, 27
- set_controller_state
 - bioscara_rviz_plugin::BioscaraPanel, 70
- set_en_btn
 - bioscara_rviz_plugin::BioscaraPanel, 70
- set_flags_for_blocking_handler
 - bioscara_joint_firmware, 19
- set_hardware_component_state

- bioscara_rviz_plugin::BioscaraPanel, 70
- set_homing_state_label
 - bioscara_rviz_plugin::BioscaraPanel, 71
- set_state_label
 - bioscara_rviz_plugin::BioscaraPanel, 71
- SETBRAKEMODE
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- setBrakeMode
 - bioscara_hardware_drivers::BaseJoint, 40
 - bioscara_hardware_drivers::Joint, 90
- SETCONTROLTHRESHOLD
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- SETCURRENT
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setDriveCurrent
 - bioscara_hardware_drivers::BaseJoint, 40
 - bioscara_hardware_drivers::Joint, 90
- setDutyCycle
 - RPI_PWM, 116
- setDutyCycleNS
 - RPI_PWM, 116
- SETHOLDCURRENT
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setHoldCurrent
 - bioscara_hardware_drivers::BaseJoint, 40
 - bioscara_hardware_drivers::Joint, 90
- setHomingOffset
 - bioscara_hardware_drivers::Joint, 92
- SETMAXACCELERATION
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setMaxAcceleration
 - bioscara_hardware_drivers::BaseJoint, 41
 - bioscara_hardware_drivers::Joint, 92
- SETMAXDECELERATION
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- SETMAXVELOCITY
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setMaxVelocity
 - bioscara_hardware_drivers::BaseJoint, 41
 - bioscara_hardware_drivers::Joint, 92
- setOffset
 - bioscara_hardware_drivers::BaseGripper, 27
- setPeriod
 - RPI_PWM, 116
- setPosition
 - bioscara_hardware_drivers::BaseGripper, 27
 - bioscara_hardware_drivers::BaseJoint, 42
 - bioscara_hardware_drivers::Gripper, 78
 - bioscara_hardware_drivers::Joint, 93
 - bioscara_hardware_drivers::MockJoint, 111
- setReduction
- bioscara_hardware_drivers::BaseGripper, 28
- SETRPM
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setServoPosition
 - bioscara_hardware_drivers::Gripper, 78
- SETUP
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 16
- setup
 - bioscara_joint_firmware, 19
 - joint.ino, 138
- setVelocity
 - bioscara_hardware_drivers::BaseJoint, 42
 - bioscara_hardware_drivers::Joint, 93
 - bioscara_hardware_drivers::MockJoint, 111
- single_trigger_controller, 21
 - CallbackReturn, 21
 - CmdType, 21
 - InterfacesNames, 22
 - MapOfReferencesToCommandInterfaces, 22
 - MapOfReferencesToStateInterfaces, 22
 - StateInterfaces, 22
 - StateType, 22
- single_trigger_controller::SingleTriggerController, 117
 - apply_command, 119
 - apply_state_value, 120
 - check_if_configured_interfaces_matches_received, 120
 - command_interface_configuration, 120
 - command_interface_types_, 125
 - command_interfaces_map_, 125
 - command_subscriber_, 125
 - create_map_of_references_to_interfaces, 121
 - get_state_interfaces_names, 121
 - initialize_state_msg, 122
 - on_activate, 122
 - on_configure, 122
 - on_deactivate, 123
 - on_init, 123
 - param_listener_, 126
 - params_, 126
 - realtime_state_publisher_, 126
 - SingleTriggerController, 119
 - state_interface_configuration, 123
 - state_interface_types_, 126
 - state_interfaces_map_, 126
 - state_msg_, 126
 - state_publisher_, 127
 - store_command_interface_types, 123
 - store_state_interface_types, 124
 - update, 124
 - update_commands, 124
 - update_dynamic_map_parameters, 125
 - update_states, 125
- SingleTriggerController
 - single_trigger_controller::SingleTriggerController, 119

- speed
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::start_homing, 16
 - 99
- split_interface_string_to_joint_and_name
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::split_interface_string_to_joint_and_name, 53
- stall_threshold
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::stall_threshold, 98
 - bioscara_joint_firmware, 19
- STALLED
 - bioscara_hardware_drivers, 14
- start
 - RPI_PWM, 116
- start_homing
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::start_homing, 54
- startHoming
 - bioscara_hardware_drivers::BaseJoint, 42
- state_interface_configuration
 - single_trigger_controller::SingleTriggerController, 123
- state_interface_types_
 - single_trigger_controller::SingleTriggerController, 126
- state_interfaces_map_
 - single_trigger_controller::SingleTriggerController, 126
- state_msg_
 - single_trigger_controller::SingleTriggerController, 126
- state_publisher_
 - single_trigger_controller::SingleTriggerController, 127
- StateInterfaces
 - single_trigger_controller, 22
- StateType
 - single_trigger_controller, 22
- stepper
 - bioscara_joint_firmware, 20
- STOP
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 17
- stop
 - bioscara_hardware_drivers::BaseJoint, 43
 - bioscara_hardware_drivers::Joint, 93
 - bioscara_hardware_drivers::MockJoint, 112
 - RPI_PWM, 117
- stop_homing
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::stop_homing, 54
- store_command_interface_types
 - single_trigger_controller::SingleTriggerController, 123
- store_state_interface_types
 - single_trigger_controller::SingleTriggerController, 124
- stp_reg_t
 - bioscara_hardware_drivers::BaseJoint, 32
 - bioscara_joint_firmware, 19
 - switch_controller_client_
 - bioscara_rviz_plugin::BioscaraPanel, 75
 - switch_controllers
 - bioscara_rviz_plugin::BioscaraPanel, 71
 - target_state_from_current
 - bioscara_rviz_plugin::BioscaraPanel, 72
 - tau
 - bioscara_joint_firmware::Lowpass, 101
 - threshold
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface::threshold, 99
 - Todo List, 3
 - TriggerInterface,
 - bioscara_joint_firmware::Lowpass, 101
 - tx_buf
 - bioscara_joint_firmware, 21
 - tx_length
 - bioscara_joint_firmware, 21
 - uErr.h
 - RETURN_ON_ERROR, 152
 - RETURN_ON_FALSE, 152
 - RETURN_ON_NEGATIVE, 152
 - ul2C.cpp
 - closeI2CDevHandle, 162
 - openI2CDevHandle, 162
 - readFromI2CDev, 163
 - writeToI2CDev, 163
 - ul2C.h
 - ACK, 155
 - closeI2CDevHandle, 155
 - MAX_BUFFER, 155
 - NACK, 155
 - openI2CDevHandle, 156
 - readFromI2CDev, 156
 - RFLAGS_SIZE, 155
 - writeToI2CDev, 156
 - ui_
 - bioscara_rviz_plugin::BioscaraPanel, 75
 - update
 - single_trigger_controller::SingleTriggerController, 124
 - update_commands
 - single_trigger_controller::SingleTriggerController, 124
 - update_dynamic_map_parameters
 - single_trigger_controller::SingleTriggerController, 125
 - update_homing_grp_state
 - bioscara_rviz_plugin::BioscaraPanel, 72
 - update_homing_state_labels
 - bioscara_rviz_plugin::BioscaraPanel, 72
 - update_state_label_and_btn
 - bioscara_rviz_plugin::BioscaraPanel, 72
 - update_state_labels_and_btns
 - bioscara_rviz_plugin::BioscaraPanel, 73

- update_states
 - single_trigger_controller::SingleTriggerController,
125
- updateState
 - bioscara_joint_firmware::Lowpass, 101
 - bioscara_joint_firmware::MovMax, 114
- uTransmission.h
 - ACTUATOR2JOINT, 158
 - DEG2RAD, 158
 - JOINT2ACTUATOR, 158
 - M_PI, 158
 - RAD2DEG, 159
- wait_while_busy
 - bioscara_hardware_drivers::BaseJoint, 43
- write
 - bioscara_hardware_drivers::Joint, 94
 - bioscara_hardware_interfaces::BioscaraArmHardwareInterface,
54
 - bioscara_hardware_interfaces::BioscaraGripperHardwareInterface,
61
- writeSYS
 - RPI_PWM, 117
- writeTol2CDev
 - ul2C.cpp, 163
 - ul2C.h, 156
- writeValue
 - bioscara_joint_firmware, 20
- x
 - bioscara_joint_firmware::Lowpass, 101